IMPULSE
Learning Center

C1 · Advanced

Milliy sertifikat · CEFR

Multilevel

B2 darajadan **C1 darajaga qadar**

Reading · Listening · Writing · Speaking

Unit 1 2 3 4 5 6 7 8 9 10 | + Answer key

HOW TO USE THIS BOOK

Every unit follows the same route — from speaking and vocabulary, through two articles, grammar and academic writing, to real **Multilevel (CEFR)** exam practice in all four skills.

Warm-up speaking
Start each unit on the cover: discuss the four questions with a partner, about 30 seconds each.

Article & Vocabulary
Match the key words to their meanings first, then read Article 1. Words in red are the unit's key vocabulary.

Match the heading
Skim each paragraph of Article 1 and choose its heading. One heading is extra.

Language development
Learn collocations as whole chunks from the table, then use them to complete the sentences.

Article 2 & Discussion
Choose the meaning of each word in bold, read the second article and discuss the questions.

Grammar for writing
Study the rules and the blue examples, then do Exercises A and B.

Academic writing
Practise one academic writing skill and finish with a short writing task.

Reading practice
A real Multilevel (CEFR) reading part. Time yourself, then learn the 30-word wordlist.

Listening practice
A Multilevel listening part — scan the QR code below for the audio. Learn the phrases & idioms.

Writing
Useful language, rules, a sample answer and your own task: letters (Part 1) or essays (Part 2).

Speaking
Multilevel speaking task with a sample answer and useful language. Record yourself and compare.

Answer key
Check your answers at the back of the book. Listening answers are added by your teacher.

EXAM PRACTICE IN EACH UNIT

Unit	Topic	Reading	Listening	Writing	Speaking
1	Conservation	Parts 1 + 2	Parts 1 + 2	Part 1.1	Part 1.2
2	Design	Part 3	Podcast + Part 3	Part 1.2	Part 2
3	Technology	Part 4	Part 4	Part 2	Part 3
4	Risk	Part 5	Part 5	Part 2	Part 3
5	Manufacturing	Part 1	Part 6	Parts 1.1 + 1.2	Part 1.1
6	Environment	Part 2	Parts 1 + 2	Part 2	Part 2
7	Architecture	Part 3	Part 3	Parts 1.1 + 1.2	Part 3
8	Energy	Part 4	Part 4	Part 2	Part 2
9	Medicine	Part 5	Part 5	Parts 1.1 + 1.2	Part 1.1
10	Ageing	Part 3	Part 6	Part 2	Part 3

AUDIO

Scan to open our Telegram channel with all the listening audio.

t.me/multilevel_express

Conservation UNIT 1

Preserving Buildings & Memories

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. Is there an old building in your city that you think should be preserved? Why?

2. Do you keep any old photos, letters, or objects that mean a lot to you?

3. Do you think older things are usually better made than new things, or is that just nostalgia?

4. If you had to choose between restoring an old building and building something new in its place, which would you choose?

UNDERSTANDING KEY VOCABULARY

Article 1

1. The sentences below are about the value we place on old buildings. Choose the correct meaning (a, b or c) for the word in bold.

1 The critic said the museum's new galleries felt genuinely **vibrant**, full of colour, noise and energy.
a dull and quiet
b full of life and energy
c old-fashioned and outdated

2 Because the building was structurally unsafe, the council finally approved its **demolition**.
a the act of knocking a building down
b the act of repairing a building
c the act of selling a building

3 The old cinema is currently closed for **renovation** and should reopen next spring.
a the process of demolishing a building
b the process of repairing and improving a building
c the process of buying a building

4 Some of the damage caused by the fire turned out to be **irreversible**.
a able to be repaired completely
b impossible to change back to how it was before
c not very serious

5 Architects are increasingly **adaptive** in their approach, redesigning old factories for completely new uses rather than knocking them down.

a willing and able to change to suit new situations

b unwilling to make any changes

c focused only on brand-new buildings

6 Only a small number of **affluent** buyers could afford apartments in the restored waterfront building.

a having very little money

b having a lot of money

c interested in local history

7 From the outside, the warehouse looked like just another **anonymous** grey building, with nothing to suggest its history.

a famous and well-known

b beautiful and colourful

c without any distinctive or identifying features

8 After years of debate, the city finally **opted for** a compromise: restoring the facade while rebuilding the interior.

a rejected

b chose, after considering the alternatives

c forgot about completely

SHOULD WE ALWAYS SAVE THE PAST?

1 Walk through the centre of almost any old city and you'll find at least one building caught between two fates: careful restoration or the wrecking ball. Local councils across the world face the same dilemma every year — should a crumbling warehouse, theatre or factory be saved, or is **demolition** sometimes the more sensible choice?

2 Supporters of conservation argue that historic buildings give a city its character. A carefully planned **renovation** can turn a neglected structure into a genuinely **vibrant** public space — a market, a gallery, a community centre — rather than letting it sit empty and decaying. Increasingly, architects favour **adaptive** reuse: instead of restoring a building to look exactly as it once did, they redesign the interior for a completely new purpose while keeping the original facade. An old power station, for example, might become an art museum, its huge turbine hall left open rather than divided into small anonymous offices.

3 Critics, however, point out that conservation is expensive. A full structural renovation can cost far more than building something new, and this often means that only **affluent** developers, or well-funded public bodies, can afford to take on such projects at all. In cities with tight housing budgets, some argue that money spent preserving one old building might have built dozens of affordable new homes instead. There is also a harder truth: some structures have decayed so badly, or were built so cheaply in the first place, that the damage is effectively **irreversible**, and no renovation, however costly, could realistically save them.

4 In practice, most cities now **opt for** a case-by-case approach rather than a single fixed rule. A building of genuine historic or architectural value, even one that looks like an **anonymous** ruin to the untrained eye, might be protected by law and carefully restored, while a structurally unsound building with no real historic significance is cleared to make way for something new. This compromise rarely satisfies everyone: preservationists complain that too much is lost, while developers complain that protection laws slow down badly needed housing and infrastructure projects.

5 What almost everyone agrees on, though, is that the decision should never be made carelessly. Once a historic building has been demolished, the choice cannot be undone — and a city that tears down its past too quickly may find, decades later, that it has erased something it can never get back.

4 Match headings A–F to paragraphs 1–5 of Article 1. There is one extra heading you do not need.

A A costly and sometimes impossible task

B Giving old buildings a completely new purpose

C A tourist's-eye view of a historic district

D Weighing up each building individually

E A decision that cannot be reversed

F Two very different fates for old buildings

MATCH THE HEADING

LANGUAGE DEVELOPMENT — Compound adjectives

VOCABULARY SKILL

Compound adjectives

A compound adjective is made of two or more words joined together, often with a hyphen, to describe a noun as a single idea — for example a **state-of-the-art museum** or an **out-of-date software**. Once you recognise the pattern, you can understand many more of them.

energy-efficient	using less power to do the same job
fast-growing	increasing in size or number quickly
long-term	relating to a period of many years into the future
one-way	allowing movement or communication in only one direction
out-of-date	old-fashioned; no longer the newest or most useful
state-of-the-art	using the most modern and advanced methods or technology
well-known	known or recognised by many people

Complete the sentences with a compound adjective from the table above.

1. The museum's _____ climate-control system uses far less power than the air-conditioning it replaced.

2. Digital storage needs are _____, so archives must plan years ahead, not just for next year.

3. Many families still own a computer running _____ software that almost nothing can open anymore.

4. The new archive building has a _____ scanning lab that can digitise thousands of documents a day.

UNDERSTANDING KEY VOCABULARY

Article 2

The sentences below are about preserving digital records and memories. Match each word in bold to its definition (a–h).

1 Firefighters worked for hours to **retrieve** important documents from the damaged archive building.
2 Digital preservation is not a single event but an ongoing **practice** that takes regular maintenance.
3 Museums are particularly concerned about the **longevity** of digital photographs compared with printed ones.
4 Software updates are not always **compatible** with files created many years earlier.
5 Old recordings stored in obscure formats are especially **vulnerable** to becoming permanently unreadable.
6 Imagine a **scenario** in which a family's entire photo collection becomes impossible to open.
7 Growing concern about lost data has **prompted** several libraries to begin converting old files.
8 The museum's collection of **memorabilia** includes tickets, letters and photographs from decades past.

a (adj) able to work together or be used together without problems
b (n) a possible situation or sequence of events that could happen
c (v) to get something back, especially from a place that is difficult to reach
d (adj) easily harmed, damaged, or lost
e (n) the length of time something lasts or continues to exist
f (n) something done regularly as a habit or routine, often to maintain a skill or condition
g (v) to cause something to happen, or make someone decide to do something
h (n) objects collected because of their connection to a particular event, period or person

Complete the sentences with the correct form of a word from either article.

1 The empty factory was completely _____ last year to make space for a car park. (demolition)
2 Poor planning increased the building's _____ to serious structural damage. (vulnerable)
3 The architect's plan involved the full _____ of the old station into a new use as a market. (adaptive)
4 Experts questioned the long-term _____ of the software with newer computer systems. (compatible)
5 A wealthy family with unusual _____ funded the museum's expensive renovation. (affluent)

WHAT WILL SURVIVE THE FUTURE?

1 A hundred years ago, a family might have kept its history in a shoebox: photographs, letters, ticket stubs and other small pieces of **memorabilia** passed down from one generation to the next. Today, that same family's photos are more likely to exist as files on a phone, a laptop, or somewhere in the cloud. This shift raises an uncomfortable question: will any of it still be readable in a hundred years' time?

2 The problem is not that digital files disappear the way paper yellows and fades. It is that they become impossible to open. File formats go out of use, storage devices stop being made, and software updates are often not fully **compatible** with files created decades earlier. A photograph saved on a floppy disk in the 1990s may now be technically intact yet completely unreachable, simply because almost no device left in existence can read it. Digital archivists sometimes compare the situation to a library where the books survive perfectly, but every key that could unlock the building has been lost.

3 This has led museums, libraries and even ordinary families to think seriously about the **longevity** of their digital collections. Good digital preservation is not a one-off task but an ongoing **practice**: files must be periodically copied onto new devices and converted into current formats before the old ones become unreadable. Archivists describe certain kinds of data as particularly **vulnerable** — handwritten family recipes typed once into an obscure app, for instance, or voice recordings saved in a format no company still supports.

4 Consider one realistic **scenario**: a researcher fifty years from now wants to **retrieve** a relative's old messages and photos to write a family history, only to discover that the storage device still exists but nothing can open it. This exact worry has already **prompted** several national libraries to begin systematically converting historic digital files into up-to-date formats before they become permanently locked away, treating the problem with the same urgency once reserved for fragile paper manuscripts.

5 The irony is hard to miss: a handwritten letter from a century ago can still be read by anyone who picks it up, while a digital photo from twenty years ago may already need expert help to open. Preserving our digital past, it turns out, requires far more constant attention than anyone expected.

GRAMMAR FOR WRITING

GRAMMAR

First and Second Conditionals — accuracy for discussing conservation choices

We use conditional sentences to talk about the results of real or imagined choices — exactly what conservation debates are about: what happens if a building or record is saved, and what would happen if it weren't.

Conditionals link a condition (the if-clause) to a result. The tense you choose signals how real or likely the situation is.

1. First conditional — a real, likely future possibility: **if + present simple, will + infinitive.**

If the council doesn't act soon, the old theatre will fall into permanent disrepair.

2. First conditional can also use may / might / can instead of will, to sound less certain about the result.

If the city invests in the museum, visitor numbers might increase significantly.

3. Second conditional — an imagined or unlikely present/future situation: **if + past simple, would + infinitive.**

If the government banned all demolition, many unsafe buildings would simply collapse instead.

4. Second conditional is also useful for polite, hypothetical opinions in formal writing.

If I were responsible for the budget, I would prioritise buildings with genuine historic value

Exercise A — Read each situation. Which sentence, a or b, is the more appropriate response?

1 A city council is genuinely considering a real, current proposal to renovate a train station next year.

a If the council approves the plan, the station will reopen by next spring.

b If the council approved the plan, the station would reopen by next spring.

2 A student is imagining a completely unrealistic idea — knocking down every old building in the world.

a If every old building disappears, we will lose an enormous amount of history.

b If every old building disappeared, we would lose an enormous amount of history.

3 An architect is giving a genuinely possible warning about a real, decaying building this year.

a If nobody repairs the roof this year, the whole building will collapse.

b If nobody repaired the roof this year, the whole building would collapse.

Exercise B — Complete the paragraph with the correct form of the verbs in brackets: first or second conditional.

Our town has an old cinema that has been empty for ten years. If the council (1) _____ (not / find) new investment soon, the building (2) _____ (probably / face) demolition within a few years. Personally, if I (3) _____ (be) in charge of the budget, I (4) _____ (turn) it into a community arts centre instead of letting it disappear. Of course, if the town (5) _____ (choose) that option, it (6) _____ (need) significant funding for repairs. Still, if nothing (7) _____ (change) soon, we (8) _____ (lose) a building that has been part of this town for over a century.

ACADEMIC WRITING

Impersonal statements — sounding objective in formal writing

ACADEMIC WRITING SKILL

Impersonal statements

Academic and formal writing usually avoids '**I think**' or '**in my opinion**' and instead uses impersonal structures — passive forms, '**It is + adjective + that…**', or general nouns like '**many people**' or '**critics**' — to sound more objective and less personal.

AVOID: *I think old buildings are more important than new developments.*

WRITE LIKE THIS: *It is often argued that old buildings are more important than new developments.*

Rewrite the personal sentences below as impersonal, formal statements.

1 I believe cities spend too much money restoring old buildings.

_____

2 I think digital records are just as important as physical ones.

_____

3 In my opinion, governments should protect more historic sites.

_____

4 I feel that families should preserve old photographs more carefully.

_____

5 I don't think every old building deserves to be saved.

_____

WRITING TASK

Write a short paragraph (80–100 words) giving your view on whether digital or physical records are more important to preserve, using at least two impersonal structures from this section.

READING PRACTICE

PART 1

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

The online retail giant Amazon has released its first ever big-budget PC game. The game is called Crucible. It is a science-fiction shooter (1)_____. It has many unusual characters that have special powers. Crucible is currently only available in North America, where people can download it from the video distribution website Steam. Amazon said the game will be (2)_____ in other parts of the world soon. Crucible was created by Amazon Game Studios and Relentless Studios and is free. Amazon is a newcomer in the video games market. It hopes to become a major producer of games. Many experts on (3)_____ games believe Crucible could be very successful. They say it could be a rival to the game Fortnite. Crucible is set in an apocalyptic world on a fictional planet in outer space. Very few people can walk around outside on this (4)_____. Players can go outside but they must hunt and kill different creatures and monsters. They must also collect a special material called "essence". This helps (5)_____ level up and become stronger. Mike Frazzini, the vice president of Amazon Games, spoke about Crucible. He said he hoped it would help people to connect with each other. He told reporters: "We want to make games that resonate with a very large audience of players." He added: "One of the reasons I've always loved games is not only is it a fun form of entertainment, but it's a way to (6)_____ with others."

PART 2

Read the texts 7-14 and the statements A-J. Decide which text matches with the situation described in the statements. Each statement can be used ONCE only. There are TWO extra statements which you do not need to use. Mark your answers on the answer sheet.

A) This café is the first thing most travelers can go.

B) Workers can go there to kick back.

C) You can go there on any part of the day.

D) This café serves free chilling drinks.

E) You can a free sandwich, internet connection and herbal teas there.

F) Prices can go lowers on Saturday and Sunday there.

G) This café has both modern and vintage look combined.

H) You can visit this café right after you have shopped.

I) You can drink your tea and coffee and enjoy an internet connection free of charge.

J) This café has food that has no meat in it.

7. Breezy Bean Café

Located beside the industrial area. Enjoy a cozy corner to relax with fresh coffee and baked goods. Indoor seating only. We have a terrace but no tables. **Menu:** Espresso, cappuccinos, organic teas, croissants, and muffins. Free refills on brewed coffee.

8. Urban Grind Café

Located in the heart of the city, just across from the park. Enjoy artisan coffee and light bites in a modern space. We have communal tables and cozy booths. Menu: Cold brew, nitro coffee, fresh juices, bagels, and granola bowls

Free iced water.

9. The Rustic Cup

Located a bit distance away from our local market, our small café combines vintage charm with a selection of freshly brewed beverages. We have outdoor tables. Menu: French press coffee, herbal teas, flavored lattes, scones, and sandwiches. Free wi-fi.

10. Cozy Corner Café

Located near the biggest local Farm, Cozy Corner offers a warm atmosphere and farm-to-table snacks. COZY READING NOOK. INDOOR SEATING. Menu: Hot chocolate, freshly brewed coffee, vegan pastries, and salads. Free hot water refills.

11. Café On The Corner

Located beside the airport, Café On The Corner is perfect for a quick coffee break or a relaxing afternoon. SEATS BY THE WINDOW. QUAINT ATMOSPHERE. HOMEMADE COOKIES, ESPRESSOS, CAPPUCINOS, ICED LATTES, PASTRIES. FREE BREWED COFFEE.

12. Sunrise Café

WE SERVE ALL DAY. RELAXED AMBIANCE AND LARGE WINDOWS. FRESH COFFEE AND SMOOTHIES. PANCAKES. PROTEIN SHAKES. +1 323 678 9012.

13. The Daily Brew

Located beside the mall, we offer a laid-back vibe with a strong coffee selection. We have plenty of seats for individuals or groups. Menu: Espresso, cappuccinos, gourmet drinks (matcha lattes), cold brew, breakfast burritos, and pastries. Free water refills.

14. Bakers & Brews

We are known for its homemade pastries and strong coffee. We have plenty of space for groups. WE HAVE SPECIAL DISCOUNTS ON WEEKEND. Menu: Mocha, flat whites, croissants, quiches, and cakes. Free refills on brewed coffee. Free Wi-fi.

📌 Wordlist

1. refill
2. light bites
3. booth
4. charm
5. beverage
6. herbal tea
7. snack
8. vegan pastries
9. farm-to-table
10. quaint
11. laid-back
12. ambiance
13. protein shake
14. gourmet
15. reading nook

LISTENING PRACTICE

PART 1

You will hear some sentences. You will hear each sentence twice. Choose the correct reply to each sentence (A, B, or C).

1. A) I am proposed.

B) Quite well, actually.

C) Likewise.

2. A) The red one.

B) I want to know when.

C) I have chosen.

3. A) For a minute.

B) Tomorrow.

C) All day.

4. A) It must be expensive.

B) I haven't bought it.

C) Yeah, it is.

5. A) I will tell Mark.

B) It's a good mark, really.

C) Oh, Ok.

6. A) I'll pick up my dog.

B) We better go home.

C) I like cats, actually.

7. A) He must be busy.

B) He has a presume.

C) I'll arrive there ASAP.

8. A) We can build it too.

B) I heard it's the gym.

C) Our school is

PART 2

You will hear someone giving a talk. For each question, fill in the missing information in the numbered space.Write ONE WORD and / or A NUMBER for each answer.

REGISTAN SQUARE TOUR

The word "Registan" means "**sandy place**".

First Madrasa - built by a renowned **9**_____ and mathematician.

- facade has tileworks with geometric **10**_____.

Second Madrasa - is built in the 17th century.

- has an artwork of two lions hunting a **11**_____

- illustrates the ambitions of the **12**_____ at the time.

Third Madrasa - not only a madrasa but also a **13**_____.

The tour guide recommends to visit Registan in the **14**_____.

WRITING PART 1.1

USEFUL LANGUAGE — informal letter/note

Function	Useful phrases
Opening	*Hi [name], / Hey [name], / Guess what...! / You'll never believe this, but...*
Reacting / feelings	*I'm really disappointed/excited/worried about... / Honestly, I couldn't believe it when I heard... / I was gutted to hear that...*
Giving an opinion	*I think they should... / In my opinion, it'd be better if... / Personally, I reckon...*
Suggesting	*Maybe we could... / Why don't we...? / It might be worth (+-ing)...*
Closing	*Let me know what you think. / Talk soon, / Best, / Take care,*

PART 1.1 PRACTICE

You are a member of the City History Museum. You received this email from the museum director.

Dear Member,

I'm writing to let you know that the City History Museum will be closed for three months starting next week for essential renovation work. We want to make the galleries more modern and accessible. We'd love to hear your ideas for new exhibits or features before we reopen. Your feedback means a lot to us.

Best wishes,

The Museum Director

Task 1.1 — Write a note to your friend, who is also a museum member. Write about your feelings about the closure and what you hope the museum will be like when it reopens. Write about 50 words.

SAMPLE ANSWER

Hi Elyor, Did you hear the museum's closing for three months for renovations? I'm a bit sad — I love visiting on weekends! Still, I hope they'll add more interactive exhibits and better lighting in the old wing. Could be worth the wait. Let me know if you've heard more news!

PART 1.1 PRACTICE 2 — now you try

You are a resident of a town where an old cinema is going to be demolished. You received this notice from the town planning office.

Dear Resident,

We are writing to inform you that the old Regal Cinema on Mill Street, closed for the past five years, will be demolished next month to make way for a new car park. The site will help ease parking congestion in the town centre. We understand some residents have fond memories of the cinema and welcome your feedback before work begins.

Yours sincerely,
Town Planning Office

Task 1.1 — Write a note to your friend, who also grew up in the area. Write about your feelings about the news and what you think should happen to the building instead. Write about 50 words.

Now write your own answer below. No model this time — use the rules and phrases above to help you.

SPEAKING PART 1.2

QUESTIONS

1. Describe what you can see in these two pictures.
2. What are some of the benefits of preserving historical and cultural heritage?
3. Why might some people be less interested in visiting heritage sites and traditional places?

SAMPLE ANSWER

*1. **In the first picture,** I can see a modern, spacious city promenade lined with trees, where people are relaxing and walking along a waterfront path with massive, sleek glass skyscrapers towering in the background. I**n the second picture, in contrast**, there is a historic, cobbled street featuring traditional architecture, including an ornate stone gateway, a minaret, and a dome, with a few people strolling past small local shops.*

*2. **I think a major benefit of** preserving historical and cultural heritage is that it helps communities maintain a strong sense of identity and a tangible connection to their past. It also provides great educational value for future generations, while often boosting the local economy by attracting visitors to historic locations like the one shown in the second picture.*

*3. **On the other hand**, some people might be less interested in visiting heritage sites because they prefer the conveniences, shopping, and fast-paced entertainment found in modern cities, much like the area in the first picture. **Additionally,** some travelers might find historical locations a bit dry or disconnected from their contemporary interests, especially if the sites lack interactive or modern exhibits.*

USEFUL LANGUAGE

Starting your comparison

Both of these pictures show…

These two pictures show…

The two situations in these pictures are…

Talking about similarities

These two pictures/situations are similar because/in that…

In both pictures we can see…

One similarity between these two pictures/situations is that…

Another similarity might be that…

One thing that these two pictures/situations have in common is that…

One thing that these two pictures share is that…

The most obvious similarity is that…

These two pictures share the fact that…

Both pictures (…) in the same way.

Talking about differences

One of the most obvious differences (between these two pictures) is that…

Another difference is that…

Something that makes these two pictures different is…

What makes these two situations (so/completely) different is…

While the first picture …, the second one…

The first picture…. However, the second one…

The first picture… whereas the second one…

Although the first picture…., the second one…

On the one hand, the first picture shows… . On the other hand, the second one…

DESIGN

LOGOS & BRANDING

UNIT 2

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. Can you think of a logo you'd recognise instantly? What makes it memorable?

2. Do you think a company's logo really matters, or is it just decoration?

3. Have you ever bought something — or avoided it — because of how it was designed or branded?

4. If you designed a logo for yourself, what would it look like?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Before choosing a designer, the company drew up a clear list of **criteria** the new logo had to meet.
2 The bold red used in the original design **aroused** such strong reactions that the team softened the colour in later versions.
3 Good design should **convey** the right message about a brand within a couple of seconds.
4 A logo with rounded shapes and bright colours can feel **evocative** of childhood and playfulness.
5 Designers increasingly choose materials and processes that are **sustainable**, avoiding anything that harms the environment.
6 Even after a full rebrand, the company decided to **retain** its original mascot, since customers loved it.
7 The first version of the logo was simple; every **subsequent** update added a little more detail.
8 A strong logo can **reinforce** a company's reputation for reliability every single time a customer sees it.

a (adj) next; happening after something else
b (n) standards used for judging something
c (v) to keep; to continue having
d (v) to make someone have a particular feeling or reaction
e (v) to express an idea so that it is understood by other people
f (adj) making a person remember or feel a particular thing
g (adj) causing little or no damage to the environment and able to continue for a long time
h (v) to provide more support for an idea and make it seem true

Complete the sentences with the correct form of a word from either article.

1. Environmental _____ has become one more factor designers must consider today. (sustainable)
2. The brand's mascot has become closely _____ with reliability in customers' minds. (associate)
3. Years of small _____ turned a cluttered original design into something clean and modern. (refined)
4. A memorable logo can offer real _____ to a company's reputation over time. (reinforce)
5. What one design _____ should a new logo satisfy above all others? (criteria)

WHAT MAKES A LOGO WORK?

1 A good logo does an enormous amount of work in a very small space. In roughly the time it takes to glance at a phone screen, it needs to **convey** who a company is, what it stands for, and why it deserves attention — through little more than a shape, a colour, and sometimes a few letters. Ask ten designers what **criteria** a logo must meet, and you'll likely get ten slightly different lists, but a few themes come up again and again.

2 Simplicity is usually near the top. Studies that show people hundreds of unfamiliar logos consistently find that the ones remembered later share simple, uncluttered designs. A cluttered logo may look impressive up close, but shrink it down to a phone icon and the detail disappears entirely.

3 A logo also needs to be distinctive enough that people don't confuse it with a competitor's, while still being **evocative** of the right feeling. A children's toy brand might choose a colourful, playful design specifically because it's meant to feel fun and nostalgic, while a bank's logo is far more likely to look serious and dependable — a mismatch between feeling and industry can quietly undermine trust before a customer reads a single word.

4 Logos also need to survive change. A brand's first design is rarely its last: as tastes shift and technology changes, most **subsequent** versions simplify rather than complicate the original. A well-known athletic brand's swoosh, for example, has barely changed in decades, even as everything around it has been transformed. **Retaining** a recognisable core while updating small details lets a brand feel current without losing the customers who grew up trusting it.

5 Increasingly, companies also think about how a logo behaves across dozens of surfaces: a shopping bag, a tiny app icon, an animated loading screen. A mark that only works printed large on a shop window is no longer enough; it has to flex to a hundred different formats without losing its identity, which is one reason so many well-known logos have been quietly simplified over the past decade.

6 None of this happens by accident. A logo that succeeds tends to **reinforce** a brand's message every single time someone sees it, gradually building trust through pure repetition. An increasing number of companies also factor in **sustainable** design choices — from the materials a logo is printed on to the energy used by an animated version online — treating environmental impact as one more design criteria, not an afterthought.

7 What has **aroused** the strongest reactions in recent years, though, hasn't been simplicity or sustainability, but change itself: even a small redesign can provoke surprisingly passionate backlash from loyal customers who feel, for a moment, that something familiar has quietly been taken away.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Doing a lot in a small space

B Why simple designs stick in memory

C The environmental cost of printing

D Matching the feeling to the industry

E Built to last without losing the core

F One mark, a hundred formats

G Building trust through repetition

H When even a small change causes a big reaction

MATCH THE HEADING

LANGUAGE DEVELOPMENT - Describing emotional responses

VOCABULARY SKILL

Verb + noun collocations

Certain verbs regularly combine with the same small set of nouns to describe emotional reactions — for example **evoke memories**, or **stir up controversy**. Learning these as whole chunks, rather than word by word, makes your language sound far more natural.

evoke	feelings, memories, emotions (mixed)
inspire	confidence, awe, admiration (positive)
arouse	interest, curiosity, suspicion (mixed)
stir up	controversy, anger, opposition (negative)
generate	excitement, interest, enthusiasm (positive)
provoke	outrage, criticism, debate (negative)

Complete the sentences with the correct collocation from the table above.

1. The unusual new logo _____ a lot of criticism online within hours of its release.

2. A well-designed workspace can _____ real creativity in a team.

3. The bright colours of the packaging _____ curiosity even in people who weren't looking to buy anything.

4. Nostalgic design choices often _____ strong childhood memories in older customers.

UNDERSTANDING KEY VOCABULARY

Article 2

2 The sentences below are about rebranding. Choose the correct meaning (a, b or c) for the word in bold.

1. The new packaging design faced strong **opposition to** its bright colour scheme from long-time customers.

a support for

b disagreement with

c confusion about

2. Fans immediately noticed that the redesigned logo **resembles** the company's very first version from decades ago.

a looks like

b is completely different from

c is an improvement on

3. Choosing a new logo colour is a **strategic** decision that can take marketing teams months to finalise.

a unplanned and rushed

b related only to cost

c carefully planned to achieve a goal

4. She is considered the **foremost** expert on colour psychology in advertising today.

a most experienced and well-known

b most controversial

c least experienced

5. After customer feedback, the designers **refined** the icon, making small improvements rather than starting over.

a completely replaced

b improved with small changes

c made more complicated

6. Many shoppers **associate** the colour green with eco-friendly products, even before reading the label.

a confuse

b mentally connect

c avoid

7. A **distinctive** logo helps a brand stand out immediately, even in a crowded market.

a easily forgettable

b easy to recognise and different from others

c expensive to produce

WHY DO BRANDS REBRAND?

1. Every few years, a familiar logo quietly disappears and a new one takes its place — sometimes to almost no reaction, and sometimes to a wave of public **opposition to** the change so loud it makes international news. Rebranding, changing a company's visual identity and sometimes its whole message, is rarely a casual decision; it's usually the result of a strategic process that can take marketing teams months, or even years, to complete.

2. Companies rebrand for different reasons. Some have simply grown up: a business that started decades ago with a busy, detailed logo often finds that a **refined**, simplified version fits a modern, mobile-first world far better than the original ever could. Others rebrand because their reputation, fairly or not, has become tangled up with something they'd rather leave behind — a fast-food chain distancing itself from concerns about its food, for instance, may quietly remove language that customers **associate** with those old worries.

3. Timing matters enormously. Some companies wait until a logo starts to look dated, no longer feeling current to a younger audience raised on newer visual trends. Others act because of a shift in what they actually do: a photocopier company that now sells a much wider range of office technology may rebrand specifically to stop customers assuming it still only makes one product.

4. New platforms create pressure too. As more of daily life moves onto small screens, logos that once worked beautifully printed large on a building or a truck can struggle to survive shrunk down to a tiny, blurry app icon. Several major tech companies have simplified their logos for exactly this reason, dropping fine detail that simply disappears at a small size.

5. Rebranding is not without real risk, however. When one soft-drink company briefly replaced its familiar packaging with a plainer, more generic design, sales dropped noticeably within weeks — shoppers, it turned out, were specifically looking for a design that **resembles** what they already knew, and the company quietly brought the original back. Elsewhere, a **foremost** example of a successful large-scale redesign is a search engine's colourful logo, which now subtly transforms itself for different products while keeping the same core shapes and colours recognisable throughout.

6. What almost every successful rebrand shares, though, is restraint: keeping just enough of the original that loyal customers still associate the new version with the brand they already trust, while making it feel fresh enough for a new generation. A logo that changes too much, too fast, risks losing the very **distinctive** identity that made it valuable in the first place — proof that in design, as in most things, sometimes less really is more.

DISCUSSION

Work with a partner. Discuss the questions.

1 Think of a logo you dislike. What would you change about it, and why?

2 Do you think companies rebrand too often, or not often enough? Why?

GRAMMAR

Defining relative clauses, omission & reduced clauses

Defining relative clauses add essential information that identifies exactly which person, thing, or place we mean — without them, the sentence would be unclear or incomplete. Unlike non-defining clauses, they use no commas, and the relative pronoun can sometimes be left out or reduced to a participle.

The rules below explain when a relative pronoun (who / which / that / where) is needed, when it can be dropped, and when the whole clause can be reduced to a participle phrase.

1. Use **who/that for people, and which/that for things,** to identify exactly which one you mean — no commas.

The designer who created the logo later became famous for minimalist branding

2. Use **whose** to show possession inside a defining clause.

A company whose logo changes too often can confuse loyal customers.

3. Use **where** for a place, in the sense of '**in/at which**' — it cannot be omitted.

This is the workshop where our designers create new prototypes.

4. The relative pronoun CAN be omitted when it is the OBJECT of the clause (not the subject).

The logo (that) the team designed last year is still in use today.

5. *The relative pronoun* ***CANNOT be omitted*** *when it is the SUBJECT of the clause.*

The designer who redesigned the logo now works for a different company.

6. *A clause with* ***which/that + verb*** *can often be reduced to a participle phrase: drop the pronoun and change the verb to* ***-ing.***

There are delays for people who travel to work on Southern Region trains. → There are delays for people travelling to work on Southern Region trains.

Exercise A — Complete the paragraph with a relative pronoun where it is needed. Leave the space blank if the pronoun can be omitted.

The designer (1) _____ created our new logo studied graphic design in Milan. She used a colour palette (2) _____ she chose after months of research. The client (3) _____ she worked with wanted something modern but timeless. The first sketch (4) _____ she showed us in April was completely different from the final version. A logo (5) _____ works well in black and white usually works everywhere. The agency (6) _____ we hired last year has redesigned dozens of well-known brands. The feedback (7) _____ the team received from focus groups shaped almost every decision. The version (8) _____ finally launched in June kept the original colours but simplified the shape.

Exercise D — Combine each pair of sentences into one, using where or a reduced participle clause as shown in the rules above.

1. The design studio is in the old power station. Our designers create new prototypes there. (where)

2. The rebrand leads to instant recognition. It cost the company millions. (participle)

3. This is the exhibition hall. Visitors can see the winning designs there. (where)

4. The new logo appeals to a younger audience. It has boosted sales significantly. (participle)

ACADEMIC WRITING

Paraphrasing — expressing other people's ideas in your own words

ACADEMIC WRITING SKILL

Paraphrasing

Paraphrasing means expressing someone else's idea in your own words, using different vocabulary and sentence structures, without changing the original meaning. It's essential when summarising a source, and if you keep any of the original wording, it must go in quotation marks.

ORIGINAL: *A good logo must also be unique so it will not be confused with the logo from another organisation.*

PARAPHRASE: *For a logo to work well, it needs a distinctive design so people don't mix it up with a rival's.*

Paraphrase the sentences below in your own words, without copying the original phrasing.

1 Logos should be flexible enough to adapt across time and placement.

_____

2 The public generally resists changes to familiar logos at first.

_____

3 More than anything else, a logo needs to tell a story.

_____

4 Some companies simply need a new logo to show that they have matured as a business.

_____

5 A distinctive logo helps a brand stand out immediately, even in a crowded market.

_____

WRITING TASK

Write a short paragraph (80–100 words) paraphrasing an opinion you've read or heard about branding or design — from an article, a friend, or this unit — without copying the original wording directly.

READING PRACTICE

PART 3

Read the text and choose the correct heading for each paragraph from the list of headings below. There are more headings than paragraphs, so you will not use all of them. You cannot use any heading more than once.

List of Headings:

A) Legacy of Sassanids.

B) Once nomadic, then an empire.

C) The fall of the Empire.

D) Commercial influence.

E) Melting pot of nations.

F) Four major beliefs.

G) The distributor of faith.

H) In the middle of great civilizations.

15. Paragraph I

16. Paragraph II

17. Paragraph III

18. Paragraph IV

19. Paragraph V

20. Paragraph VI

KUSHAN EMPIRE

I. The Kushan Empire was a significant cultural and political entity that flourished in Central Asia and Northern India between the 1st and 3rd centuries CE. Its origins trace back to the Yuezhi, a nomadic group from the steppes of Central Asia. They established control over Bactria, a Hellenistic region, blending Greek, Persian, and Indian influences. The Kushans were instrumental in fostering trade along the Silk Road, connecting the East and West. Under their rule, arts, religion, and commerce flourished, making them a vital link in the ancient world's cultural and economic exchanges.

II. The Kushan Empire spanned a vast territory, including present-day Afghanistan, Pakistan, Northern India, and parts of Central Asia. It was strategically located along major trade routes, particularly the Silk Road. The empire's geographic position allowed it to act as a bridge between the Mediterranean, China, and Indian spheres of influences. This connectivity facilitated not only trade but also the exchange of ideas, technologies, and cultural practices. The Kushans expanded their influence through military conquests and diplomacy, securing their position as a dominant power in the region for over two centuries.

III. The Kushan Empire was pivotal in the flourishing of the Silk Road, a network of trade routes connecting Asia to Europe. The empire served as a hub for the exchange of goods such as silk, spices, gemstones, and textiles. Kushan coins, widely circulated, reflect their role in trade and economic integration. The Kushans also promoted caravanserais and security along trade routes, ensuring safe passage for merchants. This vibrant trade network enriched the empire economically and culturally, fostering interactions between diverse civilizations.

IV. Kushan art is a testament to the empire's diverse cultural influences, blending Greek, Indian, Persian, and Central Asian styles. The Gandhara School of Art, which thrived during their reign, is renowned for its Greco-Buddhist sculptures, particularly depictions of the Buddha. The Kushans were patrons of various art forms, including architecture, sculpture, and coinage. These artistic expressions not only reflected their cosmopolitan society but also played a significant role in spreading Buddhist iconography and religious ideas across Asia.

V. The Kushan Empire was a melting pot of religions, including Buddhism, Hinduism, Zoroastrianism, and Greek polytheism. They were instrumental in the spread of Buddhism to Central Asia and China. Emperor Kanishka, a notable Kushan ruler, convened the Fourth Buddhist Council, which contributed to the development of Mahayana Buddhism. The empire's tolerance and support for multiple faiths fostered a rich spiritual environment, influencing religious practices far beyond its borders.

VI. The decline of the Kushan Empire began in the 3rd century CE due to internal conflicts and external invasions, particularly by the Sassanid Empire and the Gupta Empire. Despite its fall, the Kushan Empire left a lasting legacy. It played a critical role in the cultural and economic integration of the ancient world. The empire's contributions to art, religion, and trade continue to be studied and celebrated, highlighting its importance in the history of Central and South Asia.

Wordlist

1. entity ▢
2. flourish ▢
3. trace back to ▢
4. steppe ▢
5. nomadic ▢
6. blend ▢
7. establish control ▢
8. foster trade ▢
9. commerce ▢
10. vital link ▢
11. span ▢
12. trade route ▢
13. act a bridge ▢
14. spheres of influence ▢
15. facilitate ▢
16. military conquest ▢
17. dominant power ▢
18. pivotal ▢
19. serve as a hub ▢
20. reflect ▢
21. diverse ▢
22. patron ▢
23. spread ▢
24. be renowned for ▢
25. faith ▢
26. far beyond ▢
27. spiritual ▢
28. religious practice ▢
29. internal conflict ▢
30. legacy ▢

LISTENING PODCAST

TASK 1

📺 **Listen to the podcast and answer the questions.**

1. Liam says that Parkour is …

a. a type of military training.

b. a means of jumping off high buildings.

c. a way of overcoming obstacles.

d. a kind of video.

2. Liam's main job is …

a. to check the bikes are safe.

b. to clean the ramps.

c. to do demonstrations.

d. to teach young people about BMX.

3. Which area did Liam not have training in?

a. giving medical attention.

b. cooking hamburgers.

c. food hygiene.

d. basic accounting.

4.Why did Debbie volunteer the second time?

a. to be more competitive in the job market.

b. to help other people.

c. to complete her degree course.

d. to learn how to train volunteers.

5. Which sport did Debbie not volunteer to help with?

a. cricket

b. swimming

c. table tennis

d. football

6. Debbie and Liam both say that …

a. it's important to pay people to work in sport.

b. volunteers are often exploited.

c. they love volunteering.

d. volunteers are essential for sport.

TASK 2

Match the two parts of the phrase and write a–h next to the numbers 1–8.

1…….. to do	a. an experience
2…….. to overcome	b. a job
3…….. to volunteer	c. safety standards
4…….. to meet	d. to help
5…….. to organise	e. experience
6…….. to gain	f. an event
7…….. to deal with	g. the public
8…….. to share	h. an obstacle

TASK 3

📝 **Watch the video and write a summary.**

- the main topic

- the key points

- important examples or facts

Write a 100–150 word summary in your own words.

Scan to watch video

PART 3

You will hear people speaking in different situations. Match each speaker (15-18) to the description they give about another person(A-F). There are TWO EXTRA options

Speaker 1 ...

Speaker 2 ...

Speaker 3 ...

Speaker 4 ...

A) to strengthen a relationship
B) to get some personal space
C) to help deal with extensive responsibilities
D) to meet the demands of colleagues
E) to cope with unpredictability
F) to avoid unfair criticism

Phrases & Idioms Wordlist

Stretched to the limit: Pushed as far as possible; almost exhausted (regarding patience or resources).

Squirrel (oneself) away: To hide or isolate oneself in a quiet, secluded place.

Fresh-faced: Looking healthy, youthful, and full of energy.

Stunned silence: Complete quiet caused by surprise or shock.

Make a drastic decision: To make a severe, extreme, or radical choice.

Can't get enough of (something): To enjoy something so much that you want more and more of it.

Let's face it: Let us be realistic and accept the truth of a situation.

Get on someone's nerves: To annoy or irritate someone a lot.

Make one's blood boil: To cause someone to become very angry.

Self-imposed: A rule, task, or restriction that you decide to put on yourself, rather than having someone else force it on you.

Not have a clue: To be completely ignorant or unaware of something.

Live on air: Broadcasting in real-time without pre-recording.

Tear one's hair out: To show extreme frustration, stress, or anxiety.

Throw (something) into complete disarray: To cause total confusion, disorder, or chaos.

Bear in mind: To remember or consider a specific piece of information.

Get worked up: To become upset, stressed, or overly emotional about something.

WRITING PART 1.2

USEFUL LANGUAGE — Formal letter

Function	Useful phrases
Opening	*Dear Sir/Madam, / Dear [Title], / I am writing in response to your recent email regarding...*
Reacting / feelings	*I was pleased/disappointed to learn that... / I must admit I have some concerns about...*
Giving a suggestion	*I believe it would be beneficial to... / I would suggest that... / It might be worth considering...*
Requesting a reply	*I would be grateful if you could... / I look forward to hearing your thoughts on this.*
Closing	*Thank you for taking the time to consider this. / Yours faithfully, / Yours sincerely,*

PART 1.1 PRACTICE

You are a member of a local Design Society. You received this email from the society's director.

Dear Member,
I'm writing to let you know that our Design Society building will be closed for six weeks starting next month for a major renovation, including a full redesign of our exhibition space and workshop rooms. We want to create a more modern, inspiring environment for our members. During this time, we'd love to hear your ideas for how the new space could be designed, or what facilities you'd like to see included. Your feedback means a great deal to us.
Best wishes,
The Director

Task 1.1 —Write a letter to your friend, who is also a member of the Design Society. Write about your feelings and what you think the new space should include. Write about 50 words..

RULES for Task 1.2 (formal letter)

☐ Respond to the SAME message as Task 1.1 — cover the same two things (feelings + opinion/ suggestion), but in more depth.

☐ Aim for 120–150 words — several short paragraphs, not one block of text.

☐ Use a formal register throughout: no contractions, polite and impersonal phrasing.

☐ Structure it clearly: an opening line, your reaction, your suggestion(s) in detail, and a polite closing.

☐ Open with Dear Sir/Madam or Dear [Title], and close with Yours faithfully/sincerely.

☐ Use a wider range of grammar than Task 1.1 — relative clauses, passive forms, and formal linking words (furthermore, given that, in addition).

SAMPLE ANSWER

Dear Director,
Thank you for informing members about the upcoming renovation of the Design Society building. While I understand the necessity of these upgrades, I must admit I am a little disappointed that the space will be unavailable for six weeks, as I regularly attend the workshop sessions there.
That said, I very much welcome the opportunity to contribute ideas for the redesign. I would suggest including a dedicated area for digital design work, with computers and software for members interested in graphic design, alongside the traditional workshop spaces.
It would also be beneficial to install better natural lighting in the exhibition space, as this would make displayed work look far more professional. Additionally, I believe a small café area could encourage members to stay longer and socialise after sessions, which might help build a stronger sense of community.
Thank you again for considering members' feedback. I look forward to seeing the new space when it reopens.
Yours faithfully, N. Yusupova

TASK 1.2 PRACTICE — now you try, with a new message

You are a member of a local Photography & Design Club. You received this email from the club's chairperson.

Dear Member,

I'm delighted to announce that our club has been offered a new, larger space in the city's Arts Quarter, and we're planning to move next month. Before we finalise the layout, we would like to hear your suggestions for how the new space should be organised — for example, areas for printing, exhibitions, or workshops. Please let us know your thoughts.

Kind regards,
The Chairperson

Task 1.2 —Write a letter to the Chairperson. Write about your feelings about the move and what you think the new space should include. Write about 120–150 words.

SPEAKING PART 2

QUESTIONS

Describe a product or brand whose design you really admire.

You should say:

- ☐ what the product or brand is
- ☐ what you like about its design
- ☐ how the design makes you feel or behave

…and say whether you think design generally influences your own buying decisions.

SAMPLE ANSWER

*I'd like to talk about my phone, which I think has a **genuinely distinctive design**. What immediately **stands out** is how minimalist it looks — there's barely a single unnecessary line or button, and the whole thing feels incredibly smooth in your hand. What I really love about it is the way the colour and shape seem to **convey a sense of quality** without the brand needing to say a word about it.*

Using it every day actually makes me feel a bit more organised, if that doesn't sound strange — because everything about the design is so clean, I find myself wanting to keep my apps and files just as tidy. It's a small thing, but it does shape how I behave with it.

As for** whether design influences my buying decisions in general, I'd say it absolutely does, maybe more than I'd like to admit. I've definitely paid more for a product purely because it looked better, even when a cheaper option would have worked just as well. Overall, I think good design isn't just decoration — it genuinely changes how a product feels to use, and that's often **worth paying for.

USEFUL LANGUAGE

Introducing the topic:	*I'd like to talk about... / One product whose design I really admire is...*
Describing appearance:	*It's got a really sleek/minimalist/bold design... / What stands out is...*
Explaining why you like it:	*What I love about it is... / It really appeals to me because...*
Describing the effect on you:	*It makes me feel... / Every time I use/see it, I...*
Giving a wider opinion:	*In general, I think design plays a huge role in... / Personally, I tend to...*
Concluding:	*Overall, ... / To sum up, ...*

TECHNOLOGY

UNIT 3

TECHNOLOGY & DIGITAL LIFE

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. How much time do you spend on your phone every day?
2. What's one piece of technology you couldn't live without?
3. Do you think people rely on technology too much nowadays?
4. Tell me about an app or gadget you started using recently.

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–f).

1 His **hostile** response to the interviewer's questions made viewers feel genuinely uncomfortable.
2 She felt overwhelming **humiliation** when screenshots of her private messages began circulating without her permission.
3 No one should have to tolerate **abusive** comments simply for sharing an honest opinion online.
4 After weeks of unwanted messages, he decided to **withdraw** from the app completely.
5 The court's ruling set an important **precedent** that several other judges have since followed.
6 The app allows users to post under complete **anonymity**, without ever revealing their real names.

a (adj) unfriendly; showing strong dislike
b (n) shame and loss of self-respect
c (adj) rude and offensive; causing another person mental or physical harm
d (v) to stop taking part in something
e (n) an earlier action or decision that becomes an example for future decisions
f (n) the situation in which somebody's name or identity is not known

Complete the sentences with the correct form of a word from either article.

1 She described the whole experience as deeply _____. (humiliate)
2 Some comment sections online are filled with open _____ towards anyone who disagrees. (hostile)
3 Companies now face growing pressure to accept more _____ for how they handle user data. (accountable)
4 There was no _____ behind the mistake — it was simply a careless error. (malicious)
5 Local officials have promised stronger _____ of online platforms operating in the region. (regulate)

LIFE BEHIND THE SCREEN

1 Almost everyone who spends real time online eventually runs into behaviour that feels openly **hostile** — a stranger mocking a photo, a pile of insulting replies under an honest opinion, or worse. For a graphic designer we'll call Elena, it started with a single unkind comment about her portfolio and slowly grew into a stream of **abusive** messages that arrived every time she posted new work. She is far from alone. Cyber harassment, unwanted and repeated hostile behaviour carried out online, has become one of the more troubling costs of living a large part of our lives on the internet.

2 A recent survey of internet users found that well over half had witnessed this kind of behaviour directed at someone else, and roughly a third said they had experienced it personally. Even these figures likely understate the real scale of the problem, since many people never report what happens to them, either because they doubt anything will change or because they feel a private sense of **humiliation** about the whole situation.

3 Why does this behaviour happen at all? The **anonymity** the internet offers is often blamed first. When a username replaces a real face, some people say and do things they would never consider face-to-face. Anonymity also makes harassment easier to get away with, since a harasser can simply create a new account after being blocked or reported, appearing again under a different name with no history attached.

4 Researchers who have studied this behaviour, however, caution against assuming every troll fits a single profile. Some data **assembled** from harassment reports does show a small number of unusually active, deliberately **malicious** users responsible for a disproportionate share of the abuse. But a surprisingly large share of hostile comments come from ordinary users who behave badly only occasionally, often anonymously, and often without much thought about the person on the receiving end.

5 Legal responses vary enormously from one country to another. Some countries have updated older laws to specifically cover online threats and stalking; others still rely on rules written long before social media existed, leaving significant gaps. Even where laws exist on paper, relatively few offenders are ever formally **prosecuted**, since gathering evidence and identifying an anonymous account can be slow and difficult, and cases frequently cross national borders.

6 Some legal experts argue that limiting anonymity altogether would help hold people **accountable**, since it is harder to behave abusively under your own name. Others push back, pointing out that anonymity also protects people who genuinely need it — whistle-blowers, activists in hostile political environments, or simply someone quietly researching a sensitive personal question they would rather keep private. Removing anonymity for everyone, they argue, would punish the many to catch a much smaller number of bad actors, without necessarily changing their behaviour much at all.

7 For victims, in the meantime, the impact rarely feels abstract. Many describe genuine humiliation, lasting anxiety, and a reluctance to post anything at all, even long after the original comments have disappeared. Some, like Elena, eventually **withdraw** from a platform entirely rather than continue putting new work in front of an audience that might turn hostile again — a quiet, largely invisible cost that rarely appears in any survey.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A A single comment among many

B When the numbers still fall short of the full picture

C Hiding behind a different name each time

D Not every troll fits the same profile

E Laws that struggle to keep pace

F The debate over ending anonymity

G A quiet, largely invisible ending

H Legal action available to every victim

MATCH THE HEADING

LANGUAGE DEVELOPMENT - Collocations for behaviour

VOCABULARY SKILL

Verb + noun collocations

Certain verbs regularly combine with the same small set of nouns to describe behaviour and its effects — for example **exhibit hostile behaviour**, or **lose confidence**. Learning these as whole chunks, rather than word by word, makes your language sound far more

build	*a reputation, confidence*
exhibit	*behaviour (hostile/abusive)*
exprience	*abuse, pain, behaviour*
lose	*confidence, a reputation*
suffer	*abuse, pain*
take	*responsibility*

Complete each sentence with one verb (in the correct form) and one noun from the table above. In some items, more than one answer is possible.

1 When something goes wrong with an account, it helps to _____ _____ promptly and explain what happened.

2 Our platform has _____ _____ for responding quickly to reports of abuse.

3 People who _____ _____ online often decide to leave a platform altogether.

4 It isn't only strangers who _____ hostile _____ — some abuse comes from people users actually know.

5 Long-term harassment can cause people to _____ _____ that is both emotional and physical.

6 Even a single hostile comment can cause a person to _____ _____ and stop posting altogether.

UNDERSTANDING KEY VOCABULARY

Article 2

2 The sentences below are about online behaviour and its consequences. Choose the correct meaning (a, b or c) for the word in bold.

1 Company leaders should be held **accountable for** decisions that put users' personal data at risk.

a punished

b aware

c responsible

2 Authorities have promised to **prosecute** anyone who deliberately spreads someone's private photos without consent.

a search for in order to arrest

b take to court to argue the guilt of

c commit a crime against

3 The message contained **malicious** software specifically designed to steal people's banking details.

a legal but inappropriate

b untruthful

c intended to cause harm

4 No platform can completely **eliminate** the risk of personal data being leaked somewhere.

a remove entirely

b find a punishment for

c reduce slightly

5 Researchers **assembled** a database of thousands of harassment reports in order to study the problem.

a gathered

b interviewed

c attracted

6 New automated tools have been developed to **combat** the spread of fake profiles on social media.

a evaluate

b try to stop

c destroy

7 Several governments are now working to **regulate** how companies collect and store personal information.

a control

b encourage

c tax

8 Accounts that **exhibit** unusual posting patterns are often flagged automatically by the platform's systems.

a create

b accept

c show

CAN ONLINE COMMUNITIES POLICE THEMSELVES?

1 Tackling cyber harassment usually requires a two-pronged effort: preventing bad behaviour in the first place, and penalising it after the fact. So far, most attention has gone into prevention, partly because it is so difficult to **eliminate** harmful behaviour once it has already reached a victim. Anonymity again complicates things here — placing new limits on how people interact online could genuinely reduce harassment, but it risks damaging the openness that makes online communities valuable in the first place.

2 Online gaming has often been treated as an unusually difficult case, given how much casual interaction happens there and how much abuse female players in particular report. One large multiplayer game, after studying its own community, **assembled** a research team to examine chat logs from a huge sample of its regular players. The team's early expectation was that a small number of consistently badly behaved players would be responsible for most of the abuse — and while it did find a handful of genuinely malicious repeat offenders, most of the offensive messages actually came from otherwise ordinary players who simply behaved badly occasionally. Notably, disabling public chat by default cut abusive messages by close to a third, suggesting that a small structural change can meaningfully **combat** bad behaviour before it even starts.

3 Other communities have taken a different route: letting users **regulate** each other. Several platforms now allow members to rate or report one another, similar to how buyers and sellers on some online marketplaces already rate each other after a transaction. Reputation systems like this give ordinary users a visible incentive to avoid behaving in ways that would get them flagged, without requiring a company to review every single message by hand. Some services also use automated systems trained to flag accounts that **exhibit** repeated aggressive language or unusually poor spelling — patterns that, on their own, weakly predict harassing behaviour but become useful signals in combination.

4 Government regulation has been slower to catch up. Several countries have extended older laws to cover online threats and stalking more clearly, yet very few offenders are ever formally prosecuted under them, since gathering usable evidence against an anonymous account is genuinely difficult and often expensive. One notable court decision, allowing a harassment victim to identify an anonymous account holder through legal channels, set a **precedent** that several other courts have since followed. Experts frequently offer victims the same advice — simply ignore the harassment and avoid responding to it — but many victims say that, while well-meant, this advice does little to make ongoing abuse actually stop.

5 As more of everyday life moves online, addressing this behaviour matters more each year, not less. The internet is no longer a separate space we can easily walk away from; increasingly, it's where people socialise, work, and build a reputation that follows them elsewhere. Treating online harassment as a minor inconvenience, rather than behaviour with real social, professional, and even economic costs, is an attitude that is slowly, if unevenly, starting to change.

GRAMMAR FOR WRITING

GRAMMAR

Impersonal passive constructions & the passive for continuity

Writers often choose the passive voice when the agent is 'people'. As a subject, 'people' may be too general or repetitive. When reporting what people think or say, writers create a passive with no apparent agent, using two common constructions.

These constructions often appear with the verbs agree, argue, believe, claim, consider, decide, expect, say, think and understand. Note that not all of these verbs allow both patterns equally naturally.

1. **It is + said/believed/argued/thought + that + clause** — used when reporting a general opinion with no named source.

It is said that cyber harassment is most common in online spaces where anonymity is easiest to maintain.

2. **Subject + is/are + said/believed/thought/considered + to + infinitive** — the same idea, with the topic as the subject.

Stricter regulation is believed to be one of the most effective ways to reduce online abuse.

3. The passive is also useful for keeping the same subject across two clauses, making a sentence easier to follow.

Because the harassment takes place online, it is not always taken seriously.

4. When using the passive for continuity, the by-phrase is usually omitted if the agent is unclear or unimportant.

The account had been reported many times before it was finally suspended.

Exercise A — Rewrite sentences 1–4 in the impersonal passive in two ways. For sentences 5 and 6, only one construction is natural.

1 People believe that these figures underestimate the true scale of the problem.

2 People claim that anonymous accounts are responsible for most abusive comments.

3 People expect that victims of harassment will eventually stop using the platform.

4 People understand that self-regulation is the most realistic short-term solution.

5 People consider ignoring the harassment to be sensible advice for victims.

6 People have argued that these problems will require a stronger legal response.

Exercise B — Rewrite the second clause or sentence using the passive to maintain continuity. Omit the by-phrase if it is not necessary.

1 Although the account had been posting abusive comments for weeks, moderators did not suspend it until yesterday.

2 A full year after the harassment began, a journalist discovered the account holder's real identity.

3 These trolls were not very careful. Investigators found evidence of their activity quite quickly.

4 The news usually includes positive stories about technology, but in recent months, stories of online abuse have dominated the coverage.

ACADEMIC WRITING
Writing about problems and solutions

ACADEMIC WRITING SKILL
Writing about problems

When presenting problems, writers use a range of strategies to show readers why a problem matters: facts (F), statistics (S), and examples (E) — especially ones that will resonate with the reader.

Read the statements below and label the strategy the writer uses: F, S or E.

1 Nearly two-thirds of teenagers say they have witnessed online harassment firsthand.

_____

2 After a wave of abusive reviews, a small family-run business lost almost half its regular customers.

_____

3 In a recent survey, 40% of students reported experiencing unwanted contact from strangers online.

_____

4 Some companies simply need a new logo to show that they have matured as a business.

_____

5 A user we'll call GreenFox92 lost access to her savings after someone found her personal details online

_____

6 An anonymous account has been sending threatening messages to a local teacher for more than five years.

_____

7 In several documented cases, exposing someone's private information online without consent has led to real-world harm.

_____

8 The most commonly reported platform for online harassment is social media.

_____

PART 4

Read the following text for question 21-29.

PENGUINS

There are seventeen kinds of penguins in the world. They are all alike in some ways and all cannot fly. All have dark backs and white fronts and live in the Southern Hemisphere near the sea

They differ in several ways. They can be as small as 16 inches or as tall as 4 feet. They differ in the exact location they inhabit. Some live in the frigid climate of Antarctica. Others can live along the coast of forested land. Galapagos penguins live three thousand miles north of Antarctica in a tropical climate at the equator. All penguins live at or south of the equator.

Some penguins are known for yellow or orange feathers on their heads. Some have a single or double black band across their upper chest. Brush-tailed penguins have tails that are longer than any other. The emperor and the king are called the giant penguins. The little blue and the yellow-eyed are isolated types of penguins not related to any other. Sometimes penguins are named by what they look like. Sometimes they are named based on where they live

Penguins have torpedo-shaped bodies, long and slim. They can move quickly and gracefully through the sea. Their short wings and flippers move up and down swiftly. They steer with their tails and feet. Penguins have solid bones. This helps them dive. They can leap out of the water. They can travel twenty-five miles an hour for short distances. Some types of penguins may live in one area in summer and travel a good distance for the winter. The biggest penguins are the fastest swimmers and best divers. Emperors can dive as much as fifteen hundred feet below the surface of the ocean for prey, though food is much closer to them.

Penguins feed on fish, squid and krill. They have no teeth. Their tongues and roofs of their mouths are covered with stiff spines which keep slippery prey inside. A penguin comes up from below to strike a school of fish. In a few minutes a penguin can catch and eat a hundred small fish or krill.

Wherever penguins swim, the water is cold. Their coat of small, scale-like feathers protects them from the cold. Penguins can groom themselves by taking oil from a gland at the base of their tails. With their beaks they spread this oil all over their feathers. This is their way of waterproofing their bodies. A layer of stored fat under their skin also keeps them warm. It gives them extra energy when needed too. In warmer climates penguins may have bare skin on their faces to stay cooler.

Penguins could live all year in the sea but they must come ashore to mate and raise their babies. Most nests are made by the male. Both parents help raise the babies. In warm climates penguins must protect the eggs from the heat. Nests can be placed in crevices and caves along the shore. Penguins in colder climates build nests on top of soil or rocks out of pebbles. They are above ground to avoid being flooded by melting ice.

In spring more than a million king penguins raise their young on South Georgia Island in the southern part of the Atlantic Ocean. The male and female take turns holding a single egg on top of their feet. They keep it warm by holding it up to a special warm area of their belly skin called the brood patch. The other parent goes off to hunt for food.

In summary, all seventeen types of penguins in the world are alike in some ways. They are all flightless birds. They have dark backs and white fronts and live near the sea in the Southern Hemisphere.

For questions 21-24, choose the correct answer A, B, C, or D. Mark your answers on the answer sheet.

21. Which of the following are the two largest types of penguins?

A) King and emperor

B) Brush-tailed and emperor

C) Galapagos and king

D) None of the above

22. Which of the following reasons tells why penguins use the oil from the

gland at the base of their tail?

A) To protect their eyes from bright light

B) To waterproof their feathers

C) To keep their feet tough

D) To keep their beaks sharp

23. What is unusual about how the male and female emperor penguins on South Georgia Island take care of their egg?

A) They bury it in the sand.

B) The female sits on it for three weeks.

C) The male and female use a brood patch to keep the egg warm.

D) The parents leave the egg alone while they go off to hunt for food.

24. Penguins live …

A) anywhere.

B) Only in the forests of Galapagos.

C) at the equator.

D) in the frigid climate of Antarctica.

For questions 25-29, decide if the following statements agree with the information given in the text. Mark your answers on the answer sheet.

25. All penguins have black bands across their chests.

A) True B) False C) No Information

26. Penguins have sharp teeth.

A) True B) False C) No Information

27. Penguins eat small animals like fish and squid.

A) True B) False C) No Information

28. Penguins choose their partner for life.

A) True B) False C) No Information

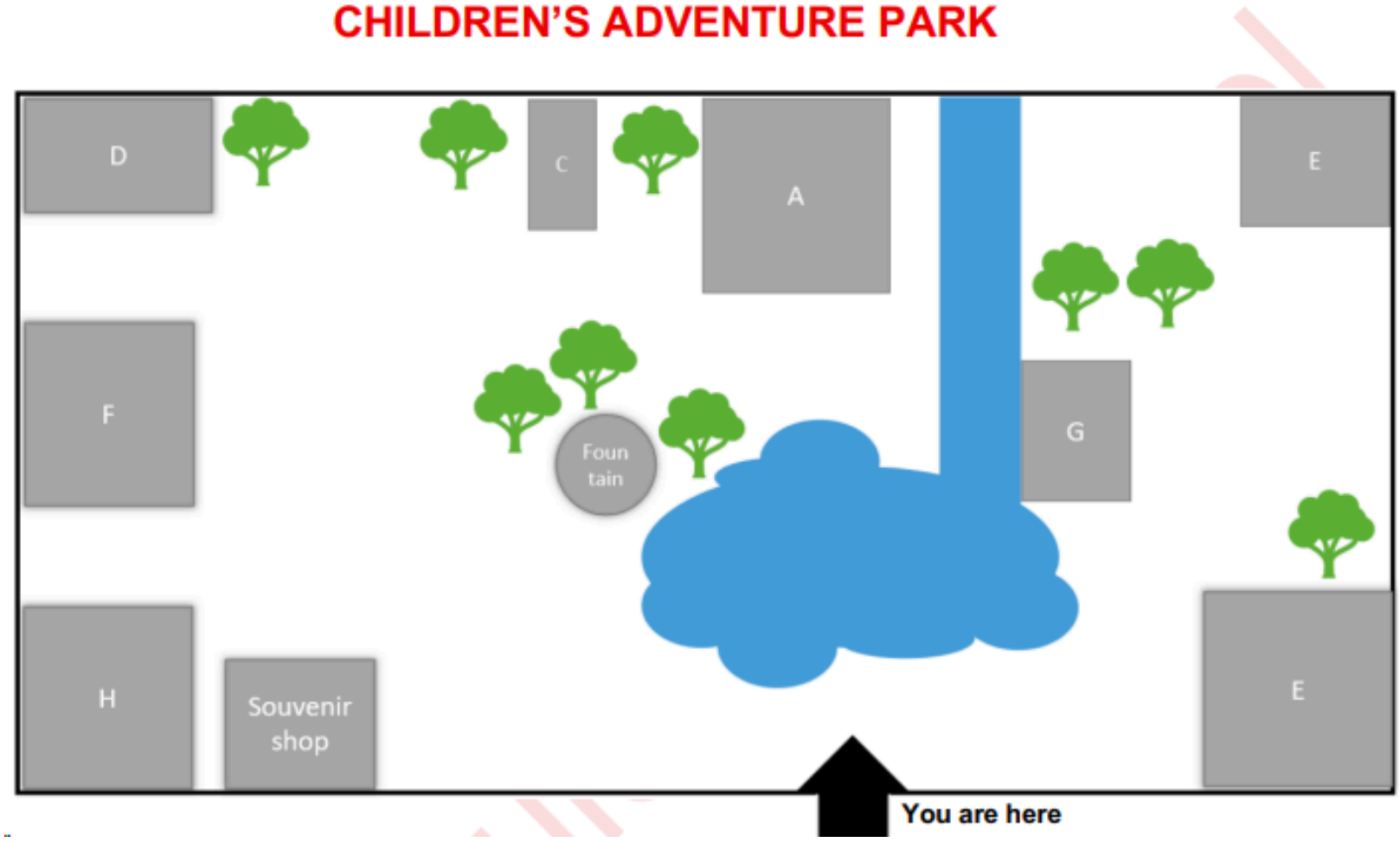
29. King penguins and emperor penguins dislike each other.

A) True B) False C) No Information

LISTENING PRACTICE

PART 4

You will hear someone giving a talk. Label the places (19-23) on the map (A-H). There are THREE extra options which you do not need to use.



19. Adventure course …

20. Restroom …

21. Swimming pool …

22. Tennis court …

23. Scary room

USEFUL LANGUAGE — Opinion essay

Function	Useful phrases
Introducing the topic	*In recent years, … has become a widely debated issue. / It is often argued that …*
Stating your position	*I would argue that … / In my view, the benefits clearly outweigh …*
Supporting a point	*This is largely because … / A clear example of this is … / Research suggests that …*
Presenting the other side	*Admittedly, … / Critics point out that … / It cannot be denied that …*
Refuting	*However, this overlooks the fact that … / While this may be true, …*
Concluding	*On balance, … / All things considered, I firmly believe that …*

THE TASK

You are writing an essay for an academic audience. Write 180–200 words.

Some people believe that social media users should be required to use their real names online, while others think anonymity should be protected. Discuss both views and give your own opinion.

PARAGRAPH PLAN

Introduction	paraphrase the question + state your position
Body paragraph 1	arguments for real names: accountability, less harassment, trust + example
Body paragraph 2	arguments for anonymity: whistle-blowers, safety, free speech + example
Conclusion	summarise both sides and restate your opinion clearly

RULES for Part 2 (essay)

☐ Answer ALL parts of the question — both views AND your own opinion.

☐ Write 180–200 words in four clear paragraphs.

☐ Use a formal, academic register: no contractions, slang or direct questions to the reader.

☐ Begin each body paragraph with a clear topic sentence.

☐ Support each point with a reason and an example; use hedging (may, tend to, it could be argued).

☐ Make sure your conclusion matches the position in your introduction.

SAMPLE ANSWER

In recent years, the question of whether people should be allowed to remain anonymous online has become a widely debated issue. While real-name policies may reduce some abusive behaviour, I would argue that anonymity is too valuable to remove completely.

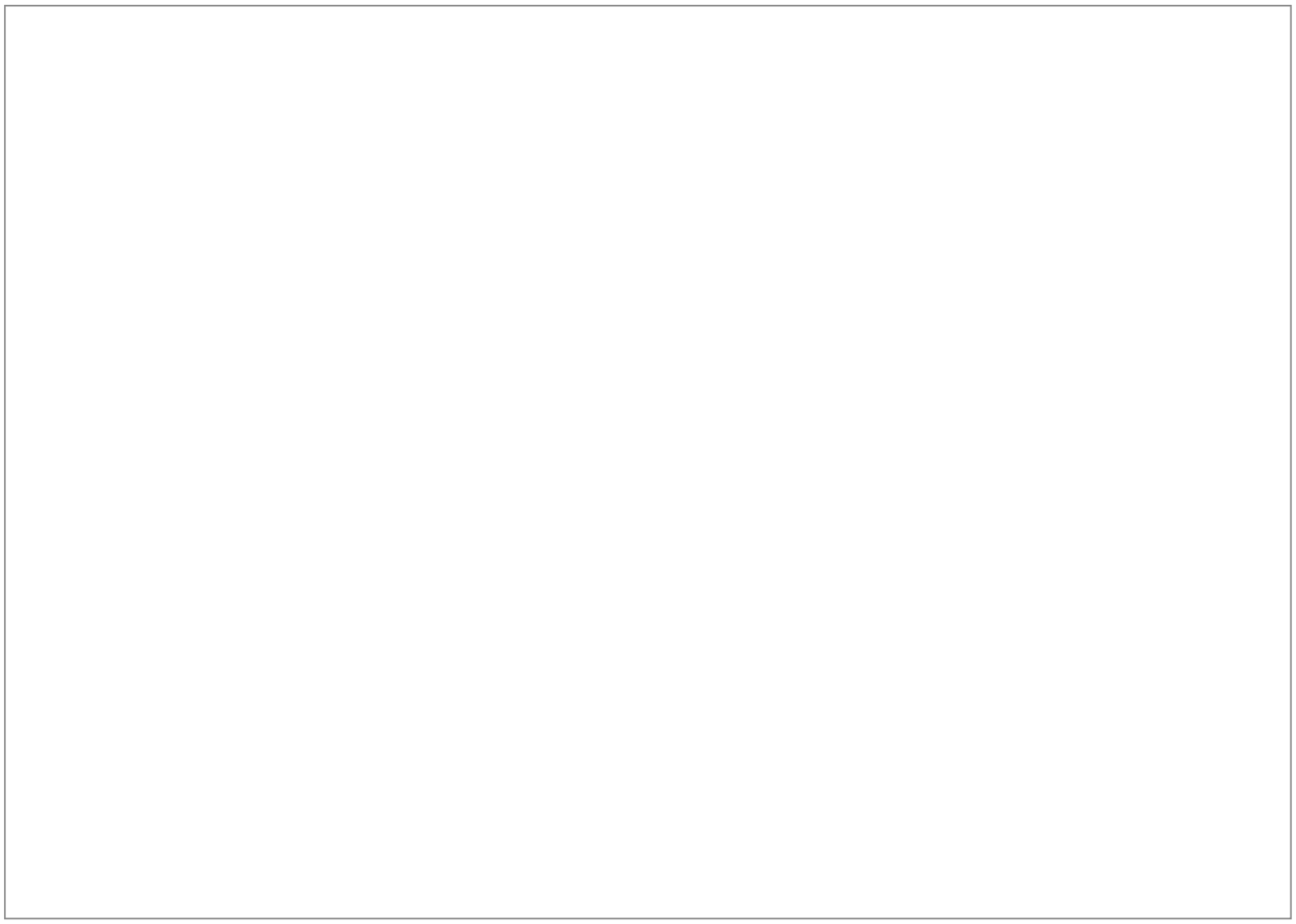
Those in favour of real names argue that people tend to behave more responsibly when their identity is visible. A user who knows that an insulting comment can be traced back to them is far less likely to post it. Furthermore, platforms where users are identified are often seen as more trustworthy, since fake accounts are harder to create.

However, this overlooks the fact that anonymity protects many vulnerable people. Whistle-blowers, victims of abuse and activists living under strict governments may depend on hidden identities to speak safely. Moreover, research suggests that a large share of online harassment comes from people who already use their real names, so removing anonymity would not solve the problem on its own.

On balance, I firmly believe that anonymity should be protected. Rather than forcing everyone to reveal their identity, platforms should invest in better moderation and faster responses to abuse.

PART 2 PRACTICE — now you try

Some people think that children under 16 should not be allowed to use smartphones at school. To what extent do you agree or disagree?



SPEAKING PART 3

STATEMENT

People should share less of their personal life online.

Discuss both sides of the statement and give your own opinion. Use the points below to help you.

FOR	AGAINST
protects privacy and personal data from companies and criminals	sharing helps people stay in touch with family and friends abroad
reduces the risk of online harassment and identity theft	useful for building a personal brand, a business or a career
less pressure to show a perfect life; better mental health	people should be free to decide what they post

SAMPLE ANSWER

***There is no doubt that** this is a topic that affects almost everyone today. **On the one hand**, sharing less online would clearly **protect our privacy**. Every photo or check-in we post can be collected by companies and, in the worst cases, used by criminals for **identity theft**. **What is more**, constantly showing a perfect life puts a lot of pressure on young people, and I think it can **damage their self-esteem**.*

***On the other hand**, social media is not only negative. My cousin lives in Korea, and thanks to her posts our whole family **feels connected** to her every day. **In addition**, many people in Uzbekistan now **build their careers** and small businesses through Instagram and Telegram, so sharing is part of their work.*

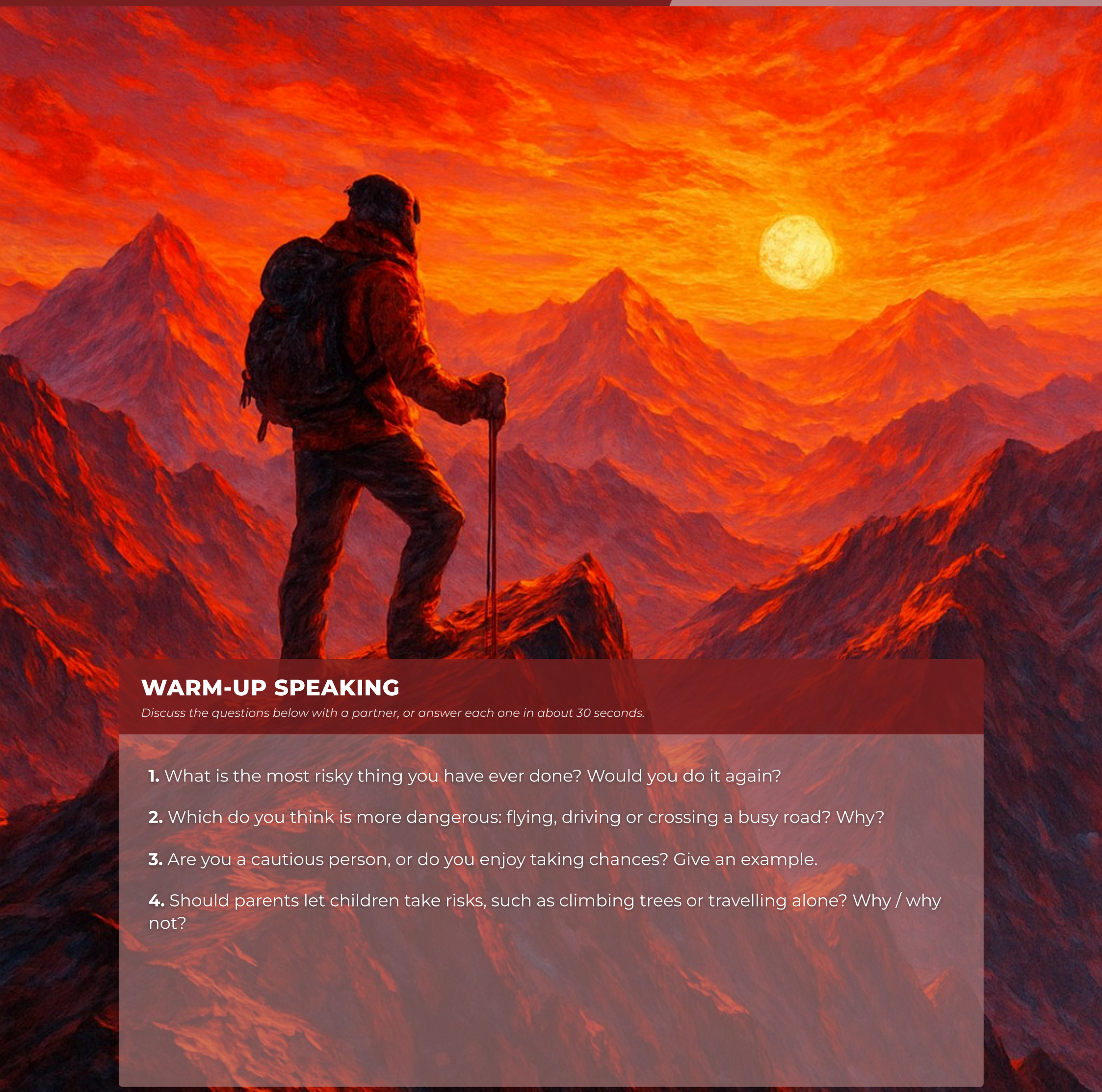
***Overall, I would say that** the answer is not to stop sharing, but to **share more wisely**. People should think twice before posting their location, their children's photos or personal problems, and they should check their privacy settings regularly. **In my view**, it is about balance rather than silence.*

USEFUL LANGUAGE	
Introducing the topic:	*There is no doubt that … / This is a topic that affects …*
Arguments for:	*On the one hand, … / One clear advantage is that …*
Adding a point:	*What is more, … / In addition, … / Not only that, but …*
Arguments against:	*On the other hand, … / Critics would say that …*
Giving an example:	*Take my cousin, for example … / A good example of this is …*
Concluding:	*Overall, I would say that … / In my view, it is about balance …*

RISK

TAKING CHANCES

UNIT 4



WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. What is the most risky thing you have ever done? Would you do it again?

2. Which do you think is more dangerous: flying, driving or crossing a busy road? Why?

3. Are you a cautious person, or do you enjoy taking chances? Give an example.

4. Should parents let children take risks, such as climbing trees or travelling alone? Why / why not?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Many tourists **perceive** the old city as unsafe at night, although crime there is actually very low.

2 Engineers calculated the **probability** of the bridge failing in a major earthquake.

3 She still has **vivid** memories of the storm that destroyed her grandparents' house.

4 For weeks before the operation, he lived with a constant sense of **dread**.

5 Investors often **overestimate** their ability to predict how the market will move.

6 After ten years without an accident, the factory managers had become **complacent** about safety.

7 Experienced nurses often make **intuitive** decisions that turn out to be remarkably accurate.

8 The city built new flood barriers to **mitigate** the effects of heavy rain.

a (adj) too satisfied with a situation, so that you stop paying attention to possible dangers

b (n) how likely it is that something will happen

c (v) to make something less harmful, serious or unpleasant

d (v) to notice or understand something in a particular way

e (adj) producing very clear, powerful images in the mind

f (v) to think that something is larger, more likely or more important than it really is

g (adj) based on feelings or instinct rather than on facts or careful reasoning

h (n) a strong feeling of fear about something that may happen

Complete the sentences with the correct form of a word from either article.

1. Many people have an exaggerated _____ of how dangerous air travel really is. (perception)

2. Statistically, a fatal fall at home is far more _____ than a shark attack. (probable)

3. Pilots are trained to follow checklists rather than trust their _____, which can fail under pressure. (intuition)

4. Regular inspections play a key role in the _____ of risks on building sites. (mitigation)

5. Experienced climbers warn that _____ is often more dangerous than the mountain itself. (complacency)

WHY DO WE FEAR THE WRONG THINGS?

1 Ask a nervous passenger why they grip the armrest during take-off, and they will rarely mention statistics. Measured by distance travelled, commercial flying is one of the safest ways to get from one place to another — far safer than the car journey to the airport. Yet millions of people **perceive** it as more dangerous than driving, and some refuse to fly at all. This gap between the risks we face and the risks we fear is not a sign of stupidity; it reflects the way the human mind evolved to make quick decisions in a world without spreadsheets or safety reports.

2 Psychologists have long argued that we judge how likely something is by how easily examples come to mind. A plane crash is reported around the world for days, complete with **vivid** images of wreckage and grieving families, while the thousands of people who die on the roads each year rarely make the headlines. Because dramatic events are easy to picture, we tend to **overestimate** how often they happen, and to underestimate the slow, quiet dangers that never become news. Shark attacks, lightning strikes and terrorist incidents all loom far larger in the imagination than in the mortality tables.

3 Certain kinds of risk seem to trigger a particularly strong reaction. Researchers use the term ‘**dread** risk’ to describe low-probability events that could kill many people at once, such as a nuclear accident or a new epidemic. Such risks feel uncontrollable and unfamiliar, and people often demand far greater protection from them than from everyday hazards that, over time, claim many more lives. Control matters too: drivers tend to feel safer than passengers partly because their hands are on the wheel, even though being in charge does little to change the odds of a collision.

4 Misjudging risk is not merely an academic curiosity; it can have serious consequences. After the attacks on the United States in September 2001, many Americans chose to drive rather than fly for months afterwards. One widely cited study estimated that the extra traffic may have led to around 1,500 additional road deaths in the following year — far more than the number of people killed on board the hijacked planes themselves. By avoiding a frightening but rare danger, people unknowingly exposed themselves to a familiar and much more common one.

5 Familiarity, in fact, is one of the main reasons we underrate everyday hazards. A large share of serious accidents happen at home — on the stairs, in the kitchen or in the bathroom — yet few people think of their own house as a dangerous place. The same is true of habits such as driving while tired, eating badly or never exercising: because the harm builds up slowly and nothing bad happened yesterday, we become **complacent**. A risk we have survived a thousand times simply stops feeling like a risk at all.

6 Numbers, unfortunately, do not always help. Many people find **probability** difficult to interpret, especially when it is expressed as a percentage. A headline announcing that a medicine ‘doubles the risk’ of a side effect sounds alarming, but if the original risk was one in ten thousand, doubling it still leaves the chance extremely small. Without knowing the starting point, readers have no way of judging whether the change matters, and frightening relative figures can lead patients to abandon treatments that would actually do them good.

7 None of this means we should ignore our feelings. Our **intuitive** sense of danger often serves us well, warning us away from a dark alley or an unstable cliff edge long before we could calculate anything. The challenge is to recognise when instinct is likely to mislead us and to check it against the evidence. Experts suggest that presenting risks in simple terms — ‘three people in a thousand’ rather than ‘0.3%’ — can **mitigate** some of these biases, helping people to fear a little less of what is rare and a little more of what is common.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Lost in the numbers

B A fear that defies the figures

C When the familiar feels safe

D Why experts usually get it wrong

E Catastrophe and control

F Training the gut, not ignoring it

G What sticks in the mind

H The hidden cost of avoiding danger

LANGUAGE DEVELOPMENT - Talking about risk

VOCABULARY SKILL

Verb + noun collocations with risk

The noun *risk* combines with a fairly fixed set of verbs, and choosing the wrong one immediately sounds unnatural — we **take a risk** or **pose a threat**, but we do not 'make a risk' or 'do a danger'. Learning these collocations as whole chunks will make your writing about safety, health and decision-making far more precise.

take / run	a risk, a chance (neutral)
pose	a risk, a threat, a danger (negative)
reduce / minimise	the risk, the danger, the impact (positive)
weigh up / assess	the risks, the odds, the options (neutral)
outweigh	the risks, the costs, the drawbacks (positive)
raise / increase	the risk, the likelihood, concern (negative)

Complete each sentence with one verb (in the correct form) and one noun from the table above. In some items, more than one answer is possible.

1. Before investing their savings, the couple sat down to _____ the _____ carefully.

2. The doctors agreed that the benefits of the operation clearly _____ the _____.

3. Rising sea levels _____ a serious _____ to low-lying coastal towns.

4. Wearing a helmet can significantly _____ the _____ of a serious head injury.

5. He knew he was _____ a _____ by leaving a secure job, but he did it anyway.

6. Lack of sleep greatly _____ the _____ of making mistakes at work.

Article 2

2 The sentences below are about risk-taking. Choose the correct meaning (a, b or c) for the word in bold.

1. Many climbers describe reaching the summit as the most **exhilarating** moment of their lives.

a extremely exciting and enjoyable
b physically exhausting
c deeply frightening

2. Psychologists link a love of fast cars, travel and loud music to a trait called **sensation-seeking**.

a a habit of avoiding strong feelings
b a need for new and intense experiences
c a tendency to care about others' feelings

3. The judge described the driver's behaviour as **reckless** and irresponsible.

a carefully planned
b simply unlucky
c showing no thought for danger or consequences

4. Professional free divers take only **calculated** risks, after months of training.

a carefully considered in advance
b very small and unimportant
c completely unnecessary

5. Her **meticulous** preparation meant that nothing was left to chance.

a rushed and last-minute
b very careful and detailed
c unusually expensive

6. After his second business failed, he started a new **venture** selling handmade furniture.

a a hobby done in free time
b a long holiday
c a new business activity that involves risk

7. Losing their biggest client was a serious **setback** for the small company.

a something that delays progress or makes a situation worse
b an unexpected success
c a formal legal agreement

8. Athletes with great **resilience** bounce back quickly from injuries and disappointments.

a physical strength
b the ability to recover quickly from difficulties
c natural talent

WHY DO SOME PEOPLE CHASE DANGER?

1. Every year, thousands of people pay large sums of money to throw themselves out of aeroplanes, climb frozen waterfalls or ski down slopes so steep that a single mistake could be fatal. To many observers, such behaviour looks **reckless**, even selfish, since rescue teams and hospitals often bear the cost when things go wrong. Yet those who take part in extreme sports rarely describe themselves as careless. Instead, they talk about focus, freedom and the **exhilarating** sense of being completely alive — feelings they say they cannot find anywhere else in modern, comfortable life.

2. Psychologists have tried for decades to explain why some people actively seek out danger while others avoid it at all costs. One influential idea is **sensation-seeking**, a personality trait describing the need for new, varied and intense experiences. People who score highly on this trait tend to become bored easily and may be drawn not only to dangerous sports but also to travel, unusual food and loud music. Research suggests that the trait is partly inherited, and that it typically peaks in late adolescence before gradually declining with age.

3. However, the stereotype of the adrenaline junkie who cares nothing for safety may be misleading. Interviews with experienced mountaineers, BASE jumpers and big-wave surfers reveal people who are, if anything, obsessed with control. They study weather patterns, check equipment repeatedly and rehearse emergency procedures until they become automatic. Their preparation is **meticulous** precisely because they understand how little room for error there is. In their own eyes, they are not gambling with their lives but taking **calculated** risks, reducing the danger as far as possible before deciding whether the uncertainty that remains is acceptable.

4. Interestingly, very similar language is used by entrepreneurs. Starting a business is rarely life-threatening, but it can put savings, careers and relationships on the line, and a large proportion of new firms close within their first five years. Successful founders often insist that they are not natural gamblers. Before launching a **venture**, they test their ideas on small groups of customers, keep their costs low and plan what they would do if things went wrong. Like climbers, they try to limit the downside while leaving room for a potentially large reward.

5. What both groups seem to share is an unusual relationship with failure. A climber who turns back a few hundred metres from the summit, or a founder whose first company collapses, may treat the experience as useful information rather than as a disgrace. Studies of serial entrepreneurs suggest that an early **setback** can prove valuable, provided that people reflect honestly on what went wrong. This kind of **resilience** — the capacity to recover, adapt and try again — may matter far more in the long run than raw courage or talent.

6. None of this means that risk-taking is always admirable. Overconfidence has killed experienced climbers and bankrupted talented business people, and luck plays a larger role in many success stories than the winners like to admit. Still, the lives of those who deliberately court danger raise an uncomfortable question for the rest of us. In a society that increasingly tries to remove every possible hazard, we may be losing opportunities to discover what we are capable of — and to learn that most fears, once faced, are smaller than they first appeared.

DISCUSSION

Work with a partner. Discuss the questions.

1 Would you ever try an extreme sport such as skydiving or ice climbing? Why / why not?

2 Do you think extreme athletes and entrepreneurs really share the same attitude to risk? Give examples.

GRAMMAR

Mixed and inverted conditionals

Conditionals do not always follow the neat 'first, second, third' patterns taught at lower levels. Mixed conditionals link a condition in one time frame to a result in another, while inverted conditionals replace if with had, should or were to create the more formal, emphatic style typical of academic and official writing.

The rules below show how to combine different time frames and how to form inverted conditionals correctly.

1. **Past condition → present result:** *If* + past perfect, *would* + infinitive. Use it when an earlier action (or lack of it) affects the situation now.

If she had taken out travel insurance, she wouldn't be paying the hospital bill herself now.

2. **Present condition → past result:** *If* + past simple, *would have* + past participle. Use it when a permanent quality or situation explains a past event.

If he were a more cautious person, he would never have invested all his savings in one start-up.

3. **Had** + subject + past participle replaces *if* + past perfect (third or mixed conditional).

Had I known the ice was so thin, I would never have walked across the lake.

4. **Should** + subject + infinitive replaces *if* in the first conditional. It sounds formal and suggests the situation is fairly unlikely — common in instructions and notices.

Should you experience any chest pain, stop exercising immediately and inform a member of staff.

5. **Were** + subject + *to* + infinitive (or **Were** + subject + adjective/noun) replaces *if* in the second conditional.

Were the government to raise the speed limit, the number of accidents would almost certainly increase.

6. **Were it not for** / **Had it not been for** + noun = 'if something didn't exist / hadn't happened'. In inverted conditionals the negative is never contracted (*Hadn't I known...* ×).

Had it not been for the safety rope, the climber would have fallen more than fifty metres.

Exercise A — Complete the paragraph with the correct form of the verbs in brackets, or with one word (*had*, *should* or *were*) where there is no verb given.

Three years ago, Dilnoza left a well-paid job at a bank to open a small bakery. (1) _____ she known how difficult the first year would be, she might never have taken the risk. If she had stayed at the bank, she (2) _____ (earn) a comfortable salary now, but she (3) _____ (not / feel) nearly as proud of her work. (4) _____ it not been for a loan from her uncle, the business would have closed within months. Today she tells young entrepreneurs: '(5) _____ you decide to start your own business, be ready for setbacks.' She also believes that if she (6) _____ (be) a more cautious person by nature, she would never have left her job in the first place. (7) _____ the economy to slow down next year, she says, she would simply adapt. And (8) _____ it not for her loyal customers, she admits, the bakery would not exist today.

Exercise B — Rewrite each sentence as an inverted conditional, beginning with the word in brackets.

1. If I had known about the storm, I would have cancelled the trip. (Had)

2. If you have any problems with your equipment, contact the instructor immediately. (Should)

3. If it weren't for his helmet, he would be in hospital now. (Were)

4. If the company invested more in safety, there would be fewer accidents. (Were)

5. If they hadn't checked the weather forecast, they would be stuck on the mountain now. (Had)

6. If it hadn't been for the rescue team, the climbers would have died. (Had)

ACADEMIC WRITING

Hedging and boosting — showing how certain you are

ACADEMIC WRITING SKILL

Hedging and boosting

Hedging means using tentative language — *may*, *might*, *tend to*, *appear to*, *is likely to*, *it could be argued that* — so that a claim is not stated more strongly than the evidence allows. Boosting does the opposite: *clearly*, *undoubtedly*, *there is strong evidence that* show confidence when the evidence is solid. Good academic writers balance the two, hedging speculative ideas and boosting well-established facts.

UNHEDGED: *Extreme sports are dangerous, and young people who do them are addicted to adrenaline.*

HEDGED: *Some extreme sports may carry a considerably higher risk of injury, and it could be argued that certain participants tend to seek out intense physical sensations.*

Rewrite the sentences below, making the claims more tentative (hedging). For sentences marked [B], make the claim more confident (boosting).

1 Social media makes teenagers take dangerous risks.

_____

2 People who do extreme sports are addicted to adrenaline.

_____

3 Most new businesses fail because their owners are reckless.

_____

4 Flying is safer than driving. [B]

_____

5 Children who are never allowed to take risks become anxious adults.

_____

6 Wearing a seat belt reduces the risk of death in a car crash. [B]

_____

WRITING TASK

Write a short paragraph (80–100 words) answering the question 'Are people today more afraid of risk than previous generations?' Use at least three hedging expressions and one boosting expression.

PART 5

Read the following text for questions 30-35.

Australia's flying foxes are 'curious, gentle and intelligent' – and often misunderstood

Endangered spectacled flying foxes and vulnerable grey-headed flying foxes are 'astonishing' animals but misinformation is rife.

One of the most spectacular sights at Adelaide's Womadelaide music festival is not on the official lineup. As dusk approaches, thousands of grey-headed flying foxes begin chattering and stretching their wings as they prepare to ascend from their roosts in Botanic Park and set out in search of food. "It's an amazing sight," says associate professor Wayne Boardman, a wildlife veterinarian and flying fox researcher at the University of Adelaide. At first a few early birds take to the skies, he says, then more and more, "diving, swooping, flapping". "As the night sky darkens, steadily the full camp take to the air – a mass of circling, squawking bats all eager to start foraging after a day of rest."

For many festivalgoers, the nightly fly-out ritual is their first time seeing the animals, also called megabats, up close. Boardman runs a "bat tent" near one of the main stages, kitted out with displays, videos and binoculars to educate visitors and address misconceptions about the flying foxes. "They are just astonishing animals," he says. "The loveliness of flying foxes, their sheer beauty, astonishing biology, the amazing anatomy – it never ceases to amaze me." Yet misinformation about the animals is rife, from well-meaning parents telling children they use sound waves to navigate (they don't), to more sinister notions including that the animals are "pests" that "don't belong here". "They came of their own volition," Boardman says. Grey-headed flying foxes first arrived in Adelaide in 2010, establishing a permanent camp on First Creek, in parklands just north of the CBD, where they have increased in number from 1,000 to about 50,000 today.

Flying foxes are nomadic, capable of travelling up to 50km at night in search of food resources such as flowering gums, figs and fruit trees. Tim Pearson, a wildlife ecologist who specialises in flying fox behaviour and communication, says the "double whammy" of destroying native forests while providing alternative food sources in farms and residential areas creates the potential for human-bat conflict and harassment, identified as a threat in recovery plans for endangered spectacled flying foxes and vulnerable grey-headed flying foxes. Populations of both species have declined in recent decades, although assessing their current status has become more challenging since the national flying fox monitoring program concluded in 2022. "The Australian attitude towards wildlife has always been positive, providing it doesn't inconvenience you," Pearson says. Unfortunately for flying foxes, in their case it often does.

Things have been getting better, Pearson says, but flare-ups still occur when practical concerns – about noise, the animals' impact on fruit growers, power outages or potential disease threats – are reported in breathless terms by the media. Efforts to disperse the bats are generally counterproductive and attempts to promote empathy and coexistence have not worked. Scientists now think "normalising" flying foxes might be a way forward, depicting them as a natural part of Australian wildlife in art, photography, books, community events and education. Despite their often visible and audible presence in cities and towns, representations of flying foxes in mainstream media and popular culture remain scarce and largely negative, according to the paper published in Australian Zoologist. Pearson, a co-author of the paper, works with tame education flying foxes at a wildlife park. Once people see these cute, furry animals up close, they realise they are "curious, gentle, intelligent" with individual personalities, he says.

Doug Gimesy, a wildlife and conservation photojournalist, says bats – flying foxes and microbats – make up nearly 20% of all mammals, but "don't get 20% of the air time". "As a group, they're often ignored, under-appreciated and misunderstood," he says. "At worst they're vilified and face a constant battle of disinformation, hate speech and are even persecuted." Gimesy began photographing flying foxes in 2016 and has since spent more than 160 days in the field, taking thousands of images that document every aspect of their lives. Through his award-winning photographs and children's book, Life Upside Down, Gimesy is hoping to show people how magnificent the animals are, "to get people to engage and ask questions and understand". Two images have been important to capture and share, he says. In one, a flying fox pokes its pink tongue into a eucalyptus blossom – because people don't realise what important pollinators they are, he says.

In the other, a pup clings to its mum as it breastfeeds in mid-flight, which usually triggers an emotional response. It's the realization "they carry their young in flight, and they're flying mammals". The paper says shifting attitudes towards species such as sharks and crocodiles, once subject to widespread persecution, offer optimism that contentious issues between flying foxes and people can be managed without maligning them as a species. In Fisher reserve, in the inner Melbourne suburb of North Fitzroy, children play on the green grass as an oversized flying fox takes wing with a magpie, a squirrel glider and pink paper planes against a purple sky. The colourful mural, by Alex Sugar, is one of several depicting flying foxes alongside other Australian wildlife and quirky elements. Sugar is fond of the bats, and they became a recurrent feature in his work after their nightly flyout over the Yarra River drew his attention. "We need to highlight them more in everyday culture and the issues that they're going through as well with their current habitat," he says.

For questions 30-33, fill in the missing information in the numbered spaces. Write no more than ONE WORD and / or A NUMBER for each question.

FLYING FOXES

Flying foxes live by moving from place to place. Their diet usually consists of 30_____, flowering gums and some other fruits.

They have two 31_____, both of which are endangered.

Scientists regard that portraying them as a part of local life them might be useful. They use the term 32_____ to emphasize it.

Although they are indeed encountered in ordinary life, people pay little attention to them, and even when they do, people see them as 33_____.

For questions 34-35, choose the correct answer A, B, C, or D. Mark your answers on the answer sheet.

34. According to the passage, once people get to know flying foxes closer ...

A) they start to believe that they don't belong in Australia.
B) they become terrified by the sight of them.
C) they started to become guilty by the fact that they themselves destroyed the species' natural habitat.
D) they start to think that flying foxes are smarter than they thought.

35. According to the passage, the flying foxes ...

A) takes care of their offspring even when they are flying.
B) eat a lot of fruits and pests like mice and rats.
C) are a favorite among the Australian children.
D) are seldom photographed as they fly too fast.

✔ Wordlist

1. ascend ▢
2. roost ▢
3. set out ▢
4. forage ▢
5. kitted out with ▢
6. misconception ▢
7. sheer ▢
8. never cease to amaze ▢
9. rife ▢
10. sinister ▢
11. of their own volition ▢
12. nomadic ▢
13. specialise in ▢
14. double whammy ▢
15. harassment ▢
16. endangered ▢
17. vulnerable ▢
18. decline ▢
19. flare-up ▢
20. breathless ▢
21. disperse ▢
22. counterproductive ▢
23. empathy ▢
24. coexistence ▢
25. scarce ▢
26. vilify ▢
27. persecute ▢
28. pollinator ▢
29. contentious ▢
30. malign ▢

PART 5

You will hear three extracts. Choose the correct answer (A, B or C) for each question (24-29). There are TWO questions for each extract. Mark your answers on the answer sheet.

Extract One

24. Many parents ...
A) don't have the time or knowledge to help with their children's homework.
B) think that schools should have the knowledge of how to help with their children's homework.
C) don't want to help with their children's homework.

25. The app ...
A) is for parents who wants the best tutor for their children.
B) is for students who want the most expensive and efficient teachers.
C) is for students who want to find a tutor.

Extract Two

26. Chris cannot attend to the study group because ...
A) he doesn’t want to.
B) he has other study groups to attend to.
C) he has a task.

27. They agreed to meet ...
A) in the library
B) in the study hall
C) in the cafeteria

Extract Three

28. What has made Clara check on Ben?
A) He missed their class that day.
B) He's missed a few classes.
C) He has a lot of essays to write.

29. What is the main way Ben's anxiety is affecting normal life?
A) He feels stupid.
B) He can't remember what day it is.
C) He doesn't want to go out.

Phrases & Idioms Wordlist

Play with fire: To do something dangerous that is likely to cause problems for you.

Push your luck: To keep taking risks after being successful, so that you may lose what you have gained.

Throw caution to the wind: To stop being careful and do something bold or risky.

Take the plunge: To finally decide to do something important or risky after thinking about it for a long time.

Stick your neck out: To say or do something that others may criticise or that could get you into trouble.

Skate on thin ice: To be in a risky situation that could easily lead to trouble.

Bite off more than you can chew: To try to do more than you are able to manage.

Better safe than sorry: It is wiser to be careful now than to regret taking a risk later.

Nothing ventured, nothing gained: You cannot achieve anything worthwhile unless you are willing to take a risk.

Put all your eggs in one basket: To depend completely on one plan or investment, so that you could lose everything.

A leap in the dark: An action whose results cannot be known in advance.

Hedge your bets: To protect yourself by supporting more than one option, so that you do not lose everything.

Tempt fate: To do something that seems to invite bad luck or disaster.

Go out on a limb: To take a position or action that is risky and that few others support.

Against all odds: Despite seeming very unlikely to succeed.

USEFUL LANGUAGE — Opinion essay

Function	Useful phrases
Introducing the topic	*It is often argued that… / In recent years, there has been growing concern that…*
Stating your opinion	*I would argue that… / In my view, … / I largely agree that…*
Hedging a claim	*It could be argued that… / This may / tends to… / There is some evidence that…*
Giving examples	*For instance, … / A case in point is… / This is particularly true of…*
Conceding a point	*Admittedly, … / While it is true that…, … / That said, …*
Concluding	*In conclusion, … / On balance, … / Overall, I believe that…*

THE TASK

You are writing an essay for an academic audience. Write 180–200 words.

Some people believe that children today are overprotected and should be allowed to take more risks in order to learn and develop. To what extent do you agree or disagree?

PARAGRAPH PLAN

Introduction	paraphrase the question + state your position clearly
Body paragraph 1	main reason(s) for your view + a specific example
Body paragraph 2	concede the opposing view — which risks are not acceptable?
Conclusion	summarise your position in new words; do not add new ideas

RULES for Part 2 (essay)

☐ Answer the question that is actually asked — 'To what extent…?' requires a clear position, not just a list of points.

☐ Aim for 180–200 words in four paragraphs; going far over or under the limit costs marks.

☐ Paraphrase the question in your introduction instead of copying it word for word.

☐ Support every main idea with a reason and a concrete example.

☐ Hedge claims you cannot prove (may, tend to, it could be argued) and boost well-established facts (clearly, there is strong evidence).

☐ Show a range of grammar — for example, a mixed or inverted conditional — and use formal linking words.

SAMPLE ANSWER

In recent years, there has been growing concern that children are being raised in an environment so carefully controlled that they rarely face any danger at all. I largely agree that young people should be given more freedom to take sensible risks, as this is likely to make them more confident and capable adults.

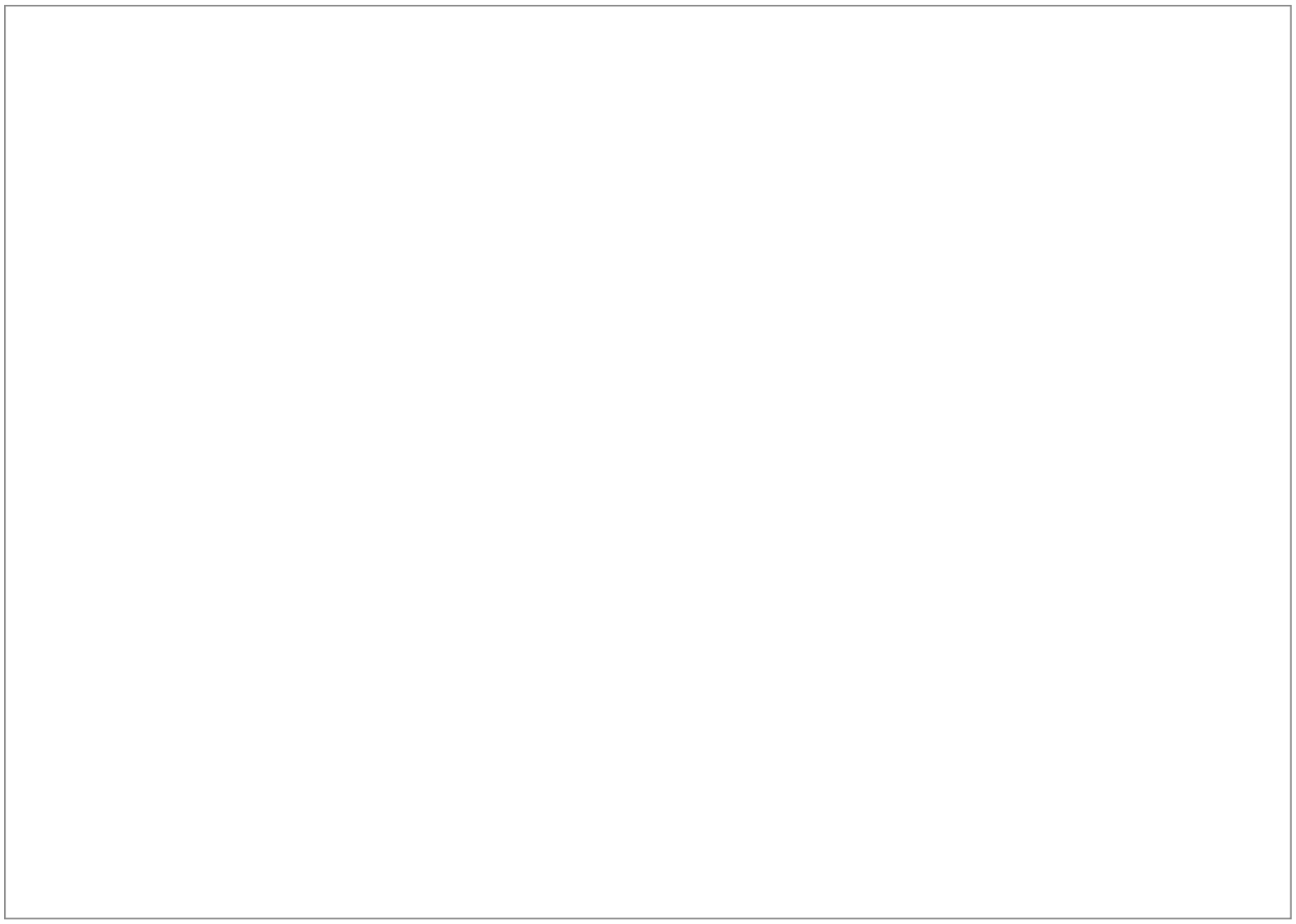
Firstly, it could be argued that children learn to judge danger only by experiencing it. A child who climbs a tree and slips tends to understand height and balance far better than one who has only been warned about them. Were parents to remove every challenge, children might never develop this practical judgement. Moreover, there is strong evidence that independent play builds resilience, since minor failures teach young people that setbacks can be overcome.

Admittedly, some risks are simply not worth taking. Busy roads, deep water and the darker corners of the internet present real dangers that children cannot be expected to assess alone, and adults clearly have a duty to protect them.

In conclusion, while sensible supervision remains essential, I believe that overprotection may do more harm than good. Had my own parents not let me make small mistakes, I would probably be far more anxious today.

PART 2 PRACTICE — now you try

Some people argue that governments should protect people from the consequences of their own risky choices, for example by banning dangerous sports or taxing unhealthy food. Others believe adults should be free to take whatever risks they choose. Discuss both views and give your own opinion.



SPEAKING PART 3

STATEMENT

People should take more risks in life.

Discuss both sides of the statement and give your own opinion. Use the points below to help you.

FOR	AGAINST
Risk-taking leads to personal growth and new opportunities	Unnecessary risks can cause injury, debt or harm to others
Most innovators and successful entrepreneurs took big chances	Not everyone can afford to fail — the consequences are unequal
Playing it safe can lead to regret and a narrow, predictable life	Careful planning often achieves more than bold gambles

SAMPLE ANSWER

*Personally, **I'm largely in favour of** the idea that people should take more risks, although **I'd add an important condition**: the risks need to be thought through.*

***The main argument in favour is that** growth rarely happens inside our comfort zone. Moving to a new city, changing careers or starting a business are all frightening, but they're also how people discover what they're capable of. **A good example is** my cousin, who left a stable job to open a small café. Had she stayed where she was, she'd probably still be wondering 'what if'. **On top of that**, many of the inventions we rely on today exist only because someone was willing to fail in public.*

***On the other hand, it's worth bearing in mind that** not everyone can afford to take chances. Someone with savings and a supportive family can bounce back from a failed venture, whereas a person supporting three children may lose everything. **There's also the point that** some risks — like dangerous driving — put other people in danger, not just the risk-taker.*

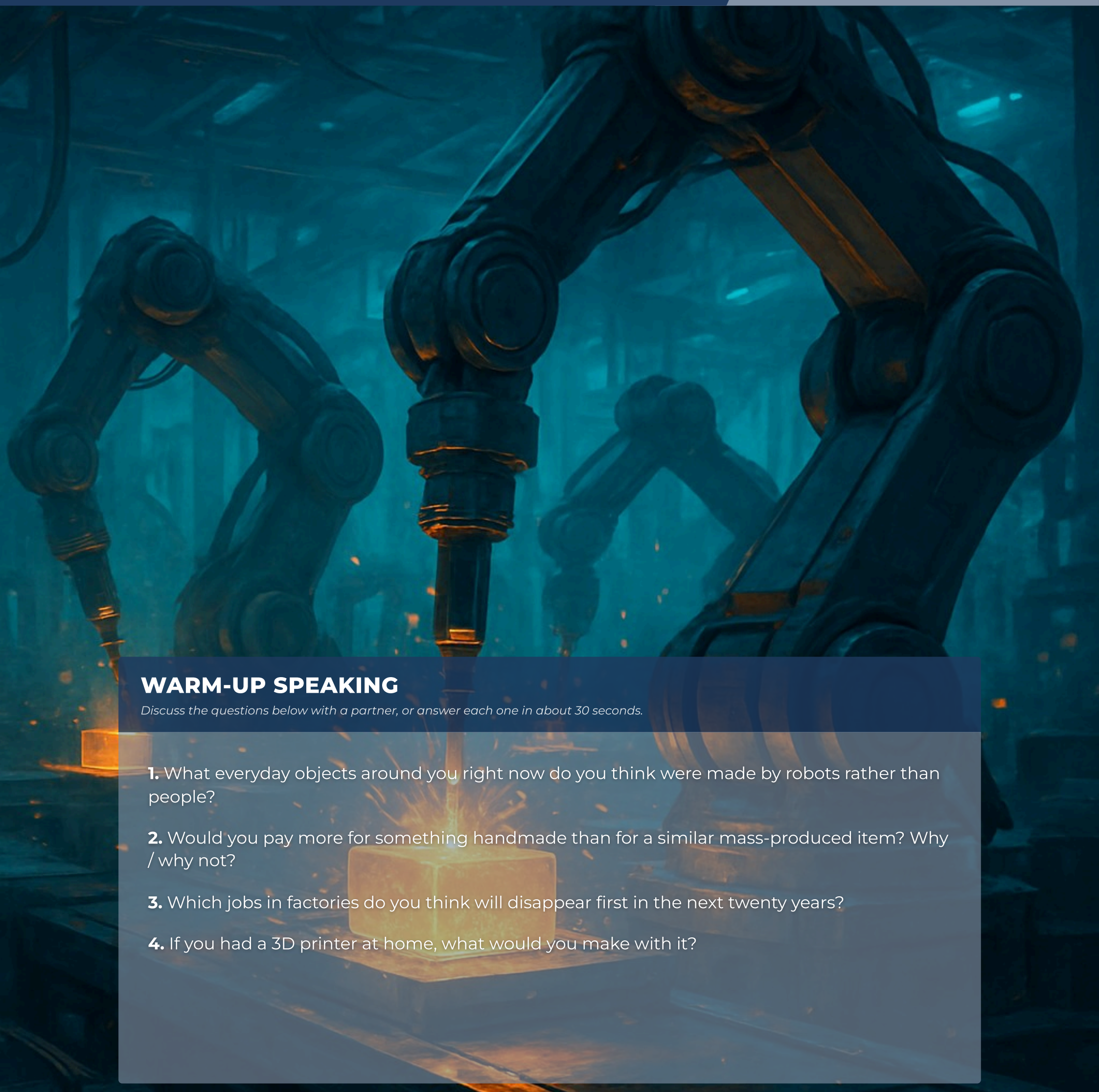
***All things considered, I'd say that** people should take more calculated risks, not reckless ones. Weighing up the odds, having a back-up plan and knowing when to walk away seem to me just as important as courage.*

USEFUL LANGUAGE	
Giving your view:	*Personally, I'm largely in favour of… / I'd have to disagree with the idea that…*
Arguments for:	*The main argument in favour is that… / One clear advantage is…*
Arguments against:	*On the other hand, it's worth bearing in mind that… / A major drawback is…*
Giving examples:	*A good example is… / Take, for instance, …*
Adding a point:	*On top of that, … / There's also the point that…*
Concluding:	*All things considered, I'd say that… / On balance, …*

MANUFACTURING

THE FUTURE OF MAKING THINGS

UNIT 5



WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. What everyday objects around you right now do you think were made by robots rather than people?

2. Would you pay more for something handmade than for a similar mass-produced item? Why / why not?

3. Which jobs in factories do you think will disappear first in the next twenty years?

4. If you had a 3D printer at home, what would you make with it?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Modern laser cutters work with such **precision** that each piece fits perfectly the first time.

2 The company **deployed** twenty new robots on the assembly line just before the busy season.

3 Almost a third of the factory's **workforce** is now made up of engineers and technicians.

4 Typewriters became almost **obsolete** once personal computers were cheap enough for every office.

5 Packing the same product into identical boxes all day is extremely **repetitive** work.

6 A single supervisor can now **oversee** three production lines from one control room.

7 Thanks to the new equipment, the plant's daily **output** rose from 500 to 800 units.

8 When the mine closed, many former miners chose to **retrain** as electricians or plumbers.

a (n) the amount of something that a person, machine or factory produces

b (adj) no longer used or needed because something newer and better exists

c (v) to learn new skills, especially in order to do a different job

d (n) the quality of being very exact and accurate

e (v) to watch and organise work to make sure it is done correctly

f (n) all the people who work in a company, industry or country

g (adj) involving doing the same thing many times, often in a boring way

h (v) put into use or into position for a particular purpose

Complete the sentences with the correct form of a word from either article.

1. Robotic arms can cut and weld metal far more _____ than even the most experienced human worker. (precision)

2. In the electronics industry, rapid _____ means that equipment bought today may be out of date within five years. (obsolete)

3. Many sports brands now offer online _____ of trainers, from the colour of the laces to the shape of the sole. (customised)

4. The large-scale _____ of cobots in small workshops has happened much faster than analysts predicted. (deployed)

5. Investors were not convinced of the long-term _____ of a neighbourhood print shop. (viable)

INSIDE THE LIGHTS-OUT FACTORY

1 On the outskirts of a mid-sized Japanese city stands a factory where the lights are almost never switched on. Inside, rows of robotic arms weld, lift and assemble components around the clock, while a handful of engineers monitor screens from a control room upstairs. Plants like this, sometimes called 'lights-out' factories, can run for weeks without a single person on the production floor. They are still rare, but they offer a glimpse of where manufacturing may be heading as robots are **deployed** in ever greater numbers across the world.

2 The appeal is easy to understand. A robot can place a microchip with a **precision** of a fraction of a millimetre, and it will do so in exactly the same way on its ten-thousandth attempt as on its first. It does not get bored, distracted or tired at the end of a long shift, and it never needs a holiday. For products where tiny errors are costly, such as car engines or medical devices, this consistency is arguably worth even more than speed, although robots usually offer that as well.

3 Until recently, only the largest manufacturers could afford this level of automation. Industrial robots were expensive, enormous and had to be kept in safety cages, well away from people. Today, prices have fallen sharply, and a new generation of smaller 'collaborative' robots, or cobots, can work safely alongside human staff. A family-run furniture workshop can now lease a robotic arm for a monthly fee, programme it using a tablet and move it between tasks in an afternoon, something that would have been unthinkable twenty years ago.

4 Unsurprisingly, the first jobs to be affected are those built around **repetitive** tasks: packing boxes, tightening the same screw, moving parts from one conveyor belt to another. In some regions, the number of people employed in these roles has dropped noticeably, even as factories produce more than ever. Economists point out that the manufacturing **workforce** has been shrinking for decades in many wealthy countries, but automation has accelerated the trend, and the people most affected are often those with the fewest qualifications.

5 At the same time, robots are creating jobs that barely existed a generation ago. Someone has to programme the machines, repair them when they break down and **oversee** entire production lines from a screen. These roles tend to be better paid and less physically demanding, but they require technical skills that many existing employees simply do not have. A growing number of companies now pay for staff to **retrain**, turning former machine operators into maintenance technicians and data analysts rather than replacing them.

6 Automation is also changing where things are made. For decades, manufacturers moved production to countries where labour was cheap. When robots do most of the work, however, wage differences matter far less, and factories can be built closer to their customers. Several European and American firms have brought production back home in recent years, arguing that a highly automated local plant can match the **output** of a much larger overseas one while cutting shipping times, transport costs and emissions.

7 Robots still have clear limits. They are excellent at doing one thing perfectly, but surprisingly poor at coping with the unexpected: a slightly bent component, an unusual order, a sudden change in design. Human workers remain far better at improvising, spotting problems and suggesting improvements. The real risk, many experts argue, is not that people will become **obsolete**, but that their skills will if they are not regularly updated. The factory of the future may be quieter and darker, but it is unlikely to be completely empty.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Bringing production back home

B A factory that works in the dark

C The jobs that disappear first

D Why robots are becoming more expensive

E Accuracy that never gets tired

F Where humans still have the edge

G Robots within reach of smaller firms

H New roles on the factory floor

LANGUAGE DEVELOPMENT - Talking about production and efficiency

VOCABULARY SKILL

Verb + noun collocations

When we describe how factories and businesses change, certain verbs combine with a small, predictable group of nouns, for example **boost productivity**, **meet demand** or **phase out machinery**. Learning these as fixed chunks helps you write about industry precisely and naturally, rather than relying on general verbs such as *make*, *do* or *change*.

boost	productivity, output, efficiency (increase)
cut	costs, waste, jobs (reduce)
streamline	processes, operations, production (make simpler and more efficient)
meet	demand, deadlines, targets (achieve what is needed)
phase out	machinery, jobs, older models (stop gradually)
scale up	production, operations, manufacturing (expand)

Complete the sentences with the correct verb from the table above. Change the form if necessary.

1. After installing the new robots, the plant managed to _____ its output by nearly 40%.

2. The company plans to _____ its oldest machines over the next three years and replace them with cobots.

3. To _____ demand during the holiday season, the factory added an extra night shift.

4. Switching to 3D-printed parts helped the workshop _____ waste, as only the material needed is used.

5. Once the prototype had been approved, the start-up began to _____ production for the international market.

6. Moving all orders onto a single digital platform has helped the firm _____ its operations and remove several unnecessary steps.

Article 2

2 The sentences below are about 3D printing and local manufacturing. Choose the correct meaning (a, b or c) for the word in bold.

1. The engineers built a simple **prototype** of the drone so they could test it before full production.

a a final version sold in shops
b a first model used for testing an idea
c a cheap copy of another product

2. The online shop sells bicycles **customised** to each rider's height and riding style.

a produced in very large numbers
b imported from another country
c changed to suit a particular person

3. The company moved from one huge plant to a more **decentralised** network of small workshops.

a controlled from one central place
b spread across many smaller local places
c closed down permanently

4. A single delay at a port can disrupt the entire **supply chain** for a product.

a the series of companies and steps involved in making and delivering goods
b a list of prices charged by suppliers
c a group of shops owned by the same company

5. Self-service checkouts have begun to **displace** cashiers in many supermarkets.

a to support
b to train
c to take the place of

6. The firm found a profitable **niche** making tools for left-handed people.

a a small, specialised area of demand
b a large general market
c a new type of factory

7. Before the strike began, the factory built up a **stockpile** of raw materials.

a a tax on imported goods
b a damaged batch of products
c a large supply kept for future use

8. Without government support, the project was not financially **viable**.

a too risky to try
b capable of succeeding in practice
c popular with young people

WILL 3D PRINTING BRING FACTORIES HOME?

1. For most of the last hundred years, manufacturing followed a simple rule: the bigger the factory, the cheaper each product. Goods were made in enormous quantities in a few locations and then shipped around the world through a long and complicated **supply chain** of ports, warehouses and lorries. 3D printing, also known as additive manufacturing, challenges that rule. Instead of cutting or moulding material, a printer builds an object layer by layer from a digital file, which means the same machine can produce a thousand identical items or a thousand different ones at roughly the same cost per piece.

2. The technology was first used mainly by engineers and designers who needed a quick **prototype** to test an idea before investing in expensive tools and moulds. A model that once took weeks to make by hand could be printed overnight, examined, improved and printed again the next day. Over time, however, printers have become faster, more reliable and able to work with a far wider range of materials, from plastics and resins to metals, ceramics and even concrete. Some aerospace companies now print finished engine parts that are lighter and stronger than conventionally produced ones.

3. Perhaps the most significant change is that products can be **customised** for individual buyers at little extra cost. Dental clinics print crowns shaped precisely to a patient's mouth; sports brands sell running shoes with soles designed around a scan of the customer's foot; hearing-aid manufacturers have almost entirely switched to printing each device individually. Products like these were once luxury items, but on-demand production is turning them into everyday ones. For businesses serving a **niche** market, such as replacement parts for vintage cars, printing can make a previously unprofitable product line suddenly worthwhile.

4. Supporters argue that 3D printing could make manufacturing far more **decentralised**. Instead of ordering a spare part from a factory on another continent, a garage or hospital might simply download the design and print it locally. Companies would no longer need warehouses holding a huge **stockpile** of parts that may never be used, which would reduce waste as well as transport emissions. During recent disruptions to global shipping, small workshops printed face shields and valves for nearby hospitals within days, showing how quickly local production can respond to a crisis.

5. The picture for jobs is mixed. On the one hand, printing could **displace** some workers in traditional factories, as well as drivers and warehouse staff whose jobs depend on moving goods over long distances. On the other hand, it may create new opportunities close to home: designers, printer technicians, materials specialists and owners of small neighbourhood print shops. Unlike earlier waves of automation, which concentrated production in fewer, larger plants, 3D printing could spread manufacturing work across many small towns and cities rather than a handful of industrial centres.

6. Even so, mass production is not about to disappear. For simple items made in millions, such as plastic bottles or screws, traditional methods remain much faster and cheaper, and printed parts can still be less durable. Many analysts therefore expect a hybrid future, in which large factories continue to produce standard goods while local printers handle spare parts, personalised products and small batches. Whether local manufacturing becomes economically **viable** on a large scale will depend on cost, regulation and skills, but for the first time in a century, the most efficient factory may not need to be the biggest.

DISCUSSION

Work with a partner. Discuss the questions.

1 Would you prefer to buy a product that was customised for you, even if you had to wait longer for it? Why / why not?

2 Do you think 3D printing will create more jobs in your town or city than it takes away? Give reasons.

GRAMMAR

The passive with modals & the causative (have / get something done)

When we describe a process, we are usually more interested in what happens to the product than in who does each step, so the passive is extremely common. Modal verbs add meaning such as possibility, obligation or advice, while the causative (have / get something done) shows that we arrange for someone else to do a job for us.

The rules below explain how to form the passive after a modal verb, and when to use have or get + object + past participle.

1. Modal + **be** + past participle describes possibility, obligation or ability in the present or future.

The parts can be printed overnight and must be checked before they are packed.

2. Modal + **have been** + past participle refers to the past — often criticism, regret or deduction.

The fault should have been noticed much earlier in the process.

3. In process descriptions, the by-phrase is usually omitted because the agent is obvious or unimportant.

The metal sheets are cleaned and then cut to size (by the workers).

4. Use **have** + object + past participle when you arrange for someone else to do a service for you.

We had the machines serviced by an outside engineer last week.

5. **Get** + object + past participle means the same but is more informal, and often suggests effort or urgency.

We need to get these orders shipped by Friday.

6. The causative can also describe something unpleasant that happened to someone, usually not by choice.

The workshop had its roof damaged in last night's storm.

Exercise A — Complete the paragraph using the words in brackets. Use the passive with a modal verb or the causative (have/get something done).

Before a new part is produced, the digital design (1) _____ (must / check) carefully by an engineer. Small errors (2) _____ (can / correct) at this stage at almost no cost. Once it has been approved, the file (3) _____ (can / send) directly to the printer. Large orders (4) _____ (might / split) between several machines to save time. Last month, the company (5) _____ (its printers / upgrade) by an outside specialist, so the process is now much faster. Earlier problems with weak parts (6) _____ (should / avoid) if the material had been tested properly. Today, every finished part (7) _____ (must / inspect) before it leaves the workshop, and customers who need parts urgently can (8) _____ (them / deliver) the same day.

Exercise B — Rewrite each sentence using the passive with a modal, or the causative with the verb in brackets.

1. Workers must clean the machines at the end of every shift.

2. A technician will repair our robotic arm tomorrow. (have)

3. Someone should have tested the software before the new line was launched.

4. We can print replacement parts on site in a few hours.

5. A local company is going to install solar panels on our factory roof. (get)

ACADEMIC WRITING

Describing a process — sequencing the stages clearly

ACADEMIC WRITING SKILL

Describing a process

When you describe how something is made, the reader needs to follow each stage in the correct order. Use sequencing language such as *initially*, *first of all*, *once…*, *after that*, *subsequently*, *at this stage*, *meanwhile* and *finally*, and combine it with the passive, since the focus is on the product rather than the people.

WITHOUT SEQUENCING: *The design is created on a computer. It is sent to the printer. The object is built layer by layer. It is removed and polished.*

WITH SEQUENCING: ***Initially**, the design is created on a computer. **Once** it has been finalised, it is sent to the printer, where the object is built layer by layer. **Finally**, it is removed and polished.*

Combine each pair of steps into one sentence, using the sequencing word or phrase in brackets.

1 The raw cotton is collected from the fields. It is cleaned to remove seeds and dirt. (initially … then)

_____

2 The fibres have been spun into thread. They are woven into cloth. (once)

_____

3 The cloth is dyed. It is washed several times to fix the colour. (after that)

_____

4 The fabric is cut into pieces. A computer checks that no material is wasted. (at this stage)

_____

5 The pieces are sewn together by machine. The labels are printed in another part of the factory. (meanwhile)

_____

6 The finished shirts are inspected and folded. They are packed and sent to shops. (subsequently)

_____

WRITING TASK

Write a short paragraph (80–100 words) describing how a product you know is made — for example bread, a cup of coffee, a T-shirt or a 3D-printed toy. Use at least four sequencing expressions and at least two passive forms.

PART 1

TEST 3

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

It is common knowledge that we have five senses. Some people believe they have six senses. They say their sixth sense is the ability to know something without using the ordinary five (1)_____ of sight, hearing, smell, touch and taste. Scientists have just discovered that some dolphins have seven senses. Researchers in Germany say bottlenose dolphins can sense electricity produced by their prey. The (2)_____ can detect the electric pulses produced by the heart beats of other sea creatures. This could help dolphins to find their next meal. Dolphins also use their excellent eyesight and sonar to sense what is around them. This seventh sense may also help them to navigate using Earth's magnetic field.

The researchers conducted a study on two dolphins named Dolly and Donna. Experiments tested what kinds of (3)_____ the dolphins could detect. Dolly and Donna could sense both kinds of electricity – direct current and alternating (4)_____. They were a lot better at picking up direct current. All organisms produce direct current electricity in water. The researchers say being able to (5)_____ this means dolphins may be able to sense the location of most kinds of fish that are near them. The dolphins could use their (6)_____ sense to hunt, even if they cannot see or hear well, or if they cannot use their other senses. The study has been published in the "Journal of Experimental Biology".

TEST 4

Read the text. Fill in each gap with ONE word. You must use a word which is somewhere in the rest of the text.

One of the most annoying things in life is an itch that won't go away. Scientists have worked for decades to find out why we (1)_____, and to find a cure. Researchers at Harvard Medical School in the USA have discovered that skin bacteria make us itch when they interact with our nerve cells. Lead researcher Liwen Deng said the (2)_____ are linked to many diseases that cause itchiness. The most common disease is eczema. Dr Leng wrote: "It's not the nicest bug to have on you, for sure. Many patients carry on their (3)_____ the very microbe we've now shown for the first time can induce itch." She added that: "Itches can be quite debilitating in patients who suffer from chronic skin conditions."

Dr Leng and her colleagues conducted their research on mice. They found an enzyme (which they dubbed V8) that was responsible for itching in the test mice. The V8 enzyme triggered a protein that made the (4)_____ itch. The scientists engineered the protein and developed a drug to stop it making the mice itch. The (5)_____ could be used to make anti-itch medication. This could block the itching process in humans. This is good news for (6)_____ sufferers. Eczema affects around 245 million people globally. As well as itchiness, sufferers often get redness on their skin and a rash. They may also get small blisters and a thickening of the skin. Eczema is twice as common in females as it is in males.

✔ Wordlist

1. common knowledge ▢
2. senses ▢
3. the ability to ▢
4. ordinary ▢
5. discover ▢
6. prey ▢
7. detect ▢
8. electric pulses ▢
9. sea creatures ▢
10. eyesight ▢
11. navigate ▢
12. magnetic field ▢
13. conduct a study ▢
14. picking up ▢
15. organisms ▢
16. location ▢
17. annoying ▢
18. for decades ▢
19. a cure ▢
20. interact with ▢
21. nerve cells ▢
22. be linked to ▢
23. itchiness ▢
24. induce ▢
25. debilitating ▢
26. chronic ▢
27. enzyme ▢
28. be responsible for ▢
29. trigger ▢
30. engineer ▢

PART 6

You will hear a part of a lecture. For each question, fill in the missing information in the numbered space.

Write no more than one word for each answer.

A WEB DEVELOPER

He studied computer science at a university.

After his graduation, he worked as a freelancer and now he works with a 30_____.

He believes that organization is crucial for developers, so he starts his work with arrangements. He usually divides his projects into 31_____.

After planning, he puts his mind on the pieces of a website which 32_____ see and interact with.

Then, he takes a break for a while. He may get a 33_____ to eat.

Debugging, he claims, needs much attention to detail. When debugging, he fixes problems in the 34_____.

Optimizing means, improving loading times and checking the efficiency of the site. To optimize a site, he may compress images or refine the code. He notes the process as 35_____.

At the end of the day, he communicates with his clients.

Phrases & Idioms Wordlist

Run like clockwork: To operate smoothly, exactly as planned and without problems.

A well-oiled machine: An organisation or team that works very efficiently.

Up and running: Working fully; in operation after being set up or repaired.

Back to the drawing board: To start planning again because the first attempt has failed.

Cut corners: To do something cheaply, quickly or carelessly, often ignoring rules or quality.

Off the shelf: Ready-made and available immediately, not designed for a particular customer.

Made to measure: Made specially to fit or suit one particular person or purpose.

Roll off the production line: To be produced in a factory, especially in large numbers.

Grind to a halt: To slow down gradually and then stop completely.

Pull the plug on (something): To stop a project or activity, usually by withdrawing money or support.

Learn the ropes: To learn the basic skills and routines of a new job.

Get your hands dirty: To do practical, physical work rather than just giving instructions.

In the pipeline: Being planned or developed and likely to happen soon.

Iron out the kinks: To solve the small problems in a plan, product or process.

A cog in the machine: A person with a small, unimportant role in a large organisation.

WRITING PART 1.2

USEFUL LANGUAGE — Formal letter

Function	Useful phrases
Opening	*Dear Sir/Madam, / Dear [Title], / I am writing in response to your recent email regarding…*
Reacting / feelings	*I was delighted/concerned to learn that… / I must admit I have mixed feelings about…*
Giving a suggestion	*I would strongly recommend that… / It might be worth considering… / Perhaps the workshop could be…*
Requesting a reply	*I would be grateful if you could… / I would appreciate further information about…*
Closing	*Thank you for taking members' views into account. / Yours faithfully, / Yours sincerely,*

PART 1.1 PRACTICE

You are a member of a local Makers' Club, a community workshop where members build and repair things. You received this email from the club's manager.

Dear Member,

I'm writing to let you know about an important change at the Makers' Club. From next month, we will be replacing half of our traditional woodworking benches with six new 3D printers and a laser cutter. We believe this will attract younger members and allow everyone to try modern manufacturing techniques. However, it does mean that there will be less space for hand tools and woodwork classes. Before we finalise the new layout, we would like to hear how you feel about this change and what training or facilities you think we should offer.

Best wishes,

The Manager

Task 1.1 — Write a letter to your friend, who is also a member of the Makers' Club. Write about your feelings about the change and what you think the club should do. Write about 50 words.

Task 1.2 — Write a letter to the manager. Write about your feelings about the change and what training or facilities you think the club should offer. Write about 120–150 words.

RULES for Task 1.2 (formal letter)

☐ Respond to the SAME message as Task 1.1 — cover the same two things (feelings + suggestions), but in more depth.

☐ Aim for 120–150 words — several short paragraphs, not one block of text.

☐ Use a formal register throughout: no contractions, slang or exclamation marks.

☐ Structure it clearly: a reason for writing, your reaction, your suggestion(s) in detail, and a polite closing.

☐ Open with Dear Sir/Madam or Dear [Title], and close with Yours faithfully/sincerely.

☐ Show a wider range of grammar than in Task 1.1 — passives with modals (could be offered, should be kept), the causative and formal linkers (furthermore, provided that, in addition).

SAMPLE ANSWER

Dear Manager,

Thank you for informing members about the planned changes to the workshop. I must admit I have mixed feelings about the news. While I am excited about the opportunity to learn 3D printing and laser cutting, I am concerned that reducing the woodworking area will discourage many long-standing members who joined the club specifically for traditional craft.

I would therefore suggest that at least a few woodworking benches should be kept, and that hand-tool classes could be offered at weekends, when demand is highest. In addition, I believe beginners' training on the new machines is essential, as most members will not have used a 3D printer before. Perhaps short evening courses could be run by experienced members.

Furthermore, it might be worth having the new equipment installed in a separate room, so that noise and dust from the woodwork area do not cause problems.

I look forward to hearing your thoughts.

Yours faithfully,

A. Karimov

TASK 1.2 PRACTICE — now you try, with a new message

You live in a neighbourhood next to a furniture factory. You received this email from the factory's director.

Dear Neighbour,

I am pleased to inform you that our factory will soon introduce a fully automated production line. As a result, the factory will operate 24 hours a day, and deliveries will arrive more frequently, including at night. We also plan to open a small training centre where local residents can learn to operate and maintain the new machines. We value our relationship with the community and would welcome your views on these plans. In particular, we would like to know whether night-time deliveries might cause problems for residents, and what kind of courses you would find most useful.

Kind regards,

The Director

Task 1.2 — Write a letter to the Director. Write about your feelings about the new plans and what you think the factory should do to support local residents. Write about 120–150 words.

SPEAKING PART 1.1

QUESTIONS

Answer the questions. Speak for about 30 seconds for each one.

1. Do you enjoy making or repairing things with your hands?
2. Is there a factory or industry that is important in your town or region?
3. Would you prefer to buy handmade or mass-produced products? Why?
4. Have you ever visited a factory or watched how something is made?
5. Do you think you would enjoy working in a factory? Why / why not?
6. What product would you most like to see made in your own country?

SAMPLE ANSWERS

Do you enjoy making or repairing things with your hands?

*Yes, **to be honest, I find it really satisfying**. My grandfather taught me to fix bicycles, so whenever something breaks at home, **I'm usually the one who has a go** at repairing it. **There's nothing quite like** seeing something work again because of your own effort.*

Would you prefer to buy handmade or mass-produced products? Why?

***It depends on what it is**, really. For everyday things like phone chargers, **I'd go for** mass-produced items because they're cheaper and more reliable. But for gifts, **I'd much rather** buy something handmade, since **it feels more personal** and you know someone has put real care into it.*

Do you think you would enjoy working in a factory? Why / why not?

***Probably not on a production line**, because I think the repetitive tasks would drive me crazy. **That said**, I'd be quite interested in a technical role, like programming robots or overseeing quality control, **since it would involve** solving problems rather than doing the same thing all day.*

USEFUL LANGUAGE	
Answering directly:	*Yes, definitely... / Not really, to be honest... / It depends on...*
Giving a reason:	*...mainly because... / ...since... / The main reason is that...*
Expressing preference:	*I'd much rather... / I'd go for... / I tend to prefer...*
Adding a personal example:	*For example, last year I... / When I was younger, I used to...*
Showing balance:	*That said, ... / On the other hand, ... / Having said that, ...*
Buying time:	*That's an interesting question... / Let me think... / I've never really thought about it, but...*

ENVIRONMENT

A CHANGING PLANET

UNIT 6

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. How much water do you think your family uses in a typical day? Could you easily use less?

2. What do you know about the Aral Sea? Why do you think its story is known around the world?

3. Which environmental problem worries you most, and why?

4. If you could bring back one animal or plant that has disappeared from your region, what would it be?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 In many dry regions, water **scarcity** now limits how much food farmers are able to grow.

2 Without **irrigation**, cotton would be almost impossible to cultivate in such a hot, dry climate.

3 Engineers **diverted** part of the river to supply a new reservoir on the edge of the city.

4 When the cannery closed, hundreds of families lost their **livelihoods** almost overnight.

5 Scientists regularly measure the **salinity** of the lake to check whether fish can still survive in it.

6 The flood had a **catastrophic** effect on the harvest, destroying almost every crop in the valley.

7 Planting trees along busy roads can help **mitigate** the effects of air pollution.

8 Managing **transboundary** rivers requires agreements between all the countries they flow through.

a (adj) causing sudden, very great damage or suffering

b (n) the supply of water to land or crops through channels or pipes

c (v) to make something harmful less severe or serious

d (n) a situation in which there is not enough of something

e (adj) crossing or existing across the borders between countries

f (n) the ways in which people earn the money they need to live

g (v) changed the direction or purpose of something

h (n) the amount of salt contained in water or soil

Complete the sentences with the correct form of a word from either article.

1. The rapid _____ of the Aral Sea shocked scientists who visited the region in the 1980s. (disappear)

2. Coastal towns were so _____ on fishing that the collapse of the industry left them with almost nothing. (depend)

3. The _____ of the wetland took more than ten years and involved hundreds of local volunteers. (restore)

4. Many experts remain _____ about whether the southern part of the sea can ever be saved. (sceptic)

5. Saxaul is valued for its remarkable _____ to drought, heat and salty soil. (resilient)

THE SEA THAT DISAPPEARED

1 Half a century ago, the Aral Sea, which lies on the border between Uzbekistan and Kazakhstan, was the fourth-largest lake on Earth, covering more than 60,000 square kilometres. Fishing fleets worked its waters, and towns along its shores depended on it for food, jobs and even a milder local climate. Today, most of that water has gone, and the Aral has become one of the most frequently quoted examples of what water **scarcity** caused by human activity can look like. Its story is a warning that reaches far beyond Central Asia.

2 The decline did not stem from drought or natural change. From the 1960s onwards, planners decided that the region's two great rivers, the Amu Darya and the Syr Darya, should feed a vast expansion of **irrigation** for cotton and other thirsty crops. Thousands of kilometres of canals were dug, and so much water was **diverted** that only a small fraction of the rivers' flow ever reached the sea. Many of the canals were unlined, so huge volumes were lost to evaporation and seepage long before they arrived at the fields.

3 As the shoreline retreated, in places by more than a hundred kilometres, the fishing industry collapsed. Ports such as Moynaq in Karakalpakstan found themselves stranded in the desert, their rusting ships now a grim attraction for visitors. Tens of thousands of people who had depended on the sea lost their **livelihoods**, and many younger residents left to look for work elsewhere. For those who stayed, the economic damage was accompanied by a painful sense that a whole way of life had disappeared within a single generation.

4 The environmental effects were equally severe. As the volume of water fell, its **salinity** rose sharply, eventually becoming too high for most native fish species to survive. The exposed seabed, now often called the Aralkum Desert, is covered in salt and the residue of agricultural chemicals that once washed into the sea. Strong winds lift this material into dust storms that can travel hundreds of kilometres, damaging crops and, according to many health studies, contributing to respiratory illness among local people.

5 Not everything has been lost, however. In 2005, Kazakhstan completed the Kok-Aral Dam, which separated the smaller northern part of the sea from the larger southern basin. Water levels in the North Aral rose noticeably within a few years, salinity fell, and commercially valuable fish returned, allowing a modest fishing industry to restart. The project showed that even after **catastrophic** damage, carefully targeted engineering can bring about real improvement, although the southern part of the sea has continued to shrink.

6 On the Uzbek side, attention has turned to the dried seabed itself. Large-scale planting of saxaul, a hardy desert shrub with deep roots, aims to hold the salty soil in place and **mitigate** the dust storms that blow across the region. International organisations have supported these efforts alongside projects to improve health care, drinking water and local employment. Few experts expect the southern sea to return, but stabilising the land around it is widely seen as an achievable goal.

7 The wider lesson concerns how Central Asia shares its water. The Amu Darya and Syr Darya are **transboundary** rivers, flowing through several countries whose needs do not always match: upstream states value water for hydropower in winter, while downstream farmers need it most in summer. Population growth and the gradual melting of mountain glaciers add further pressure. Efficient drip irrigation, repaired canals and closer regional cooperation are widely seen as the most realistic way to prevent the mistakes of the past from being repeated.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Salt and dust on the wind

B Once a giant among lakes

C One river system, many nations

D Tourism as the key to recovery

E Rivers redirected to the fields

F Holding the desert in place

G Ships stranded in the sand

H A dam that brought the fish back

LANGUAGE DEVELOPMENT - Describing environmental problems and solutions

VOCABULARY SKILL

Verb + noun collocations

Writing about the environment relies heavily on fixed verb + noun combinations — for example **pose a threat**, or **take measures**. Choosing the right verb for each noun makes your writing sound precise and academic, while the wrong choice (*make measures*, *do a threat*) immediately sounds unnatural to an expert reader.

pose	a threat, a risk, a challenge (negative)
deplete	resources, reserves, supplies (negative)
take	measures, steps, action (neutral)
raise	awareness, concerns, questions (mixed)
reverse	the damage, a decline, a trend (positive)
conserve	water, energy, habitats (positive)

Complete the sentences with the correct verb from the table above, in the correct form.

1. Decades of over-irrigation have seriously _____ the region's groundwater reserves.

2. Rising salt levels _____ a serious threat to the few fish species that remain in the lake.

3. The documentary helped to _____ awareness of the crisis among young people across the country.

4. Several governments have already _____ measures to reduce water losses from old, leaking canals.

5. Ecologists hope that replanting native trees will _____ the damage caused by decades of logging.

6. Switching to drip systems allows farmers to _____ water without reducing the size of their harvests.

Article 2

2 The sentences below are about restoring nature. Choose the correct meaning (a, b or c) for the word in bold.

1. The once-rich farmland had become so **degraded** that almost nothing would grow there.

a damaged and lower in quality
b protected by law
c recently ploughed

2. Beavers are often described as a **keystone** species because so many other animals depend on the habitats they create.

a extremely rare
b having a central role that others depend on
c dangerous to humans

3. The **reintroduction** of the lynx to the forest was planned carefully over several years.

a removal from an area
b the act of bringing back to a place where it used to live
c official count of the population

4. Removing a single predator can set off a **cascade** of changes throughout the food chain.

a a series of connected effects
b a sudden flood of water
c a carefully controlled experiment

5. Supporters argue that nature often recovers best with minimal human **intervention**.

a observation from a distance
b financial support
c involvement in order to change or control something

6. **Sceptics** question whether rewilding can ever work in densely populated countries.

a people who doubt that something is true or will work
b people who pay for a project
c people who study wild animals

7. The marsh supports remarkable **biodiversity**, from rare insects to migrating birds.

a the number of tourists visiting a place
b the variety of plant and animal life in an area
c the quality and richness of the soil

8. Forests with many different tree species tend to be more **resilient** when disease or drought strikes.

a easily damaged
b expensive to maintain
c able to recover quickly from difficulties

BRINGING NATURE BACK

1. For most of the twentieth century, conservation meant protecting what was left: drawing a line around a forest or wetland and keeping people out. Rewilding takes a more ambitious approach. Instead of simply preserving fragments of wild land, it aims to restore whole ecosystems that have been **degraded** by farming, logging or development, and then allow natural processes to take over. The idea has spread rapidly over the past three decades, from national parks in North America to former farmland in Europe, and it now attracts support from scientists, landowners and governments alike.

2. The best-known example comes from Yellowstone National Park in the United States, where grey wolves had been hunted to local extinction by the 1920s. Their **reintroduction** in 1995 is often credited with triggering a **cascade** of ecological change. With predators back, elk began to avoid certain valleys and riverbanks, allowing willow and aspen to recover in places. Researchers still debate how large these effects have really been, since drought, hunting and other factors also played a part, but the case transformed public thinking about how predators shape the landscapes around them.

3. In Britain, attention has focused on a smaller animal: the beaver, which disappeared from the country several centuries ago. Beavers are widely regarded as a **keystone** species, because the dams they build create ponds and wetlands that many other creatures depend on. Trial releases have shown that these dams can slow the flow of water after heavy rain, reducing flood peaks downstream, while also trapping sediment and pollutants. In 2022, wild beavers in England were given legal protection, a clear sign of how far official attitudes have shifted.

4. Rewilding is not only about reintroducing famous animals. At the Knepp Estate in southern England, the owners abandoned intensive farming around the turn of the century after years of financial losses. Rather than planting trees or managing habitats closely, they kept human **intervention** to a minimum, letting free-roaming cattle, pigs and deer graze the land as wild herbivores once would have done. Within two decades, the former fields had turned into a patchwork of scrub, grassland and young woodland, and rare birds such as nightingales and turtle doves had returned in significant numbers.

5. The approach has its critics, however. **Sceptics** point out that rewilding can conflict with food production at a time when the global population is still growing, and some farmers worry that returning predators will attack their livestock. In the Netherlands, a project at Oostvaardersplassen attracted strong public criticism when large numbers of grazing animals died during a harsh winter, raising difficult questions about how 'wild' a managed landscape should really be. Supporters respond that rewilding works best when local communities are involved from the start and share in its benefits.

6. Despite these debates, the case for restoring damaged land is growing stronger. Healthy ecosystems store carbon, filter water and protect soils, and areas rich in **biodiversity** tend to be more **resilient** when they face drought, disease or extreme weather. In a world where climate change is putting pressure on every landscape, rewilding offers something rare in environmental news: evidence that damage can, at least partly, be undone. It will never replace the need to cut emissions, but it suggests that nature, given enough space and time, is often capable of healing itself.

DISCUSSION

Work with a partner. Discuss the questions.

1 Would rewilding be possible in your country? Which animals or landscapes could realistically be restored?

2 Is it more important to restore damaged ecosystems or to protect the ones that are still healthy? Why?

GRAMMAR

Nominalisation and cause–effect structures

Academic writers often turn verbs and adjectives into nouns (nominalisation), which allows them to pack more information into a sentence and to focus on processes rather than on people. This noun-based style combines naturally with formal cause–effect expressions such as result in, stem from, owing to and as a consequence of.

The rules below explain how to form noun phrases from verbs and adjectives, and which cause–effect structures are followed by a noun, a clause or a new sentence.

1. **Nominalisation**: change a verb or adjective into a noun (*shrink → shrinkage, scarce → scarcity, reintroduce → reintroduction*) and use it as the subject or object of the sentence.

The sea shrank rapidly, which shocked scientists. → The rapid shrinkage of the sea shocked scientists.

2. Use **result in / lead to / give rise to** + noun when the CAUSE comes first and the EFFECT follows the verb.

Poor maintenance of the canals resulted in huge water losses.

3. Use **stem from / result from / arise from** + noun when the EFFECT comes first and the CAUSE follows the verb. Do not confuse *result in* (→ effect) with *result from* (← cause).

Many of the region's health problems stem from salt and dust storms.

4. **Owing to / due to / because of** are followed by a noun phrase, not a full clause. Before a clause, use *owing to the fact that*.

Owing to the rise in salinity, most native fish disappeared. / Owing to the fact that salinity rose, most native fish disappeared.

5. **As a consequence of / as a result of** + noun introduce a cause within the sentence; **As a consequence, / Consequently,** (+ comma) begin a new sentence that describes an effect.

As a consequence of the dam's construction, water levels rose. Fish returned to the lake. Consequently, the fishing industry restarted.

6. Nominalisation makes these structures possible: a clause with a verb becomes a noun phrase that can follow *result in*, *owing to*, etc.

Because wolves were reintroduced, the elk changed their behaviour. → The reintroduction of wolves led to a change in elk behaviour.

Exercise A — Complete the paragraph with the words and phrases in the box. Use each one once: *stem from, led to, Owing to, The disappearance, Consequently, result in, thanks to, reduction*.

Many of Central Asia's water problems (1) _____ decisions made more than sixty years ago. The rapid expansion of irrigation (2) _____ a dramatic fall in the volume of water reaching the Aral Sea. (3) _____ high evaporation rates, unlined canals lost a large share of their water before it arrived at the fields. (4) _____ of most native fish was a direct result of rising salinity. The fishing industry collapsed. (5) _____, thousands of people moved away in search of work. Dust storms from the dry seabed continue to (6) _____ health problems for local residents. Since 2005, the North Aral has partly recovered (7) _____ the Kok-Aral Dam. A further (8) _____ in water losses could be achieved by repairing old canals across the region.

Exercise B — Rewrite each sentence using a noun phrase (nominalisation) and the structure in brackets.

1. Rivers were diverted for cotton farming, so the Aral Sea shrank. (result in)

2. Many people in the region are ill because the air is polluted with salt and dust. (stem from)

3. Because rainfall decreased sharply, the harvest was poor. (owing to)

4. Wolves were reintroduced, and as a result the elk changed their behaviour. (as a consequence of)

5. Beavers build dams, which means that there is less flooding downstream. (lead to)

ACADEMIC WRITING

Writing about causes and effects — showing how events are connected

ACADEMIC WRITING SKILL

Writing about causes and effects

Cause-and-effect writing explains why something happens and what follows from it. A strong paragraph makes the direction of each relationship clear, avoids overstating certainty (*may contribute to* rather than simply *causes*), and often shows a chain in which one effect becomes the cause of the next.

CAUSE → EFFECT: *Large-scale irrigation reduced the flow of both rivers, which in turn led to the shrinking of the sea.*

EFFECT ← CAUSE: *The collapse of the fishing industry stemmed largely from the sharp rise in salinity.*

Identify the cause and the effect in each sentence. Then rewrite the sentence so that the effect comes first, using a different cause–effect expression.

1 Rising temperatures are causing mountain glaciers in the Pamirs to melt more quickly.

2 Overgrazing by sheep and goats has led to serious soil erosion on the hillsides.

3 Because many old canals leak, farmers downstream often receive too little water.

4 The return of beavers has created new wetlands for frogs, insects and birds.

5 Plastic waste dumped in rivers eventually results in the pollution of the sea.

6 Owing to a long period of drought, several villages have had to rely on deliveries of drinking water.

WRITING TASK

Write a short paragraph (80–100 words) explaining the causes and effects of one environmental problem in your city or region — for example air pollution, water shortages or waste. Use at least three cause–effect expressions from this unit, and show how one effect can lead to another.

READING PRACTICE

PART 2

Read the texts 7-14 and the statements A-J. Decide which text matches with the situation described in the statements.
Each statement can be used ONCE only. There are TWO extra statements which you do not need to use.
Mark your answers on the answer sheet.

7. You and your family want to go on a vacation to a place that has something for everyone.
8. You and your partner want to go on a romantic honeymoon.
9. You and your friends want to go on a trip to a place that has a vibrant life and you want to share your rooms.
10. You and your colleagues want to go on a place with many interesting activities and gatherings.
11. You and partner want to go on a luxury honeymoon. You want somewhere with a private place to get sunburned.
12. You and your family want to go on a vacation to a place that has a lot of background.
13. You and your friends want to go on a trip to a place that has many things to do and enjoy.
14. You and your friends want to go on a trip to enjoy yourself and get acquainted with local life and culture.

LIST OF RESORTS

A) ORGANIC RESORT
This resort is located in a national park and is known for its wildlife sightings. It offers guided safaris, bird-watching tours, and nature walks. The resort also has a restaurant that serves organic food. It offers private rooms and suites.

B) HOT DATE RESORT
This resort is located in a serene and picturesque setting, perfect for couples looking for a romantic escape. It offers private villas with stunning views, a spa, and a fine-dining restaurant.

C) SUNSHINE RESORT
This resort is located in a tropical destination and is known for its sunny weather. It offers private beaches and rooms for lonely and family travelers.

D) UNIQUE SOCIETY
This resort is located in a vibrant city and is known for its cultural experiences. It offers cooking classes, dance lessons, and language classes. The resort also has a restaurant that serves local cuisine. It offers shared rooms and private suites.

E) BUSY RESORT
This resort offers a range of team-building activities, including a ropes course, a zip line, and a rock-climbing wall. It also has a conference center and meeting rooms for business purposes. The resort offers private rooms and suites.

F) TIMELESS RESORT
This resort is located in a historic town and is known for its rich cultural heritage. It offers guided tours of the town's landmarks and museums. The resort also has a restaurant that serves traditional cuisine. It offers private rooms and suites.

G) PARTY CITY
This resort is located in the heart of a bustling city and is known for its lively nightlife. It has a rooftop bar, a nightclub, and a pool party every weekend. The resort offers shared rooms and private suites.

H) ESCAPEDE RESORT
This resort is located in a scenic location and is known for its outdoor activities. It offers hiking trails, mountain biking, and kayaking. The resort also has a zip line and a bungee jumping platform. It offers shared rooms and private suites.

I) QUALIFIED REST
This resort is known for its opulence and extravagance. It offers private villas with infinity pools, a private beach, and a Michelin-starred restaurant. The resort also has a spa and a fitness center.

J) FUN RESORT
This resort has a variety of activities for all ages, including a water park, mini-golf, and a game room. It also has a kids' club and babysitting services for parents who want some alone time. The resort offers spacious family rooms and suites.

✔ Wordlist

1. vacation ▢
2. honeymoon ▢
3. vibrant ▢
4. colleagues ▢
5. gatherings ▢
6. get acquainted with ▢
7. national park ▢
8. wildlife sightings ▢
9. guided safaris ▢
10. bird-watching ▢
11. organic food ▢
12. suites ▢
13. serene ▢
14. picturesque ▢
15. romantic escape ▢
16. stunning views ▢
17. tropical destination ▢
18. cultural experiences ▢
19. local cuisine ▢
20. team-building ▢
21. conference center ▢
22. historic town ▢
23. cultural heritage ▢
24. landmarks ▢
25. bustling ▢
26. lively nightlife ▢
27. scenic location ▢
28. opulence ▢
29. extravagance ▢
30. babysitting services ▢

LISTENING PRACTICE

PART 1–2

PART 1. You will hear some sentences. You will hear each sentence twice. Choose the correct reply to each sentence (A, B, or C).
Mark your answers on the answer sheet.

1.
A) I think so. It should be.
B) No, I think that's all that I need.
C) Sure, no problem.

2.
A) Sure, no problem.
B) When does it happen?
C) Why, we are in no rush!

3.
A) You should take a class.
B) Read the manual.
C) There's nothing we can do.

4.
A) Should we cross it too?
B) Why did he do that?
C) When did it happen?

5.
A) I'll try again next time.
B) Ok, I'll pick them up.
C) Of course, I fed them today.

6.
A) I live with my grandparents.
B) We went to party with my family yesterday.
C) Yes, We do. Always.

7.
A) That is her sister.
B) Where?
C) Marie is so sad today.

8.
A) Ok, I will
B) Yes, he remembers
C) No, it can't be

PART 2. You will hear someone giving a talk. For each question, fill in the missing information in the numbered space.
Write ONE WORD and / or A NUMBER for each answer.

LONDON TAXIS AND PRIVATE HIRE VEHICLES

Methods of payment: 9_____, or with credit and debit cards

Where to stop a cab: in the 10_____, or at cab ranks

Tariff 2: Monday – Friday: 8 pm – 10 pm; 11_____ & Sunday: 6 am – 10 pm

Tariff 3: every night: 10 pm – 6 am & on 12_____ holidays

Requirements: taxi drivers to complete a 13_____ examination

Notice: Private hire vehicles must always be 14_____ either at an operator's office, or by phone

Phrases & Idioms Wordlist

Carbon footprint: The total amount of greenhouse gases produced by a person, organisation or activity.

The tip of the iceberg: A small, visible part of a much larger problem that is mostly hidden.

A drop in the ocean: An amount so small compared with what is needed that it makes almost no difference.

Nip (something) in the bud: To stop a problem at an early stage, before it can grow or develop.

Can't see the wood for the trees: To be so focused on small details that you fail to understand the overall situation.

Weather the storm: To survive a difficult period without serious damage.

Muddy the waters: To make a situation more confusing or harder to understand.

Dead in the water: Having failed, or having no chance of succeeding or making progress.

Out of the woods: Free from danger or difficulty (often used in the negative: *not out of the woods yet*).

Reap what you sow: To experience the consequences, usually bad, of your own earlier actions.

Turn over a new leaf: To change your behaviour and start acting in a better, more responsible way.

Take a leaf out of someone's book: To copy something good that another person does.

The point of no return: The stage at which it is no longer possible to stop or reverse a process.

Bone dry: Completely dry, with no moisture at all.

Make hay while the sun shines: To take advantage of a good situation while it lasts.

USEFUL LANGUAGE — Opinion essay

Function	Useful phrases
Introducing the topic	*In recent years, there has been growing concern about… / It is often argued that…*
Stating your position	*I firmly believe that… / In my view, … / I would argue that…*
Explaining causes and effects	*This stems largely from… / As a consequence, … / This, in turn, results in…*
Giving examples	*A clear example of this is… / This can be seen in the case of…*
Conceding the other side	*Admittedly, … / While it is true that…, … / Some would counter that…*
Concluding	*In conclusion, … / On balance, I believe that… / To sum up, …*

THE TASK

You are writing an essay for an academic audience. Write 180–200 words.

Some people believe that environmental problems such as water shortages are too large for individuals to solve, and that only governments and international organisations can make a real difference. To what extent do you agree or disagree?

PARAGRAPH PLAN

Introduction	paraphrase the question + state your position (e.g. partly agree)
Body paragraph 1	why governments and organisations are essential — scale, laws, infrastructure + an example
Body paragraph 2	why individuals still matter — everyday habits, combined impact, public pressure
Conclusion	summarise both sides and restate your opinion in different words

RULES for Part 2 (essay)

☐ Answer the exact question — say clearly how far you agree, and keep this position consistent throughout.

☐ Aim for 180–200 words in four clear paragraphs: introduction, two body paragraphs, conclusion.

☐ Begin each body paragraph with a topic sentence, then support it with a reason and a specific example.

☐ Use a formal, impersonal register: no contractions, no slang, and avoid addressing the reader directly.

☐ Show a range of grammar and vocabulary — nominalisation, cause–effect expressions and topic collocations.

☐ Paraphrase the question in your introduction rather than copying it word for word.

SAMPLE ANSWER

In recent years, there has been growing concern about environmental problems such as water shortages and the loss of natural habitats. Some people argue that these issues are simply too large for ordinary citizens to tackle and that responsibility lies entirely with governments. While I accept that state action is essential, I would argue that individuals also have a meaningful role to play.

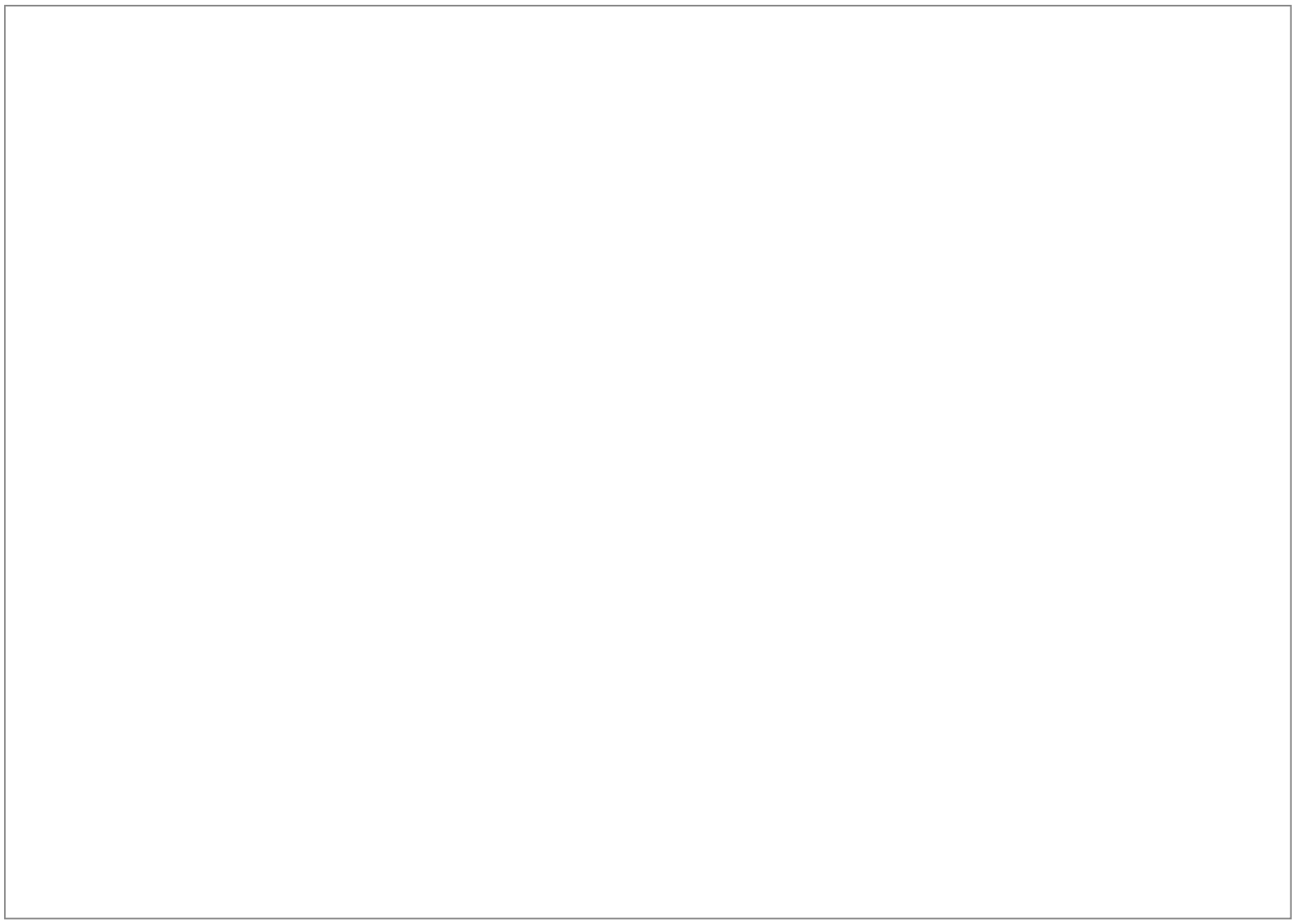
Admittedly, the sheer scale of many environmental crises means that only governments can address their root causes. The shrinking of the Aral Sea, for instance, stemmed from national decisions to divert entire rivers for irrigation, and only large-scale investment, such as modernising canals or building dams, can reverse damage of this kind. Similarly, laws on pollution and water use carry a weight that no single person can match.

Nevertheless, individual choices are far from insignificant. When millions of households waste less water or reduce their consumption, the combined effect can be considerable. Moreover, citizens who raise awareness and vote for greener policies put pressure on their leaders, which in turn results in stronger government action.

In conclusion, although governments and international bodies must lead the response to major environmental problems, individuals are by no means powerless. Real progress depends on both working together.

PART 2 PRACTICE — now you try

Rewilding projects, which return damaged land to nature, are becoming increasingly popular. However, some people argue that land should instead be used to grow food for a rising global population. Discuss both views and give your own opinion. Write 180–200 words.



SPEAKING PART 2

QUESTIONS

Describe a natural place you know that has changed over time.

You should say:

☐ where the place is and what it used to be like

☐ how it has changed

☐ what has caused these changes

...and say how you feel about these changes and whether anything should be done about them.

SAMPLE ANSWER

*I'd like to talk about a river near my grandparents' village in the Fergana Valley, which **holds a special place in my heart** because I spent almost every summer there as a child. Back then, it was wide and fast-flowing, and in the evenings the whole village seemed to gather along its banks to swim, fish or simply escape the heat.*

*Over the past ten years or so, though, it has **changed almost beyond recognition**. In late summer the water level drops so low that you can practically walk across it, and the fish my grandfather used to catch have become quite rare. Some of the trees along the bank have also been cut down to make room for new houses.*

*As far as I understand, the changes **stem mainly from** a combination of factors: more water is taken upstream for farming, the population has grown, and summers seem to be getting hotter and drier. **As a consequence**, there's simply less water to go round.*

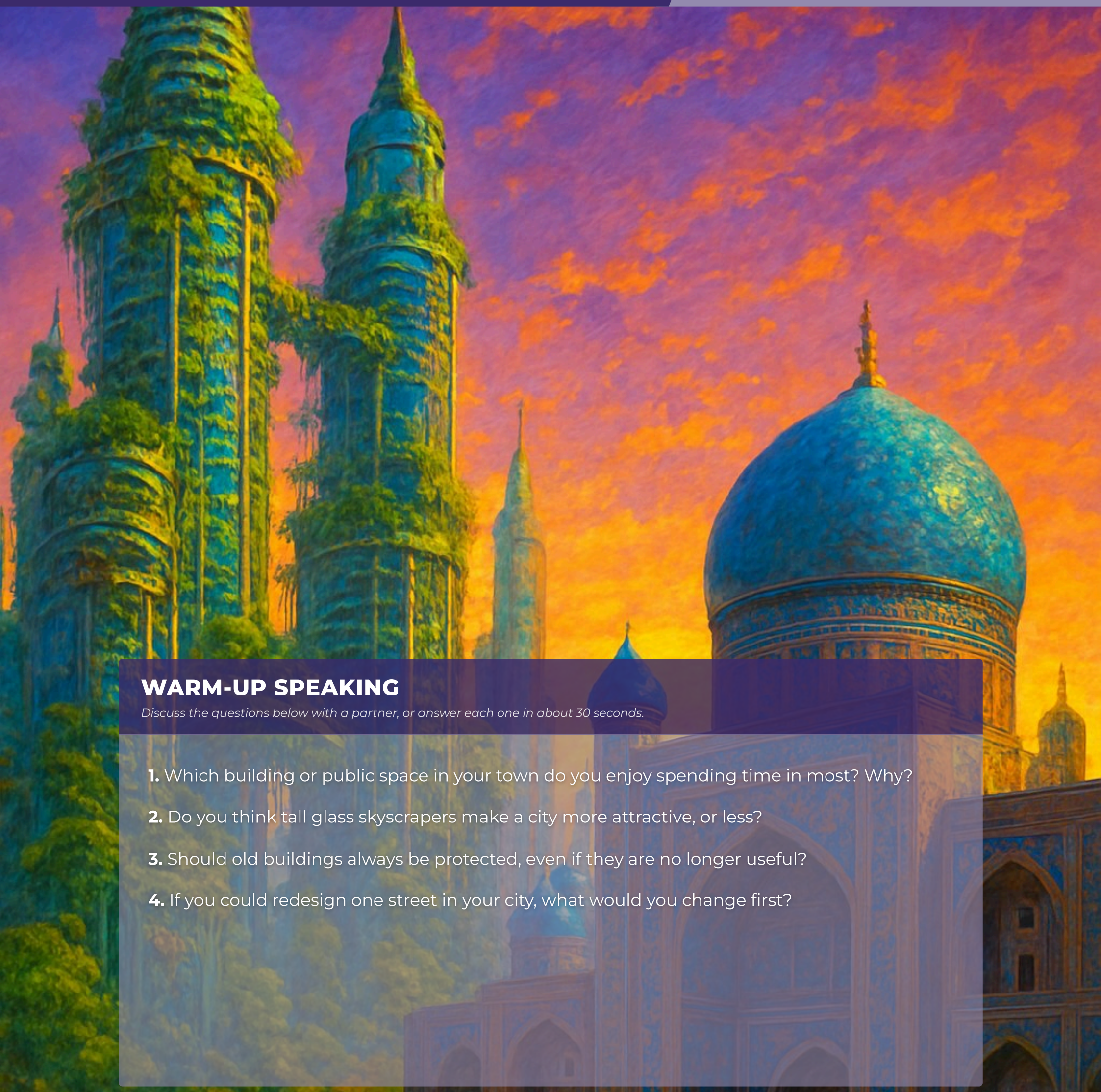
*To be honest, I find it quite sad, because the river used to be such a central part of village life. I do think something can be done, though — for example, repairing irrigation channels so less water is wasted, and planting trees along the banks again. **It might only be a drop in the ocean**, but if local people work together, I'm hopeful the river could recover at least partly.*

USEFUL LANGUAGE	
Introducing the topic:	*I'd like to talk about... / A place that has changed a lot over the years is...*
Describing the place:	*Back then, it was... / What I remember most clearly is... / It used to be...*
Describing change:	*It has changed beyond recognition... / Compared to when I was a child, it's now...*
Explaining causes:	*This stems mainly from... / The main reason seems to be... / As a consequence, ...*
Expressing feelings:	*To be honest, I find it quite sad/worrying... / It really bothers me that...*
Concluding / looking ahead:	*I do think something can be done... / Overall, I'm hopeful that...*

ARCHITECTURE

CITIES OF TOMORROW

UNIT 7



WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. Which building or public space in your town do you enjoy spending time in most? Why?

2. Do you think tall glass skyscrapers make a city more attractive, or less?

3. Should old buildings always be protected, even if they are no longer useful?

4. If you could redesign one street in your city, what would you change first?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Rapid urban **sprawl** has swallowed up much of the farmland on the edges of the growing city.

2 The new housing estate was criticised for lacking basic **amenities** such as shops, clinics and playgrounds.

3 Last year the council turned the main shopping street into a **pedestrian** zone, closing it to all traffic.

4 Rather than knock the office block down, the owners decided to **retrofit** it with solar panels and better insulation.

5 Poor **ventilation** in crowded classrooms can leave pupils tired and unable to concentrate.

6 Surrounded by low buildings and old plane trees, the small square has a surprisingly **intimate** atmosphere.

7 A growing body of research links regular access to green space with improved mental **well-being**.

8 During rush hour, the city's **congested** roads can turn a ten-minute journey into an hour-long ordeal.

a (adj) so full of traffic or people that movement is difficult

b (v) to add new equipment or features to a building that was not originally built with them

c (n) the uncontrolled spread of a city into the countryside around it

d (n) the state of feeling healthy, comfortable and happy

e (adj) for or used by people walking, not vehicles

f (n) the movement of fresh air into and through a room or building

g (n) useful facilities, such as shops or parks, that make a place pleasant or convenient to live in

h (adj) small, quiet and comfortable in a way that feels personal and welcoming

Complete the sentences with the correct form of a word from either article.

1. Low buildings and narrow lanes give the old quarter a sense of _____ that many modern districts lack. (intimacy)

2. Tourists increasingly look for _____ experiences rather than polished reconstructions of the past. (authentic)

3. Improving the energy _____ of older homes is one of the cheapest ways for a city to cut its emissions. (efficiency)

4. The plan to _____ the old cinema was abandoned after a long public campaign to save it. (demolish)

5. The bazaar street was _____ in 2019 and is now closed to cars, buses and delivery vans during the day. (pedestrianised)

WHAT MAKES A PLACE LIVEABLE?

1 Ask people to describe a place they genuinely enjoy spending time in, and very few will mention a record-breaking skyscraper or a vast glass shopping mall. Far more often, they describe a shaded square, a lively street market or a quiet courtyard where neighbours stop to talk. Architects and planners call this quality 'liveability': the extent to which a building or public space supports the everyday lives of the people who use it. It is rarely a matter of size or spectacle. What makes a place liveable is usually the sum of many small, carefully considered details.

2 One of the most influential ideas in modern urban design is 'human scale' — the principle that buildings and streets should be measured against the human body rather than the motor car. Someone walking at five kilometres an hour notices doorways, windows, textures and trees; a driver at sixty notices almost nothing. Streets lined with buildings of three to six storeys, with shops and cafés at ground level, tend to feel **intimate** and safe, whereas a plaza surrounded by blank forty-storey towers can feel exposed and empty, however impressive it looks on an architect's model.

3 For much of the twentieth century, cities were redesigned around traffic. Wide roads were cut through old neighbourhoods, and streets became places to pass through rather than places to be. Today, many cities are reversing that trend. Turning **congested** shopping streets into **pedestrian** zones was once dismissed as bad for business, yet study after study has found that people on foot linger longer and, over time, spend more than drivers who park, shop and leave. Cleaner air and quieter streets, it turns out, are good for trade as well as for health.

4 Liveability also depends on what happens inside buildings. So-called green buildings aim to use as little energy as possible while keeping the people inside them comfortable. Many achieve this by borrowing ideas that traditional builders understood centuries ago: thick walls that store heat during the day, courtyards that draw cool air through rooms, and windows positioned for natural **ventilation** instead of relying entirely on air conditioning. Not only do such features cut energy bills, but they also create healthier indoor environments, since stale air and poor light are closely linked to tiredness and headaches.

5 Perhaps surprisingly, many architects now argue that the greenest building is the one that already exists. Constructing a new building releases enormous amounts of carbon, from producing cement and steel to transporting materials to the site. Choosing to **retrofit** an existing structure — adding insulation, solar panels, efficient heating and better windows — often achieves similar energy savings at a fraction of the environmental cost. As a result, demolition, once seen as the obvious first step in any redevelopment project, is increasingly treated as a last resort.

6 Liveability is equally a question of distance. Low-density urban **sprawl**, with housing estates miles away from shops and workplaces, forces residents to drive for almost every errand and leaves those without a car isolated. In response, planners in cities from Paris to Melbourne have promoted the idea of the 'fifteen-minute city', in which schools, clinics, parks and other everyday **amenities** are within a short walk or bike ride of home. The aim is not to stop people travelling further, but to make sure they no longer have to.

7 Finally, there is the question of nature. Research consistently shows that access to trees, parks and water improves physical health and mental **well-being**, lowers stress and even reduces summer temperatures in densely built areas. Yet the most liveable places are rarely designed by experts alone. It is often residents, through years of small adjustments — a bench added here, a vegetable garden there — who turn a well-planned space into a genuinely loved one. A building can win every award going and still fail if nobody wants to spend time there.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Everything within a short walk

B Buildings that breathe

C Not the tallest, but the most welcoming

D Why skyscrapers are disappearing from city skylines

E Putting people before cars

F Nature as a public health measure

G Design measured by the human body

H The greenest building is the one already standing

LANGUAGE DEVELOPMENT - Describing urban change

VOCABULARY SKILL

Verb + noun collocations

When we write about how cities change, certain verbs combine naturally with a small group of nouns — for example **preserve the character** of an area, or **tackle congestion**. Learning these as fixed chunks, rather than word by word, helps you describe planning and architecture precisely and sound more natural in academic writing.

restore	a building, a monument, a facade (to its original condition)
preserve	heritage, the character, the atmosphere (of a place)
demolish	a building, a tower block, a structure
revitalise	a neighbourhood, a city centre, a run-down area
tackle	congestion, pollution, urban sprawl
enhance	well-being, quality of life, the appeal (of an area)

Complete the sentences with the correct verb from the table above. Use the correct form.

1. After years of decline, a new market and cultural centre helped to _____ the run-down neighbourhood.

2. Strict planning rules were introduced to _____ the historic character of the old town.

3. The city hopes that the new metro line will _____ congestion on its busiest roads.

4. Craftsmen spent three years carefully _____ the tiled dome of the madrasa.

5. Planting more trees is one of the cheapest ways to _____ residents' quality of life.

6. The council decided to _____ the unsafe tower block rather than spend millions repairing it.

Article 2

2 The sentences below are about historic cities. Choose the correct meaning (a, b or c) for the word in bold.

1. Many families moved out of the **dilapidated** apartment block after its roof began to leak.

a modern and expensive
b old and in very poor condition
c small and overcrowded

2. Visitors value the **authenticity** of the old quarter, where craftsmen still work exactly as their grandparents did.

a the quality of being real and original
b the high cost of maintaining something
c popularity with tourists

3. The **demolition** of the old bathhouse angered residents who had campaigned for years to save it.

a careful repair
b detailed study
c the act of knocking down a building

4. Only the original stone **facade** was kept; everything behind it was completely rebuilt.

a the front wall of a building
b the roof of a building
c the foundations of a building

5. Rising rents caused by **gentrification** forced many long-standing shopkeepers to close.

a government control of land and housing
b the slow decline of a poor neighbourhood
c wealthier people moving into an area, pushing out poorer residents

6. New hotels have begun to **encroach** on the protected area around the ancient mosque.

a protect and support
b gradually move into and take over
c decorate and improve

7. Local residents see themselves as the **custodians** of a history that belongs to everyone.

a occasional visitors
b people responsible for looking after something
c harsh critics

8. Critics complained that the restored street looked **sterile** — more like a museum than a living neighbourhood.

a lacking life and character
b clean and healthy
c noisy and crowded

OLD WALLS, NEW CITIES

1. Walk through the old centre of almost any Central Asian city and you are walking through layers of history: mosques and madrasas, covered bazaars, bathhouses and caravanserais that once served merchants travelling the Silk Road. Cities such as Samarkand and Bukhara attract visitors from all over the world precisely because so much of this past is still standing. Yet these same cities are home to growing populations who need housing, roads, schools and jobs. How to balance the protection of the past with the demands of the present is one of the most difficult questions facing planners today.

2. Supporters of new construction point out that not every old building is worth saving. Many historic quarters contain **dilapidated** houses with crumbling walls, poor sanitation and no proper heating, and the families living in them often want modern homes rather than romantic surroundings. Repairing such buildings can cost far more than replacing them, and in some cases **demolition** is the only safe option. A city, they argue, is not a museum; it must be allowed to change if it is to remain a good place in which to live and work.

3. Conservationists respond that once a historic building is lost, it is lost for ever. They also warn against restoration that goes too far. When ancient structures are rebuilt with new materials and bright, perfectly uniform tiles, they may look impressive in photographs, but they lose much of their **authenticity**. Some developers keep only the historic **facade** of a building and replace everything behind it with offices or hotel rooms — a compromise that critics compare to keeping the cover of a book while throwing away all of its pages.

4. A further concern is what happens to the people who actually live in historic areas. Successful restoration often attracts tourists, boutique hotels and expensive restaurants, and with them comes **gentrification**: property prices rise, long-standing residents sell up or are priced out, and local workshops and corner shops are replaced by souvenir stalls. The result can be an old town that is beautifully preserved but strangely **sterile**, crowded with visitors during the day and almost empty of residents at night, its streets quiet once the tour groups have gone.

5. International bodies such as UNESCO try to manage these pressures by protecting not only individual monuments but also the 'buffer zones' around them. Within these zones, the height and style of new buildings are strictly controlled, so that modern development does not **encroach** on the historic skyline. Many experts believe the most effective approach is to concentrate large new projects — business districts, apartment towers, transport hubs — outside the historic core, where they can grow freely without overshadowing the monuments that give the city its identity.

6. In the end, preservation and development need not be opposites. Some of the most successful projects give old buildings new purposes: a caravanserai converted into a crafts centre, a former bathhouse reopened as a gallery, a merchant's house turned into a small guesthouse run by the family that owns it. When local communities are treated as the true **custodians** of their heritage, rather than as obstacles to tourism, historic centres can remain what they were always meant to be — living places, not open-air museums.

DISCUSSION

Work with a partner. Discuss the questions.

1 Think of an old building in your town or city. Should it be restored, given a new purpose, or replaced? Why?

2 Who should decide what happens to a historic city centre — residents, architects, the government or tourists? Why?

GRAMMAR

Cleft sentences & fronting for emphasis

Writers often rearrange a sentence to draw the reader's attention to its most important idea. Cleft sentences split one clause into two parts so that the key information is placed in focus, while fronting moves a word or phrase to the beginning of a sentence, sometimes with inversion of the subject and auxiliary.

The rules below show the most common ways to add emphasis in formal and academic writing.

1. **What-clefts:** What + clause + **is/was** + the focus. Use them to highlight an action, a thing or an idea.

What makes a square feel welcoming is the sense of enclosure created by low buildings.

2. **It-clefts: It is/was** + the focus + **that/who** + the rest of the clause. Use them to emphasise one particular person, thing, time or reason, often in contrast to another.

It was the residents, not the planners, who saved the old bathhouse from demolition.

3. **All / The thing / The reason (why) / The place (where)** can replace *What* to give a slightly different emphasis.

All the old quarter really needs is careful, regular maintenance.

4. **Negative and limiting adverbials** (Not only, Never, Rarely, Seldom, Little) can be fronted for emphasis. The subject and auxiliary are then inverted.

Not only do green roofs reduce energy bills, but they also help to prevent flooding.

5. **Only + time expression** (Only after / Only when / Only then) is followed by inversion in the main clause, not in the time clause.

Only when the facade was cleaned did its original colours become visible.

6. **Fronted adjective or prepositional phrase + be/verb + subject** is used in descriptive writing to emphasise a feature.

Particularly striking is the blue-tiled dome that rises above the market.

Exercise A — Complete the paragraph with ONE word in each gap.

(1) _____ most visitors notice first in the old city is the colour of the domes. However, it is not the size of the monuments (2) _____ impresses them most, but the narrow, shaded streets around them. Not only (3) _____ these streets offer protection from the sun, but they (4) _____ channel cool breezes between the houses. Rarely (5) _____ modern planners designed a district that works so well in the summer heat. Only after the restoration work had begun (6) _____ engineers realise how cleverly the original walls had been built. (7) _____ the residents really want now is more trees and fewer cars. The reason (8) _____ the area feels so intimate is that no building is taller than the trees around it.

Exercise B — Rewrite each sentence for emphasis, beginning with the words in brackets.

1. The narrow streets give the old city its unique character. (What)

2. Local volunteers rescued the abandoned caravanserai. (It was)

3. The new park reduces traffic noise, and it also attracts wildlife. (Not only)

4. We understood the value of the old bazaar only after it had been demolished. (Only after)

5. Developers have seldom shown such respect for a historic site. (Seldom)

6. I love the way the afternoon light falls across the courtyard. (What)

ACADEMIC WRITING

Comparing and contrasting — showing similarities and differences

ACADEMIC WRITING SKILL

Comparing and contrasting

When you compare and contrast, you show how two places, ideas or approaches are similar or different. Use expressions such as *similarly*, *likewise* and *in much the same way* to show similarity, and *whereas*, *while*, *by contrast* and *conversely* to show difference. Note that *whereas* and *while* join two clauses in one sentence, while *by contrast*, *conversely* and *similarly* usually begin a new sentence and are followed by a comma.

COMPARING: *Trees cool the streets beneath them.* ***In much the same way****, green roofs lower the temperature of the buildings they cover.*

CONTRASTING: ***Whereas*** *many new districts are designed around the car, historic centres were built for people on foot.*

Join each pair of ideas into one or two sentences, using the expression in brackets to compare or contrast them.

1 Old city centres were built for walking. / Modern suburbs are designed around the car. (whereas)

2 Restoration preserves a building's history. / New construction allows modern standards of comfort and safety. (conversely)

3 Parks improve residents' mental health. / Tree-lined streets reduce stress. (in much the same way)

4 Glass towers are expensive to cool in summer. / Traditional mud-brick houses stay cool naturally. (by contrast)

5 Samarkand depends heavily on tourism to its historic monuments. / Bukhara relies on its heritage to attract visitors. (similarly)

6 Some planners favour high-rise housing to save land. / Others argue that low-rise districts are more liveable. (while)

WRITING TASK

Write a paragraph (100–120 words) comparing and contrasting a historic district and a modern district in a city you know well. Use at least four different expressions for comparing and contrasting.

READING PRACTICE

PART 3

Read the text and choose the correct heading for each paragraph from the list of headings below. There are more headings than paragraphs, so you will not use all of them. You cannot use any heading more than once.

List of Headings:

A) An ancient place

B) Holy water

C) Famous industry

D) Ajanta in the modern day

E) Resisting the heat

F) Things about Mumbai

G) City of contrasts

H) Modern languages' ancestor

15. Paragraph I
16. Paragraph II
17. Paragraph III
18. Paragraph IV
19. Paragraph V
20. Paragraph VI

INDIA

I. The world's first university was established in India in 700 BC, where more than 10,500 students from around the world studied more than 60 different subjects. The training was conducted in Sanskrit. Nowadays, Sanskrit is considered to be the mother of all the European languages. Its elements can be traced in most of them.

II. The most frequently visited place in India is Delhi, the capital city of the country. It seems to be divided into two parts. New Delhi is a modern city with hi-tech architecture, tourist attractions and modern facilities. Old Delhi is several centuries old, with narrow streets, ancient churches and noisy bazaars. Lots of people live in slums without the most essential facilities like toilets and running water.

III. People all over the world know of India's greatest river, the Ganges. The river looks extremely beautiful in the morning light. It is the subject of thousands of prayers. It has become a common ritual for pilgrims to gather by the holy river in the mornings. It is believed that bathing in the Ganges helps to cure all kinds of diseases and can improve your life in general.

IV. Indian climate differs from one part of the country to another. The air is cool and fresh in the mountains, but in the plains it is usually very hot and dry, so even a short trip can be very hard for an inexperienced traveller. That is why tourists are always glad to have a rest at one of the numerous stations in the hills. There they can enjoy a nice cup of real Indian tea, the most refreshing drink in such a climate.

V. India is not all history. It is also famous for its Bollywood industry, which produces different films. Indian films have always had their own style; they are rich in music and dancing. Though the industry is still developing, many Indian film stars have received Oscar prizes and are known throughout the world. The home of Bollywood is Mumbai, also known as a busy port and the country's commercial centre.

VI. There is an unusual treasure hidden in the Indian mountains. These are striking ancient temples cut in the rocks. For example, the Buddhist cave temples at Ajanta were built at least 2,000 years ago. Inside the caves on the walls, tourists can see animal images and pictures of people's lives. Some researchers believe that the interpretation of the images can help us to learn a lot about people's lives in ancient times.

✔ Wordlist

1. establish ▢
2. conduct (training) ▢
3. be considered to be ▢
4. be traced in ▢
5. frequently visited ▢
6. hi-tech architecture ▢
7. tourist attraction ▢
8. modern facilities ▢
9. bazaar ▢
10. slum ▢
11. essential facilities ▢
12. running water ▢
13. be the subject of ▢
14. prayer ▢
15. ritual ▢
16. pilgrim ▢
17. holy ▢
18. cure (a disease) ▢
19. differ from ... to ... ▢
20. plains ▢
21. inexperienced ▢
22. numerous ▢
23. refreshing ▢
24. be rich in ▢
25. throughout the world ▢
26. commercial centre ▢
27. treasure ▢
28. striking ▢
29. cut in the rocks ▢
30. interpretation ▢

PART 3

You will hear people speaking in different situations. Match each speaker (15-18) to the description they give about another person (A-F). There are TWO EXTRA options which you do not need to use. Mark your answers on the answer sheet.

15. Speaker 1 ...
16. Speaker 2 ...
17. Speaker 3 ...
18. Speaker 4 ...

A) A programmer.
B) A supermarket assistant.
C) A nurse.
D) A fast-food cook.
E) A business manager.
F) A restaurant manager.

Phrases & Idioms Wordlist

Lay the foundations (for something): To create the basic conditions from which something can develop and succeed.

Concrete jungle: A city area full of large, unattractive buildings, with little greenery or open space.

Bricks and mortar: Buildings, especially seen as property or a physical shop rather than an online business.

Hit the roof: To become suddenly very angry.

Go through the roof: (of prices or numbers) To rise very quickly to a very high level.

Under one roof: All together in the same building or organisation.

Built to last: Made so strongly or well that it will survive for a long time.

Hit a brick wall: To reach a point where you cannot make any further progress.

The writing is on the wall: There are clear signs that something is going to fail or come to an end.

Rome wasn't built in a day: Important or ambitious projects take a long time to complete.

A stone's throw (from): A very short distance away; very close to.

Set in stone: Fixed and impossible to change.

The cornerstone (of something): The most important part, on which everything else is based.

Off the beaten track: Far away from the places that most people visit; quiet and little-known.

Paint the town red: To go out in the city and celebrate in a lively, noisy way.

WRITING PART 1.2

USEFUL LANGUAGE — Formal letter

Function	Useful phrases
Opening	*Dear Sir/Madam, / Dear Chairperson, / I am writing in response to your recent email regarding…*
Reacting / feelings	*I was deeply concerned to learn that… / I must admit I was delighted/disappointed to hear that…*
Comparing options	*Whereas a car park would only…, a restored square would… / Conversely, … / In much the same way, …*
Giving a suggestion	*I would strongly suggest that… / What the area really needs is… / It might be worth considering…*
Requesting a reply / Closing	*I would be grateful if you could pass on… / I look forward to hearing… / Yours faithfully, / Yours sincerely,*

PART 1.1 PRACTICE

You are a member of your neighbourhood's Residents' Association. You received this email from the association's chairperson.

Dear Member,

I am writing to inform you that the city council has announced plans to redevelop the old market square in our neighbourhood. According to the proposal, several of the historic trading houses around the square would be demolished to make room for a multi-storey car park and a modern shopping centre. The council has agreed to consider residents' views before making a final decision next month. We would therefore like to hear how you feel about the plans and what you think should happen to the square instead. Please send us your thoughts as soon as possible.

Best wishes,

The Chairperson

Task 1.1 — Write a letter to your friend, who also lives in the neighbourhood. Write about your feelings about the plans and what you think should happen to the square. Write about 50 words.

Task 1.2 — Write a letter to the Chairperson. Write about your feelings about the plans and what you think should happen to the square instead. Write about 120–150 words.

RULES for Task 1.2 (formal letter)

☐ Respond to the SAME message as Task 1.1 — cover the same two things (your feelings + your suggestion), but in more depth.

☐ Aim for 120–150 words — several short paragraphs, not one block of text.

☐ Use a formal register throughout: no contractions, no slang, polite and impersonal phrasing.

☐ Structure it clearly: an opening line, your reaction, your suggestion(s) with reasons, and a polite closing.

☐ Open with Dear Sir/Madam or Dear [Title], and close with Yours faithfully/sincerely.

☐ Show a wider range of grammar than in Task 1.1 — for example a cleft sentence (What the square needs is…), fronting (Not only would this…) and comparing/contrasting linkers (whereas, conversely).

SAMPLE ANSWER

Dear Chairperson,

Thank you for informing members about the council's plans to redevelop the old market square. I must admit that I was deeply disappointed to learn that several of the historic trading houses may be demolished, as they give our neighbourhood much of its character.

While I understand that parking is a genuine problem in the area, I do not believe that a multi-storey car park is the right solution. What the square really needs is careful restoration. I would suggest that the trading houses be repaired and converted into small shops, cafés and workshops for local craftspeople. Not only would this preserve our heritage, but it would also attract visitors and support local businesses.

Furthermore, it might be worth considering a car park on the edge of the district, connected to the square by a pleasant pedestrian street with trees and benches.

I would be grateful if you could pass these suggestions on to the council. I look forward to hearing about the outcome.

Yours faithfully,

D. Rakhimova

TASK 1.2 PRACTICE — now you try, with a new message

You live on a new housing estate on the edge of your city. You received this email from the city's Urban Planning Department.

Dear Resident,

We are pleased to inform you that an area of unused land next to your estate has been set aside for a new public space. Before construction begins in the spring, we would like residents to help us decide how the space should be designed — for example, as a park, a sports area, a children's playground or a community garden. Please let us know your views and any facilities you would like to see included.

Kind regards,

Urban Planning Department

Task 1.2 — Write a letter to the Urban Planning Department. Write about your feelings about the news and what you think the new space should include. Write about 120–150 words.

SPEAKING PART 3

STATEMENT

Historic buildings should be preserved at all costs, even if this slows down a city's development.

Discuss both sides of the statement and give your own opinion. Use the points below to help you.

FOR	AGAINST
Once a historic building is destroyed, it can never be replaced.	Many old buildings are unsafe, unhealthy or too expensive to repair.
Heritage sites attract tourists and bring income to local people.	Growing populations urgently need modern housing, roads and services.
Old buildings give a city its identity and a sense of continuity.	Over-protection can turn a living city into a sterile museum.

SAMPLE ANSWER

*I'd like to begin by saying that I **partly agree** with this statement. **On the one hand**, historic buildings are irreplaceable. **What makes** cities like Samarkand and Bukhara so special **is** that their past is still visible in their streets, and once a madrasa or a caravanserai has been knocked down, no amount of money can bring it back. **On top of that**, heritage is a major source of income, because tourists travel thousands of kilometres to see these places.*

***On the other hand**, I think the phrase 'at all costs' goes too far. **Not every old building is worth saving** — some are genuinely dilapidated and unsafe, and the families living in them deserve proper heating, clean water and modern facilities. **Whereas** a monument of world importance should obviously be protected, an ordinary run-down block from the last century is a very different case. **Besides**, a city that is never allowed to change risks becoming sterile — a beautiful museum where nobody actually lives.*

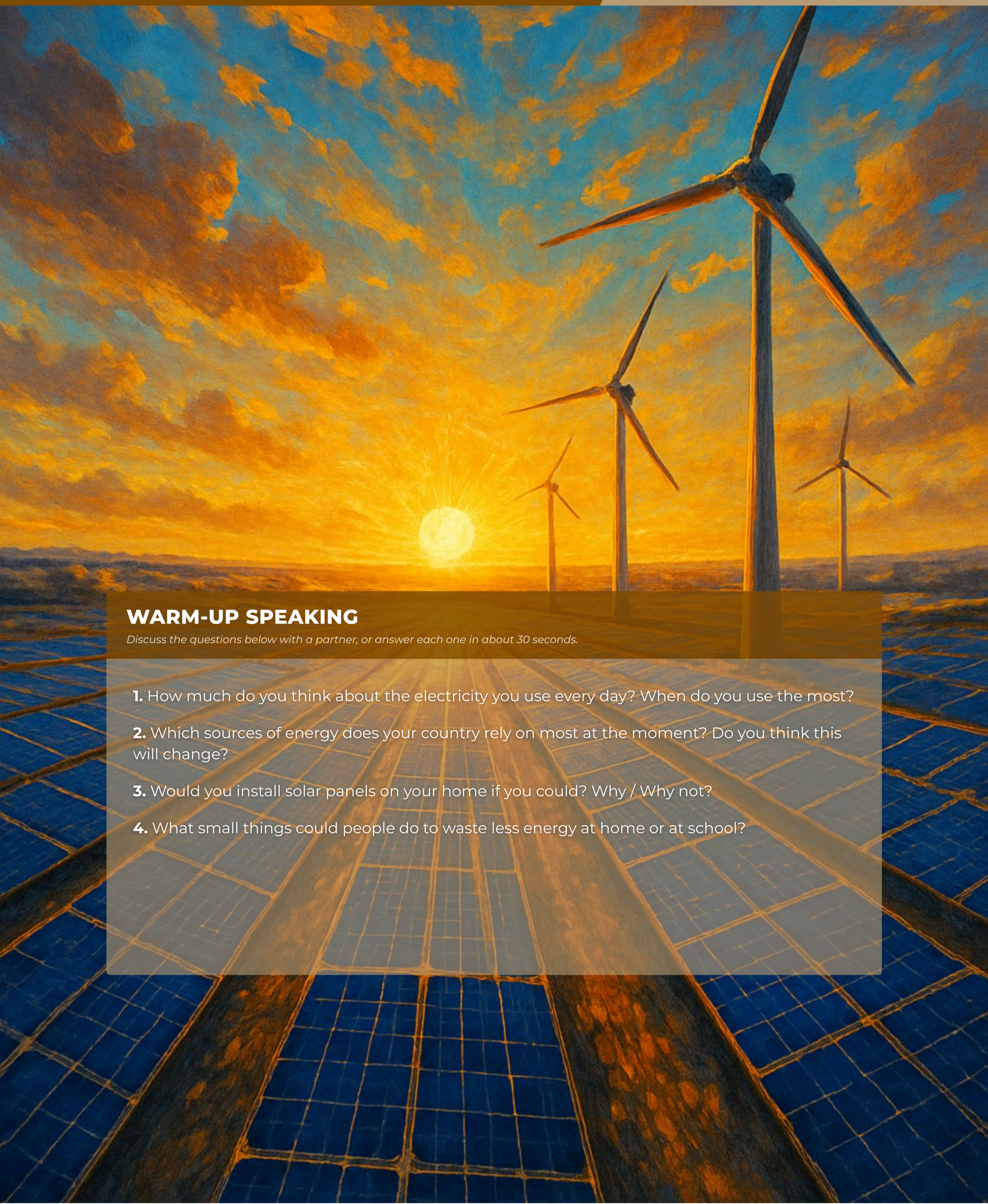
***On balance**, I believe the best solution is a compromise. **It is** the most important monuments **that** should be protected strictly, while large new projects are built outside the historic core. Old buildings can also be given new purposes, such as galleries or guesthouses, so that they stay alive. **All things considered**, development and preservation can go hand in hand if they are planned carefully.*

USEFUL LANGUAGE	
Giving your overall view:	*I partly/strongly agree that… / I'm not entirely convinced that…*
Presenting one side:	*On the one hand, … / What makes … so special is… / One strong argument in favour is…*
Presenting the other side:	*On the other hand, … / Having said that, … / It could be argued that…*
Comparing and contrasting:	*Whereas…, … / By contrast, … / In much the same way, …*
Adding a point:	*On top of that, … / Besides, … / Not only…, but also…*
Concluding:	*On balance, … / All things considered, … / The best solution, in my view, is…*

ENERGY

POWERING THE FUTURE

UNIT 8



WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. How much do you think about the electricity you use every day? When do you use the most?

2. Which sources of energy does your country rely on most at the moment? Do you think this will change?

3. Would you install solar panels on your home if you could? Why / Why not?

4. What small things could people do to waste less energy at home or at school?

UNDERSTANDING
KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Moving from paper files to a fully digital system was a difficult **transition** for many older employees.

2 Farmers in the valley have learnt to **harness** the fast-flowing river to irrigate their fields.

3 The village has only **intermittent** internet access, so the connection often drops for hours at a time.

4 The new stadium has a seating **capacity** of 40,000, making it the largest in the region.

5 Investors were unconvinced that the plan was commercially **viable** and refused to fund it.

6 Without government **subsidies**, many small farms would be unable to compete with cheap imports.

7 Hundreds of electric buses have been **deployed** across the capital over the past two years.

8 The airline plans to **phase out** its oldest aircraft and replace them with more efficient models.

a (v) to bring something into use or move it into position so that it can be used effectively

b (n) the process of changing from one state, system or situation to another

c (adj) able to work successfully; practical and likely to succeed

d (phr v) to stop using or providing something gradually, in stages

e (n) money paid by a government to help reduce the cost of something

f (adj) stopping and starting repeatedly; not continuous or regular

g (n) the maximum amount that something can contain or produce

h (v) to control a natural force or resource and make use of it

Complete the sentences with the correct form of a word from either article.

1. Several banks questioned the long-term _____ of the project and refused to lend the company any money. (viability)

2. The _____ of solar power means that it must be combined with other sources or with storage. (intermittency)

3. Families who _____ their lofts and walls properly can cut their heating bills by up to a third. (insulate)

4. Sharp _____ in the price of gas made it almost impossible for small businesses to plan ahead. (fluctuations)

5. In the early years, rooftop solar panels were heavily _____ by the state to encourage people to buy them. (subsidised)

THE AGE OF SUN AND WIND

1 For most of the twentieth century, generating electricity meant burning something — coal, oil or gas — and very few people imagined that sunlight and moving air could ever seriously compete. Yet over the past fifteen years, the world has entered a genuine energy **transition**, and it has happened far faster than even the most optimistic forecasts predicted. In several countries, solar and wind now provide more power than coal on a typical sunny or breezy day, and new records are broken every year, often with surprisingly little public attention.

2 The single biggest reason is cost. The price of solar panels has fallen by roughly ninety per cent since 2010, while wind turbines have become larger, more efficient and far cheaper to run. For years, renewable projects depended on generous government **subsidies** to attract investors, and critics argued that they would never survive without them. Today, in many markets, a new solar farm is the cheapest way to produce electricity, which means that clean energy has become commercially **viable** on its own terms rather than simply a political choice.

3 Nowhere is the potential clearer than in sunny, wide-open regions. Countries across Central Asia, the Middle East and North Africa receive intense sunshine for most of the year and have large areas of flat, sparsely populated land. Governments there have begun to **harness** this natural advantage with vast solar parks covering thousands of hectares, often built in partnership with foreign investors. Some projects are designed not only to supply domestic cities but, in time, to export electricity to neighbouring states, turning land once dismissed as empty into a valuable economic asset.

4 However, sun and wind share a fundamental weakness: they are **intermittent**. Panels produce nothing at night, output drops on cloudy days, and even the windiest coastline experiences calm spells that can last for a week. Demand for electricity, by contrast, does not stop when the weather changes; hospitals, factories and homes need a reliable supply around the clock. Grid operators must therefore keep backup power available, and matching an unpredictable supply with constant demand is arguably the central problem of the renewable age.

5 Wind power, meanwhile, is moving out to sea. Offshore turbines benefit from stronger and steadier winds than those on land, and because they are far from homes, they can be built on an enormous scale. A single modern turbine can now stand taller than many skyscrapers, with blades longer than a football pitch, and its generating **capacity** is many times that of the machines installed twenty years ago. A handful of these giants can power an entire town, although building them in deep, stormy waters remains expensive and technically demanding.

6 Producing clean electricity is only half the task; it also has to reach the people who use it. The best solar and wind sites are frequently hundreds of kilometres from major cities, so new high-voltage power lines must be **deployed** across mountains, farmland and national borders. Such projects can take a decade to approve, and they sometimes face fierce resistance from residents who object to their visual impact, or from farmers who are reluctant to give up productive land for endless rows of panels.

7 Despite these obstacles, many governments have now committed to **phase out** coal entirely within the next two decades, and some have set dates to end the sale of new petrol and diesel cars. The shift brings real social challenges, particularly for communities whose jobs have depended on mining for generations. Yet it is also creating entirely new industries, from panel installation to turbine maintenance. The question is no longer whether the world will move to renewable energy, but how quickly — and how fairly — that move can be managed.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Bigger turbines further out at sea

B A turning point for global energy

C The problem of unreliable weather

D The hidden dangers of nuclear power

E Leaving fossil fuels behind

F Turning empty land into power stations

G When clean power became the cheaper option

H Cables, land and local resistance

LANGUAGE DEVELOPMENT - Talking about energy production and use

VOCABULARY SKILL

Verb + noun collocations

Texts about energy rely on a small set of verbs that combine with particular nouns — for example **generate electricity**, **meet demand** or **cut emissions**. Learning these as fixed chunks, rather than choosing verbs word by word, will make your writing sound more precise and more academic.

generate	electricity, power, heat, income
meet	demand, targets, needs, a deadline
cut / reduce	emissions, bills, consumption, waste
harness	the power of the sun / wind / water, energy
store	energy, electricity, surplus power, heat
phase out	coal, fossil fuels, petrol cars, old equipment

Complete the sentences with the correct verb from the table above. Use the correct form.

1. Once it is finished, the new offshore wind farm will _____ enough electricity for around 400,000 homes.

2. Without more batteries, the grid may struggle to _____ demand on cold, still winter evenings.

3. Many European countries intend to _____ their remaining coal-fired power stations before 2035.

4. Engineers are still looking for affordable ways to _____ the enormous power of ocean waves.

5. Simple changes, such as switching off devices left on standby, can help families _____ their energy bills.

6. Pumped hydro plants allow operators to _____ surplus power for days or even weeks until it is needed.

Article 2

2 The sentences below are about using and saving energy. Choose the correct meaning (a, b or c) for the word in bold.

1. After an excellent harvest, the farmers sold their **surplus** grain to neighbouring regions.

a an amount that is more than what is needed
b an amount that is less than expected
c an amount kept for emergencies

2. Household energy **consumption** tends to rise sharply during the coldest months of the year.

a the production of something
b the amount of something that is used
c the cost of something

3. The owners decided to **retrofit** their 1970s house with a heat pump and new windows.

a to sell a property after improving it
b to knock down and rebuild completely
c to add new parts or equipment to something that already exists

4. Train tickets are much more expensive at **peak** times, when most people travel to work.

a the busiest or highest
b the quietest or lowest
c the most unpredictable

5. Thick **insulation** in the roof keeps the house warm in winter and cool in summer.

a a type of heating system
b material that stops heat passing through
c a design for large windows

6. The value of the currency continued to **fluctuate** throughout the year, worrying investors.

a to remain completely stable
b to rise slowly but steadily
c to change frequently and irregularly

7. Free public transport for students is a strong **incentive** to leave the car at home.

a something that encourages a person to do something
b something that prevents a person from doing something
c a rule that a person must obey

8. The city introduced new parking charges in an attempt to **curb** traffic in the centre.

a to encourage or increase
b to limit or control
c to measure or record

KEEPING THE LIGHTS ON

1. If the great challenge of renewable energy is that the sun does not always shine, the obvious solution is to save electricity when there is too much and release it when there is too little. On a bright spring afternoon, solar farms can produce such a large **surplus** that wholesale prices briefly fall below zero and some users are effectively paid to consume power. A few hours later, as people return home, cook dinner and switch on their televisions, the same grid may be struggling to cope. Storage is the bridge between those two moments.

2. The most familiar form of storage is the lithium-ion battery, the same technology found in phones and electric cars. Over the past decade, huge battery installations, some the size of several football pitches, have been connected to grids around the world. They can respond to a sudden drop in supply within a fraction of a second, far faster than any conventional power station. Their main limitation is duration: most can deliver power for only a few hours, which is enough to cover the evening **peak** but not a windless week in winter.

3. For longer periods, engineers are turning to older and more unusual ideas. Pumped hydro, which uses spare electricity to pump water uphill into a reservoir and then releases it through turbines when needed, still provides the vast majority of the world's stored energy. Other projects store heat in tanks of molten salt, compress air in underground caverns or convert excess power into hydrogen. None of these methods is perfect, and costs vary considerably, but together they could make a grid built on sun and wind as reliable as the fossil-fuel systems it replaces.

4. Storage is only one part of the answer; the other is managing demand more intelligently. A so-called smart grid uses digital meters and automated systems to track how much electricity is being used, minute by minute. Because wholesale prices can **fluctuate** dramatically during the day, smart meters allow suppliers to charge less when power is plentiful and more when it is scarce. Customers then have a financial **incentive** to run washing machines, dishwashers or car chargers overnight or at midday, spreading demand more evenly and reducing the need for expensive backup plants.

5. Perhaps the cheapest unit of energy, however, is the one that is never used. In many countries, homes account for around a quarter of total energy **consumption**, much of it lost through badly sealed windows, thin walls and uninsulated roofs. Improving **insulation** can cut heating bills significantly, and governments increasingly offer grants to help owners **retrofit** older buildings with double glazing, heat pumps or rooftop panels. Small habits matter too: lowering the thermostat by a single degree, or switching devices off rather than leaving them on standby, makes a noticeable difference over a year.

6. None of these changes is dramatic on its own, and that is precisely their strength. The energy system of the future is likely to combine large-scale storage, flexible smart grids and millions of efficient homes, each playing a modest role. Policies that **curb** waste while rewarding households for using power at the right time may prove just as important as building new wind farms. In other words, keeping the lights on in a renewable world depends not only on engineers and governments, but also on the everyday choices made in ordinary kitchens and living rooms.

DISCUSSION

Work with a partner. Discuss the questions.

1 Which of the ideas in the article — batteries, smart meters or better insulation — would make the biggest difference in your country? Why?

2 Should people pay more for electricity at peak times, or is this unfair to families who cannot change their routines?

GRAMMAR

Future forms for predictions and degrees of certainty

When writers make predictions, they rarely present every future event as equally certain. Academic and journalistic texts use a range of future forms to show how confident the writer is, whether something has already been planned, and whether an action will be in progress or completed at a particular point in the future.

The rules below show how to choose the future form that matches your degree of certainty and the timing of the event.

1. Use **will / won't + infinitive** for general predictions based on opinion or knowledge. Add **probably, almost certainly** or **definitely** to show how sure you are (after *will*, before *won't*).

Solar power will almost certainly become cheaper, but it probably won't replace gas completely in the next decade.

2. Use **be likely / unlikely to + infinitive** (or **It is likely that ...**) to express probability in a more cautious, academic way.

Electricity demand is likely to double by 2050, and it is unlikely that coal will meet this demand.

3. Use **be bound to / be certain to + infinitive** when you are very confident that something will happen, often because it is logical or inevitable.

As battery prices fall, more households are bound to install storage at home.

4. Use **be set to + infinitive** for something that is planned, arranged or strongly expected to happen soon — very common in news reports.

The country's largest solar park is set to open next spring.

5. Use the **future continuous (will be + -ing)** for an action that will be in progress at a particular time in the future.

By the end of the decade, millions of cars will be charging from the grid every night.

6. Use the **future perfect (will have + past participle)** for an action that will be completed before a certain point in the future, often with *by* or *by the time*.

By 2040, most European countries will have closed their last coal-fired power station.

Exercise A — Complete the paragraph with a suitable future form of the verbs in brackets. Use the word or form given in brackets where there is one.

Most experts agree that the global energy system (1) _____ (change / likely) more in the next twenty years than in the last hundred. According to several forecasts, by 2035 solar power (2) _____ (overtake / future perfect) coal as the world's largest source of electricity. As production increases, the price of batteries (3) _____ (fall / bound) even further. In our own region, a huge new solar park (4) _____ (open / set) next year, and this time next year it (5) _____ (supply / future continuous) power to around half a million homes. However, fossil fuels (6) _____ (not disappear / will) overnight. In 2040, many countries (7) _____ (still use / future continuous) gas to heat their homes, and the world as a whole (8) _____ (reach / unlikely) its climate targets without much faster progress.

Exercise B — Rewrite each sentence using the word or form in brackets so that the meaning stays the same.

1. It is almost certain that electricity prices will rise again this winter. (bound)

2. The government plans to open three new wind farms next year. (set)

3. There is a good chance that more people will buy electric cars in the future. (likely)

4. The country will finish closing all its coal plants at some point before 2040. (future perfect — By 2040, …)

5. At 7 p.m. tomorrow, the battery will be in the middle of supplying power to the house. (future continuous)

ACADEMIC WRITING

Presenting advantages and disadvantages — writing a balanced argument

ACADEMIC WRITING SKILL

Balanced argument and signposting

In a balanced argument, the writer presents both the advantages and the disadvantages of an issue fairly before reaching a conclusion. Signposting phrases such as *One major benefit is …*, *On the other hand, …* and *A further drawback is …* guide the reader clearly from one side of the argument to the other.

ADVANTAGE: *One clear benefit of rooftop solar panels is that, once installed, they produce electricity at almost no cost.*

DISADVANTAGE: *On the other hand, a significant drawback is that their output falls sharply at night and on cloudy days.*

Each sentence below presents one side of an argument. Write a second sentence that presents the opposite side, using a suitable signposting phrase (e.g. *however*, *on the other hand*, *a major drawback is*, *one advantage is*).

1 Wind farms produce no carbon emissions once they are up and running.

_____

2 Large solar parks can bring jobs and investment to rural areas.

_____

3 Home batteries are still very expensive for the average family.

_____

4 Smart meters give customers detailed information about their energy use.

_____

5 Building new high-voltage power lines can take more than ten years.

_____

6 Electric cars are much cheaper to run than petrol or diesel cars.

_____

WRITING TASK

Write a balanced paragraph (80–100 words) on the advantages and disadvantages of installing solar panels on homes in your country. Use at least three signposting phrases and finish with a sentence that gives your overall view.

PART 4

Read the following text for question 21-29.

PIRATES

There have always been people who robbed ships in the open sea. We know them as pirates or corsairs. Many of them were sailors who lost their jobs, others were just young men and women looking for adventure. Thousands of pirates were active from 1650–1720. These years are sometimes known as the 'Golden Age' of piracy. Probably the most famous pirates of this period are Blackbeard, Henry Morgan, William 'Captain' Kidd, and the Barbarossa brothers.

The "Jolly Roger" is the well-known pirate flag. There are a number of theories about its origin. It is thought that the first pirates used a red flag, which was common in sea attacks. A red flag told other ships that there would be no pity and no prisoners would be taken. This red flag was called Joli Rouge (pretty red) in French. Most probably, English people pronounced it in their own way "Jolly Roger". Another theory says that "old Roger" was another name for the devil.

Another popular image from the pirate world is a parrot, sitting on the captain's shoulder. Well, there's no historical proof of that. No doubt, pirates often travelled to faraway exotic lands and brought unusual animals as souvenirs with them. Parrots were particularly popular because they were colourful, could be taught to speak, and were easier to care for than, say, monkeys. However, you can't imagine a pirate fighting with a bird on the shoulder. This image possibly began with the book "Treasure Island". In the book, the ever famous Long John Silver carried a parrot as a pet.

Before joining a ship each pirate signed an agreement called "Articles". This explained the rights and laws on board the ship, how much a person was paid, what to do in case of emergency, etc. For example, playing cards for money was not allowed. For breaking the rules a pirate would be shot or left on a desert island with a bottle of water and a gun.

A pirate ship was organized democratically. After a successful attack they divided the treasures equally, only the captain and the cook got more. They sometimes decided together which ship to attack or where to sail next. Most pirates were paid four times more than they had when they were sailors. But on land they were criminals and lots of money was offered to those who caught them.

However, for some time the English government supported pirates. There were lots of pirates in the sixteenth century when England was at war with Spain and the Dutch Republic. Queen Elizabeth I ordered them to rob Spanish ships and bring gold, silver and jewels to the country. Sir Francis Drake was one of the most famous pirates. He brought lots of treasures to England and also was the first Englishman to sail all the way around the world. Because of this, he was made a knight by Queen Elizabeth I.

For questions 21-24, choose the correct answer A, B, C, or D. Mark your answers on the answer sheet.

21. “Golden age” of piracy lasted ...
A) for a century
B) for 70 years
C) until XIX century
D) until James Kidd died

22. A red flag symbolizes ...
A) French pirate ships
B) the first pirates that used them
C) no kindness, no prisoners
D) Jolly Roger - the devil

23. Parrots, according to the author ...
A) sat on captain’s shoulder
B) were nothing more than souvenirs to them
C) were taught to train monkeys
D) has possibly become fashionable after the book - “Treasure Island”

24. The pirates who decide whom to attack next were ...
A) criminals
B) boardsmen and captains
C) captains and cooks
D) sailors

For questions 25-29, decide if the following statements agree with the information given in the text. Mark your answers on the answer sheet.

25. The flag "Jolly Roger" meant death for ships' sailors and passengers.
A) True
B) False
C) No information

26. The name for the flag "Jolly Roger" is of English origin.
A) True
B) False
C) No information

27. Parrots brought luck to pirates at sea.
A) True
B) False
C) No information

28. In the past pirates did some jobs for the English queens and kings.
A) True
B) False
C) No information

29. Francis Drake was the richest English pirate.
A) True
B) False
C) No information

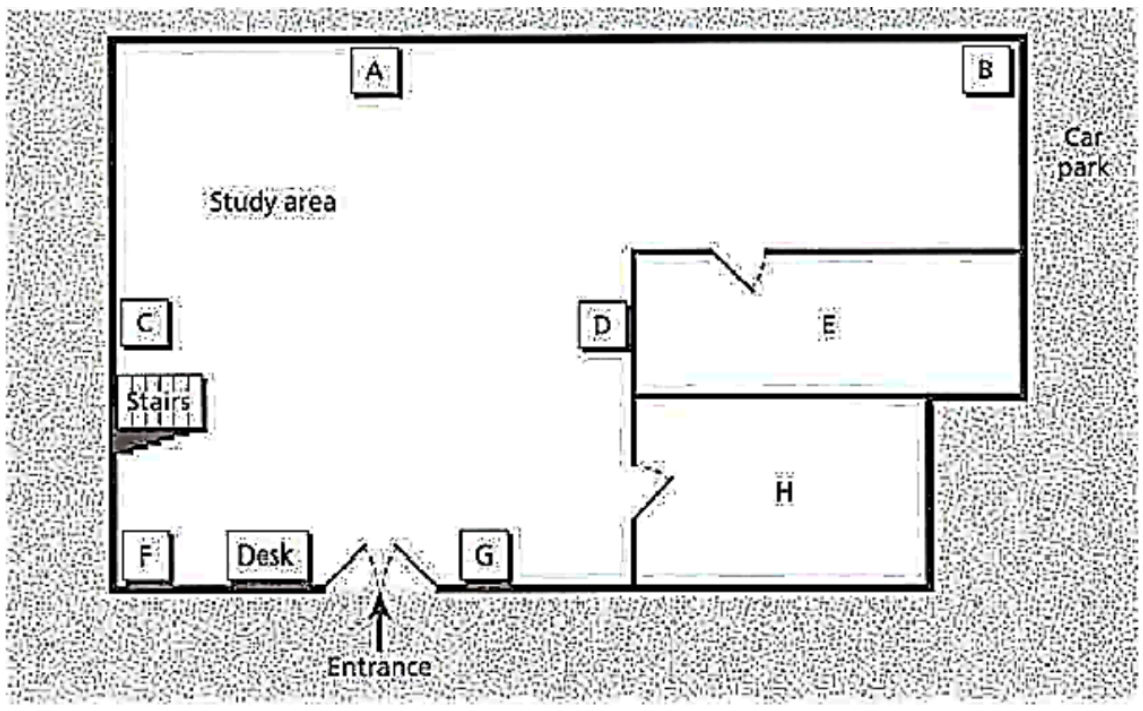
✔ Wordlist

1. rob ▢
2. the open sea ▢
3. corsair ▢
4. look for adventure ▢
5. be active ▢
6. Golden Age ▢
7. theory ▢
8. origin ▢
9. pity ▢
10. take prisoners ▢
11. pronounce ▢
12. devil ▢
13. historical proof ▢
14. faraway ▢
15. exotic ▢
16. souvenir ▢
17. care for ▢
18. sign an agreement ▢
19. on board ▢
20. in case of emergency ▢
21. break the rules ▢
22. desert island ▢
23. democratically ▢
24. divide equally ▢
25. treasure ▢
26. criminal ▢
27. support ▢
28. be at war with ▢
29. jewels ▢
30. be made a knight ▢

PART 4

You will hear someone giving a talk. Label the places (19-23) on the map (A-H). There are THREE extra options which you do not need to use.
Mark your answers on the answer sheet.

PLAN OF LEARNING RESOURCE CENTRE (GROUND FLOOR)



19. Newspapers
20. Computers
21. Photocopier
22. Café
23. Sports books

Phrases & Idioms Wordlist

Run out of steam: To lose energy or enthusiasm and be unable to continue with the same effort.

Recharge your batteries: To rest and relax so that you have energy again.

Full of beans: Very lively, cheerful and full of energy.

Burn the candle at both ends: To work or stay active from early morning until late at night, leaving yourself exhausted.

Burn out: To become completely exhausted, physically or mentally, after working too hard for too long.

Get a second wind: To suddenly find new energy to continue after feeling tired.

Firing on all cylinders: Working or performing at full power and as well as possible.

Power through (something): To continue doing something difficult with determination until it is finished.

Running on empty: Continuing to work or function even though you have no energy or resources left.

Pull the plug on (something): To stop a project or activity suddenly, usually by stopping its funding.

Blow a fuse: To suddenly become very angry and lose your temper.

On the same wavelength: Having similar ideas and understanding each other well.

A live wire: A very lively, energetic and enthusiastic person.

Switch off: To stop paying attention or stop thinking about something, especially in order to relax.

Spark off (something): To cause something to start, especially a reaction, argument or series of events.

USEFUL LANGUAGE — Advantages and disadvantages essay

Function	Useful phrases
Introducing the topic	*In recent years, … / It is widely accepted that … / A growing number of countries have …*
Presenting advantages	*One major benefit of … is that … / The most obvious advantage is … / A further point in its favour is …*
Presenting disadvantages	*On the other hand, … / However, a significant drawback is … / Critics point out that …*
Adding points	*Furthermore, … / In addition, … / What is more, …*
Making predictions	*… is likely to … / … is bound to … / By 2040, … will have …*
Concluding / giving your view	*On balance, I believe … / Overall, the benefits clearly outweigh the drawbacks … / To sum up, …*

THE TASK

You are writing an essay for an academic audience. Write 180–200 words.

Many governments are now spending large amounts of money on solar and wind farms instead of traditional power stations. What are the advantages and disadvantages of this trend? Give your own opinion.

PARAGRAPH PLAN

Introduction	paraphrase the question + say that the trend has both benefits and drawbacks
Body paragraph 1	advantages: environment, falling costs, energy independence + examples
Body paragraph 2	disadvantages: intermittent supply, cost of storage and power lines, land use
Conclusion	weigh up both sides + give your overall opinion clearly

RULES for Part 2 (essay)

☐ Answer ALL parts of the question — advantages, disadvantages AND your own opinion.

☐ Write 180–200 words in four clear paragraphs: introduction, two body paragraphs and a conclusion.

☐ Use a formal, academic register: no contractions, no slang and no direct questions to the reader.

☐ Begin each body paragraph with a clear topic sentence and use signposting phrases to move between ideas.

☐ Support each main point with a reason or an example, and use cautious future forms (is likely to, is set to) for predictions.

☐ Make your conclusion consistent with your argument — do not introduce new ideas at the end.

SAMPLE ANSWER

In recent years, a growing number of governments have chosen to invest in solar and wind farms rather than power stations that burn coal or gas. While this shift offers clear benefits, it also raises a number of practical concerns.

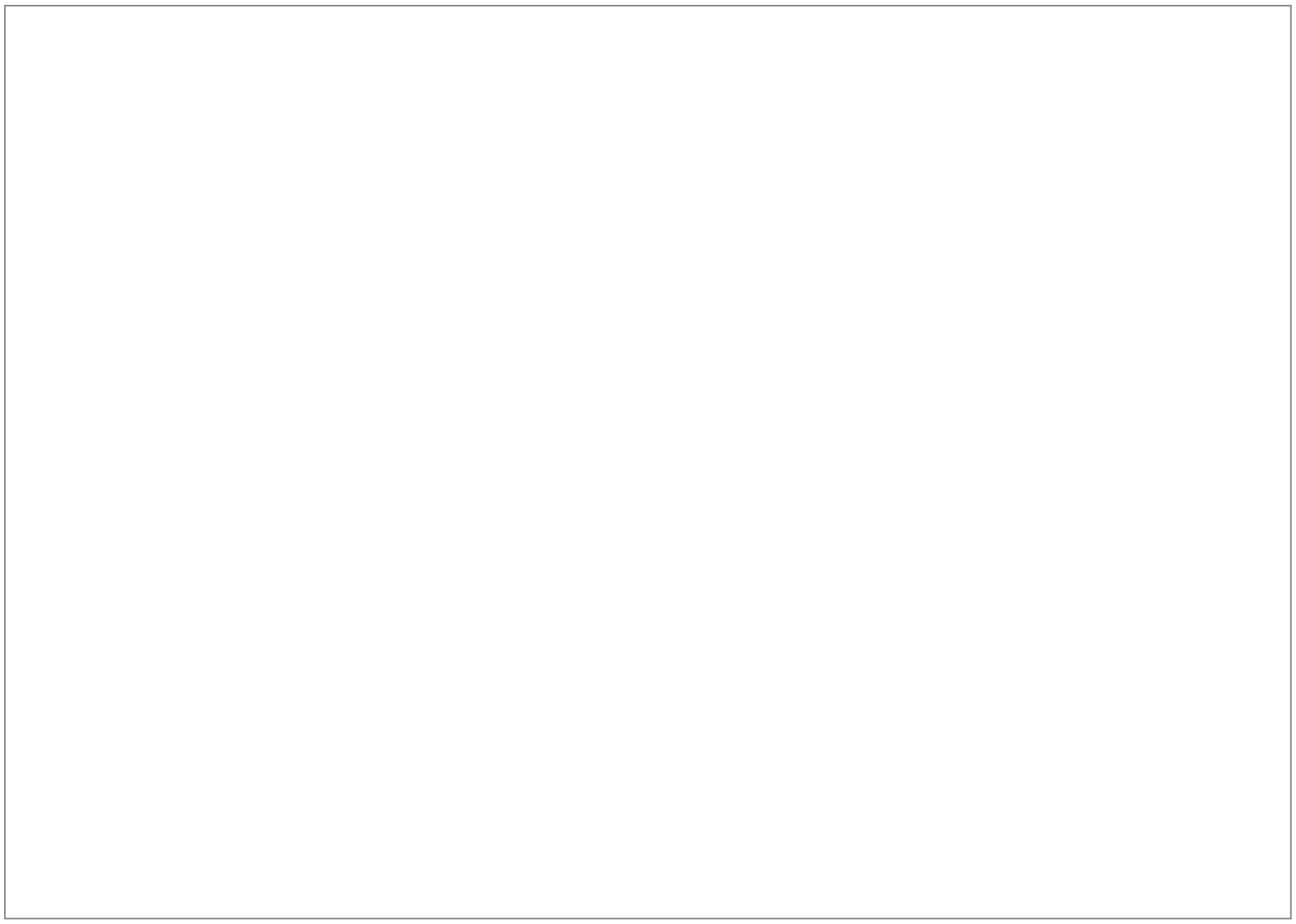
The most obvious advantage is environmental. Renewable sources produce virtually no greenhouse gases once they are operating, so they are bound to play a central role in tackling climate change. Furthermore, the cost of solar panels and turbines has fallen dramatically, which means that clean electricity is now often cheaper than power from fossil fuels. Sunny countries in particular are likely to benefit, as they can generate energy locally instead of importing expensive fuel.

On the other hand, there are significant drawbacks. Sun and wind are intermittent, so countries must also invest in batteries and new power lines, which adds considerably to the overall cost. In addition, large projects can occupy huge areas of land and sometimes face strong opposition from local residents.

On balance, I believe the advantages clearly outweigh the disadvantages. Although the transition will not be easy, governments that invest wisely in storage as well as generation will have built a cleaner and more secure energy system within a generation.

PART 2 PRACTICE — now you try

Some people believe that individuals can do very little to reduce energy use and that only governments and large companies can make a real difference. Discuss both views and give your own opinion.



SPEAKING PART 2

QUESTIONS

Describe a time when you or your family tried to save energy at home.

You should say:

- ☐ when and why you decided to save energy
- ☐ what changes you made
- ☐ how easy or difficult it was to keep to these changes

...and say whether you think individuals can really make a difference to the amount of energy a country uses.

SAMPLE ANSWER

*I'd like to talk about a winter about two years ago, when my family **made a real effort to cut our energy bills**. The trigger was a particularly cold month when our electricity and gas bill was almost double what we had expected, and my father announced, quite dramatically, that things had to change.*

*The first thing we did was **sort out the insulation** — we sealed the gaps around our old windows and put thick curtains up in the living room. We also **got into the habit of** switching devices off at the wall instead of leaving them on standby, and we replaced every old bulb with LED lights. My younger brother was put in charge of checking that lights were off in empty rooms, and to be honest, he took the job **far more seriously than any of us expected**.*

*Some changes were easy, but others were harder to stick to. Turning the heating down by a couple of degrees, for example, led to a few arguments, especially in the mornings! Still, after a couple of months the bills had **dropped noticeably**, which kept everyone motivated.*

*As for whether individuals can make a difference, I'd say **on their own, probably not much**, but millions of households doing the same things certainly can. **At the end of the day**, I think government policy and personal habits have to work together.*

USEFUL LANGUAGE	
Introducing the topic:	*I'd like to talk about a time when... / One experience that comes to mind is...*
Explaining the reason:	*The main reason was... / What really prompted us to change was...*
Describing changes:	*The first thing we did was... / We also got into the habit of...*
Talking about difficulty:	*Some changes were easy, but... / It was quite hard to stick to...*
Describing results:	*After a while, we noticed that... / The bills dropped noticeably...*
Giving a wider opinion:	*On their own, individuals... / At the end of the day, I think...*

MEDICINE

HEALTH & SCIENCE

UNIT 9

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. When you feel ill, do you go to a doctor straight away, or wait to see if you get better on your own? Why?

2. Have you ever had a medical appointment by phone or video? How did it compare with seeing a doctor in person?

3. Why do you think some people expect to be given antibiotics every time they have a cold?

4. Would you trust a computer program to diagnose an illness? Why / Why not?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 Several new varieties of the virus have **emerged** since the first cases were reported last winter.

2 Elderly people and young children are particularly **susceptible** to infections during the winter months.

3 A single **mutation** in a gene can be enough to change the way a cell responds to a drug.

4 Doctors are advised not to **prescribe** strong painkillers unless other treatments have already failed.

5 The laboratory tested more than 2,000 chemical **compounds** before finding one that killed the bacteria.

6 The plan to build a second hospital was abandoned because it was not financially **viable**.

7 All new vaccines must pass **rigorous** safety checks before they can be given to the public.

8 Tax breaks were introduced as an **incentive** for companies to invest in medical research.

a (adj) able to work successfully; practical and realistic

b (n) a substance formed from two or more chemical elements combined

c (v) to appear or become noticeable, especially for the first time

d (n) something that encourages a person or organisation to do something

e (adj) easily affected, harmed or infected by something

f (adj) extremely careful and thorough, following strict rules

g (v) to officially tell someone to take a particular medicine or treatment

h (n) a permanent change in the genetic material of a living thing

Complete the sentences with the correct form of a word from either article.

1. Patients who fail to complete a course of antibiotics may contribute to the growing _____ of bacteria. (resistant)

2. Video calls have made it far more _____ for people in rural areas to get medical advice. (convenience)

3. The _____ of the new scanning software was tested on more than 10,000 images. (accurate)

4. The rapid _____ of bacteria means that a resistant strain can spread within weeks. (reproduce)

5. Genetic factors may increase a person's _____ to certain infections. (susceptible)

THE RACE AGAINST RESISTANT BACTERIA

1 When penicillin entered hospitals in the 1940s, infections that had routinely killed patients suddenly became treatable within days. For several decades afterwards, it seemed that bacterial disease had been more or less defeated. Today, however, that confidence looks badly misplaced. Strains of bacteria that no longer respond to common antibiotics have **emerged** on every continent, and infections once considered minor — a scratch that turns septic, a urinary infection, a case of pneumonia — are becoming harder, and in some cases impossible, to treat.

2 The process behind this is, in principle, simple natural selection. Every time an antibiotic is used, the bacteria most **susceptible** to it die first, while any that happen to carry a protective **mutation** survive and multiply. Because bacteria reproduce so quickly, a resistant population can build up in a matter of weeks. Worse still, many bacteria can pass resistance genes directly to one another, so a defence that appears in one species may eventually spread to others that have never been exposed to the drug at all.

3 Human behaviour has accelerated this process considerably. In many countries, doctors under pressure from anxious patients **prescribe** antibiotics for colds and flu, even though these are viral illnesses that antibiotics cannot touch. Patients, in turn, often stop taking their tablets as soon as they feel better, leaving the toughest bacteria alive. Meanwhile, vast quantities of antibiotics are given to healthy farm animals simply to speed up their growth, creating ideal conditions for resistance to develop far away from any hospital ward.

4 The obvious answer would be to invent new drugs, yet this has proved remarkably difficult. Most antibiotics in use today belong to families discovered more than half a century ago. Researchers screening thousands of natural and synthetic **compounds** frequently find that promising candidates are either too toxic for humans or simply variations on drugs that bacteria have already learnt to resist. As one microbiologist put it, the easy discoveries were made long ago; what remains is slow, expensive and often disappointing work.

5 Even when a candidate looks **viable**, the road to the pharmacy is long. A new medicine must pass through laboratory testing, followed by several phases of clinical trials involving progressively larger groups of volunteers. Each stage is designed to answer a specific question — whether the drug is safe, what dose works, and whether it performs better than existing treatments — and the whole process typically takes ten years or more. This **rigorous** approach protects patients, but it also means that only a small fraction of the compounds that enter testing ever reach the market.

6 Here lies the central economic problem. A company that spends a decade developing a cancer drug or a medicine for a long-term condition can expect patients to buy it for years. A new antibiotic, by contrast, is likely to be held in reserve by doctors and used only as a last resort, precisely to slow the development of resistance. Low sales mean that pharmaceutical firms have little financial **incentive** to invest, and several large companies have withdrawn from antibiotic research altogether.

7 Governments and health organisations are now experimenting with alternative models, such as paying companies a fixed annual fee for access to a new antibiotic regardless of how often it is used. Scientists, meanwhile, are exploring options ranging from viruses that attack specific bacteria to artificial intelligence that can scan millions of molecules in days. None of these is a quick fix, but experts broadly agree on one point: unless prescribing habits change alongside the science, any new drug will eventually meet the same fate as the old ones.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A Why companies are walking away

B Survival of the toughest

C A long and costly journey to the pharmacy

D A miracle cure that is losing its power

E Vaccines: the complete answer

F Habits that speed up the problem

G Fresh approaches, but an old lesson

H The search for new candidates

LANGUAGE DEVELOPMENT - Talking about research and treatment

VOCABULARY SKILL

Verb + noun collocations

Medical and scientific writing relies heavily on fixed verb + noun combinations — for example **conduct a trial** or **pose a threat**. Choosing the wrong verb (*do a trial*, *make a threat*) is one of the most common errors at advanced level, so it is worth learning these combinations as complete chunks rather than word by word.

conduct	research, a study, a clinical trial
undergo	surgery, treatment, tests
develop	a vaccine, a drug, resistance
make	a diagnosis, a full recovery, progress
raise	concerns, awareness, questions
pose	a threat, a risk, a challenge

Complete the sentences with the correct verb from the table above. Use the correct form.

1. The research team plans to _____ a large clinical trial involving 3,000 volunteers next year.

2. Despite the seriousness of her injuries, she is expected to _____ a full recovery.

3. The overuse of antibiotics in farming continues to _____ a serious threat to public health.

4. Before the operation, all patients must _____ a series of blood tests.

5. Several doctors have _____ concerns about the reliability of the new diagnostic app.

6. Some bacteria can _____ resistance to a new drug within just a few years.

Article 2

2 The sentences below are about technology in health care. Choose the correct meaning (a, b or c) for the word in bold.

1. Most of the clinic's **consultations** now take place by phone, so patients rarely need to visit in person.

a meetings with a doctor to get advice or treatment

b written reports about a patient's health

c payments made for medical care

2. For most patients, the benefits of the new medicine far **outweigh** its minor side effects.

a are equal to

b are greater or more important than

c are hidden by

3. The hospital **deployed** extra staff to the emergency department during the flu season.

a dismissed

b trained

c moved into position so they could be used

4. The company's safety record has come under close **scrutiny** after several complaints from patients.

a careful and detailed examination

b public praise

c legal protection

5. The survey was criticised for its **bias**, since it only asked people who already supported the plan.

a a lack of clear questions

b an unusually large sample

c an unfair preference that distorts the results

6. Hospital managers must be held **accountable** for decisions that put patients at risk.

a praised publicly

b responsible and expected to explain their actions

c protected from criticism

7. Experienced nurses often rely on **intuition** to sense when a patient's condition is getting worse.

a written guidelines

b advice from colleagues

c a feeling based on experience rather than conscious reasoning

8. The surgeon's calm explanation gave the family some **reassurance** before the operation.

a detailed technical information

b comfort that reduces worry or doubt

c official permission

THE DOCTOR WILL SEE YOU NOW — ON SCREEN

1. For most of medical history, seeing a doctor meant being in the same room. That assumption changed almost overnight during the COVID-19 pandemic, when clinics around the world switched to video and telephone **consultations** to keep patients and staff apart. Many expected the change to be temporary. Instead, remote care has become a permanent feature of many health systems, and it is now being joined by a second, potentially more disruptive technology: artificial intelligence that can help identify disease from scans, photographs and test results.

2. The benefits of telemedicine are easy to see, particularly for people who live far from a hospital or find travelling difficult. A farmer in a remote village can show a rash to a specialist hundreds of kilometres away; a parent can get advice about a feverish child without spending half a day in a crowded waiting room. For routine follow-up appointments and repeat prescriptions, supporters argue, the convenience and savings clearly **outweigh** the drawbacks of not meeting face to face.

3. AI diagnostic tools promise something different: speed and consistency. Systems trained on millions of medical images can now spot signs of certain eye diseases, skin cancers and lung conditions, in some studies matching the performance of experienced specialists. Unlike humans, they do not get tired or distracted at the end of a long shift. In countries with too few radiologists, hospitals have **deployed** such software to sort scans so that the most urgent cases are seen first, cutting potentially dangerous delays.

4. Yet these tools have come under increasing **scrutiny**. An algorithm is only as good as the data it learns from, and if that data comes mainly from one population, the system may perform poorly for others. Researchers have found evidence of **bias** in some programs designed to detect skin conditions, which were far less accurate on darker skin because few such images had been included in their training. Errors of this kind could deepen existing inequalities in health care rather than reduce them.

5. There are also difficult questions of responsibility. If an AI system misses a tumour, who should be held **accountable** — the doctor who relied on it, the hospital that bought it, or the company that built it? Current regulations were largely written for human professionals and conventional medical devices, and lawmakers in many countries are still debating how to adapt them. Patients, too, have concerns about privacy, since both video calls and AI systems depend on large amounts of sensitive personal data being stored and shared.

6. Most doctors see the future as a partnership rather than a replacement. Software may be better at detecting patterns in images, but a clinician's **intuition**, built over years of examining patients, can pick up on details that never appear in a scan — a hesitation in someone's voice, or a symptom the patient forgot to mention. And for many people, especially those facing a serious diagnosis, the **reassurance** of a calm conversation with another human being remains an essential part of being cared for.

DISCUSSION

Work with a partner. Discuss the questions.

1 Would you be happy for your first appointment with a doctor to take place by video call? Why / Why not?

2 If an AI system makes a wrong diagnosis, who do you think should be responsible, and why?

GRAMMAR

Reporting verbs & reported speech in academic writing

Academic writers constantly report what other researchers have said, found or argued. The reporting verb you choose shows not only what a source says but also how strong the evidence is and how far you agree with it. Impersonal structures such as it has been found that allow you to report findings without naming a specific person.

The rules below show the main groups of reporting verbs, the patterns that follow them, and how the choice of verb changes the meaning of a sentence.

1. Neutral verbs — **state, report, note, explain, observe** + *that* — simply present what a source says without judging it.

The authors note that resistance rates have risen sharply over the last decade.

2. Tentative verbs — **suggest, indicate, propose, imply** — show that the evidence points towards something but is not conclusive.

The data suggest that shorter courses of antibiotics may be just as effective.

3. Strong verbs — **argue, claim, maintain, insist** — present a position the source is defending. **Claim** often signals that the writer has doubts about it.

The manufacturer claims that its app can detect skin cancer as accurately as a specialist.

4. **Acknowledge, admit, concede** show that a source accepts a point, often one that weakens its own argument.

Supporters of telemedicine acknowledge that video calls are unsuitable for some examinations.

5. Some verbs take different patterns: **warn** (sb) **against** + *-ing*, **urge** sb **to** + infinitive, **recommend / suggest** + *-ing* or *that* + (should) + base form.

Experts urge doctors to prescribe antibiotics only when absolutely necessary.

6. Impersonal passives — **It has been found / shown / suggested that…** or **X is thought / believed to…** — report findings without naming a source. In academic writing, tenses do not usually move back if the finding is still true.

It has been shown that AI tools perform less well on groups that are under-represented in their training data.

Exercise A — Complete the paragraph with the correct form of the reporting verbs in the box: acknowledge, argue, claim, conclude, find, point out, suggest, warn.

A recent review of AI diagnostic tools (1) _____ that, in controlled studies, several systems performed as well as experienced specialists. However, the authors (2) _____ that most of these studies used high-quality images that rarely reflect real hospital conditions. One technology company (3) _____ that its software 'will make radiologists unnecessary within ten years', a view that most doctors reject. Critics (4) _____ that replacing human judgement entirely would be both unsafe and unethical. Even enthusiastic developers (5) _____ that their systems sometimes make errors that no human would make. Patient groups have (6) _____ hospitals against introducing such tools without proper testing. Overall, the evidence (7) _____ that the technology works best as a second opinion rather than a replacement, and the review (8) _____ that further research in real clinical settings is urgently needed.

Exercise B — Rewrite each sentence using the impersonal structure in brackets, as shown in rule 6.

1. Researchers have found that resistant infections lengthen hospital stays. (It has been found that…)

2. Experts believe that overuse in farming is a major cause of resistance. (Overuse in farming is believed…)

3. Many studies have shown that video consultations save patients time and money. (It has been shown that…)

4. Doctors think that the new drug is safer than older treatments. (The new drug is thought…)

5. Some researchers have suggested that hospitals should share patient data more openly. (It has been suggested that…)

ACADEMIC WRITING

Referring to sources — reporting research findings accurately

ACADEMIC WRITING SKILL

Referring to sources

In academic writing you must show clearly where information comes from, using phrases such as *According to…*, *X found that…* or *As X points out,…*. Reporting a finding accurately — without exaggerating it, changing its meaning or presenting it as your own idea — is the most effective way to avoid plagiarism.

SOURCE: *Smith & Patel (2021): Of 400 patients surveyed, 72% said that a video consultation had fully met their needs.*

REPORTED: *According to Smith and Patel (2021), almost three-quarters of the patients they surveyed felt that a video consultation had been sufficient.*

Rewrite each piece of information as one sentence that refers to its source. Use a different reporting phrase each time (According to..., X found that..., X claims that..., As X points out,...).

1 Lee (2020): Patients who stop taking antibiotics early are more likely to need a second course of treatment.

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2 A hospital report (2023): Waiting times for scan results fell by 30% after an AI system began sorting scans.

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3 Professor Okafor (interview, 2022): 'Without new business models, the supply of new antibiotics will simply dry up.'

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4 Garcia et al. (2019): Most skin-cancer algorithms were trained mainly on images of lighter skin.

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5 A technology company's website: 'Our app is just as accurate as a qualified dermatologist.'

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6 A national survey of family doctors (2024): Most doctors support video calls for follow-up appointments, but not for first visits.

WRITING TASK

Write a short paragraph (80–100 words) on whether AI should be used to diagnose illness. Refer to at least two sources from this unit — the articles or the sentences above — using a range of reporting verbs, and make it clear which ideas belong to others and which are your own.

PART 5

Read the following text for questions 30-35.

IS GRAFFITI ART OR CRIME?

The term graffiti derives from the Italian graffito meaning 'scratching' and can be defined as uninvited marking or writing scratched or applied to objects, built structures and natural features. It is not a new phenomenon: examples can be found on ancient structures around the world, in some cases predating the Greeks and Romans. In such circumstances it has acquired invaluable historical and archaeological significance, providing a social history of life and events at that time. Graffiti is now a problem that has become pervasive, as a result of the availability of cheap and quick means of mark-making.

It is usually considered a priority to remove graffiti as quickly as possible after it appears. This is for several reasons. The first is to prevent 'copy-cat' emulation which can occur rapidly once a clean surface is defaced. It may also be of a racist or otherwise offensive nature and many companies and councils have a policy of removing this type of graffiti within an hour or two of it being reported. Also, as paints, glues and inks dry out over time they can become increasingly difficult to remove and are usually best dealt with as soon as possible after the incident. Graffiti can also lead to more serious forms of vandalism and, ultimately, the deterioration of an area, contributing to social decline.

Although graffiti may be regarded as an eyesore, any proposal to remove it from sensitive historic surfaces should be carefully considered: techniques designed for more robust or utilitarian surfaces may result in considerable damage. In the event of graffiti incidents, it is important that the owners of buildings or other structures and their consultants are aware of the approach they should take in dealing with the problem. The police should be informed as there may be other related attacks occurring locally. An incidence pattern can identify possible culprits, as can stylised signatures or nicknames, known as 'tags', which may already be familiar to local police. Photographs are useful to record graffiti incidents and may assist the police in bringing a prosecution. Such images are also required for insurance claims and can be helpful in cleaning operatives, allowing them to see the problem area before arriving on site.

There are a variety of methods that are used to remove graffiti. Broadly these divide between chemical and mechanical systems. Chemical preparations are based on dissolving the media; these solvents can range from water to potentially hazardous chemical 'cocktails'. Mechanical systems such as wire-brushing and grit-blasting attempt to abrade or chip the media from the surface. Care should be taken to comply with health and safety legislation with regard to the protection of both passers-by and any person carrying out the cleaning. Operatives should follow product guidelines in terms of application and removal, and wear the appropriate protective equipment. Measures must be taken to ensure that run-off, aerial mists, drips and splashes do not threaten unprotected members of the public. When examining a graffiti incident it is important to assess the ability of the substrate to withstand the prescribed treatment. If there is any doubt regarding this, then small trial areas should be undertaken to assess the impact of more extensive treatment.

A variety of preventive strategies can be adopted to combat a recurring problem of graffiti at a given site. As no two sites are the same, no one set of protection measures will be suitable for all situations. Each site must be looked at individually. Surveillance systems such as closed-circuit television may also help. In cities and towns around the country, prominently placed cameras have been shown to reduce anti-social behavior of all types including graffiti. Security patrols will also act as a deterrent to prevent recurring attacks. However, the cost of this may be too high for most situations. A physical barriers such as a wall, railings, doors or gates can be introduced to discourage unauthorized access to a vulnerable site. However, consideration has to be given to the impact measures have on the structure being protected. In the worst cases, they can be almost as damaging to the quality of the environment as the graffiti they prevent. In others, they might simply provide a new surface for graffiti.

One of the most significant problems associated with graffiti removal is the need to remove it from surfaces that are repeatedly attacked. Under these circumstances, the repeated removal of graffiti using even the most gentle methods will ultimately cause damage to the surface material. There may be situations where the preventive strategies mentioned above do not work or are not a viable proposition at a given site. Anti-graffiti coatings are usually applied by brush or spray leaving a thin veneer that essentially serves to isolate the graffiti from the surface.

Removal of graffiti from a surface that has been treated in this way is much easier, usually using low-pressure water which reduces the possibility of damage. Depending on the type of barrier selected it may be necessary to reapply the coating after each graffiti removal exercise.

For questions 30-33, fill in the missing information in the numbered spaces. Write no more than ONE WORD and / or A NUMBER for each question.

Ancient graffiti is of significance and records the (30)_____ history of details life for that period. The police can recognize newly committed incidents of graffiti by the signature which is called (31)_____ that they are familiar with. Operatives ought to comply with relevant rules during the operation, and put on the suitable protective (32)_____. Removal of graffiti from a new type of coating surface can be much convenient of using (33)_____.

For questions 34-35, choose the correct answer A, B, C, or D. Mark your answers on the answer sheet.

34. Which statement is true concerning the removal of graffiti?
A) cocktail removal can be safer than water treatment
B) small patch trial before applying large scale of removing
C) Chemical treatments are the most expensive way of removing
D) mechanical removals are much more applicable than Chemical treatments.

35. Which of the following preventive measures against graffiti are mentioned effectively in the passage?
A) organise more anti-graffiti movement in the city communities
B) Build a new building with material repelling to water
C) installing more visible security cameras
D) Provide a whole new surface with a chemical coat

✔ Wordlist

1. derive from
2. uninvited
3. phenomenon
4. predate
5. acquire
6. invaluable
7. archaeological significance
8. pervasive
9. a priority
10. emulation
11. deface
12. offensive
13. deterioration
14. social decline
15. eyesore
16. robust
17. utilitarian
18. culprit
19. bring a prosecution
20. insurance claim
21. dissolve
22. hazardous
23. abrade
24. comply with
25. legislation
26. withstand
27. deterrent
28. unauthorised access
29. viable proposition
30. veneer

PART 5

You will hear three extracts. Choose the correct answer (A, B or C) for each question (24-29). There are TWO questions for each extract.
Mark your answers on the answer sheet.

Extract One

24. What do we learn about the writer of the play?
A) He died.
B) He is dying.
C) He nearly died.

25. What do the speakers agree on?
A) The writer is quite predictable.
B) The writer's style has changed direction.
C) The play wasn't as funny as some of the writer's earlier plays.

Extract Two

26. According to Pablo Jenson
A) a variety of different shops tends to boost sales.
B) traders that have something in common can boost each other's sales.
C) butchers are the most successful traders.

27. Jenson's theory
A) advertised for similar retailers to open shops in one particular area.
B) works better for bakers and butchers than for other kinds of retailers.
C) seems to have been proved to be credible.

Extract Three

28. The Costa de la Cruz
A) is being spoiled by developers.
B) is the cheapest area of Spain in which you can buy a holiday home.
C) is close to Portugal.

29. According to Chris Mercer
A) there is a danger that the Costa de la Cruz will lose its appeal.
B) the Costa de la Cruz can be developed without the area being damaged.
C) the government has put a stop to the Costa de la Cruz being developed further.

Phrases & Idioms Wordlist

Under the weather: Feeling slightly ill or not as well as usual.

On the mend: Recovering after an illness or injury.

A clean bill of health: An official statement that someone is healthy, or that something is in good condition.

Back on one's feet: Healthy and active again after being ill or having problems.

As fit as a fiddle: In excellent physical health.

Run down: Tired, weak and more likely to become ill, usually because of overwork or stress.

Take a turn for the worse: To suddenly become more ill or a situation becoming much worse.

Out of the woods: No longer in danger or serious difficulty, especially after an illness.

A bitter pill to swallow: An unpleasant fact or situation that is difficult to accept.

Kill or cure: Describing a risky treatment or action that will either solve a problem completely or make it much worse.

Prevention is better than cure: It is easier and wiser to stop a problem happening than to deal with it afterwards.

Just what the doctor ordered: Exactly what is needed to make someone feel better or improve a situation.

Go under the knife: To have a surgical operation.

Burn the candle at both ends: To work or stay active from early morning until late at night, so that you become exhausted.

Have a frog in one's throat: To have difficulty speaking clearly because your throat feels rough or blocked.

WRITING PART 1.2

USEFUL LANGUAGE — Formal letter

Function	Useful phrases
Opening	*Dear Sir/Madam, / Dear Dr [Surname], / I am writing in response to your recent email regarding...*
Reacting / feelings	*I was pleased to learn that... / While I welcome..., I must admit I have some concerns about...*
Giving a suggestion	*I would suggest that... / It would be beneficial to... / It might be worth considering...*
Requesting a reply	*I would be grateful if you could... / I would appreciate further information about...*
Closing	*Thank you for taking the time to consider students' views. / Yours faithfully, / Yours sincerely,*

PART 1.1 PRACTICE

You are a student at a university. You received this email from the head of the university's Student Health Centre.

Dear Student,

I'm writing to let you know about an important change to our service. From next month, the Health Centre will no longer offer walk-in appointments. Instead, all students will need to book through our new online system, and most first appointments will take place as video consultations with a nurse or doctor. Face-to-face appointments will still be available whenever a physical examination is needed. We believe this will reduce waiting times considerably. We would welcome your views on the change and any suggestions for making the new system work well.

Best wishes,

Dr A. Karimova

Head of the Student Health Centre

Task 1.1 — Write a letter to your friend, who is also a student at the university. Write about your feelings about the change and what you think the Health Centre should do. Write about 50 words.

Task 1.2 — Write a letter to the Head of the Student Health Centre. Write about your feelings about the change and what suggestions you have. Write about 120–150 words.

RULES for Task 1.2 (formal letter)

☐ Respond to the SAME message as Task 1.1 — cover the same two things (feelings + suggestions), but in more depth.

☐ Aim for 120–150 words — several short paragraphs, not one block of text.

☐ Use a formal register throughout: no contractions, polite and impersonal phrasing.

☐ Structure it clearly: an opening line, your reaction, your suggestion(s) in detail, and a polite closing.

☐ Open with Dear Sir/Madam (close with Yours faithfully) or Dear Dr Karimova (close with Yours sincerely).

☐ Use a wider range of grammar than Task 1.1 — reporting verbs, passive forms, and formal linking words (firstly, in addition, given that).

SAMPLE ANSWER

Dear Dr Karimova,

Thank you for informing students about the planned changes to the Health Centre's appointment system. I welcome the aim of reducing waiting times, as I have often waited over an hour for a walk-in appointment. However, I must admit that I have some concerns about relying so heavily on video consultations.

Firstly, not all students have a quiet, private place to discuss health problems, particularly those who share rooms in halls of residence. I would therefore suggest providing a few private booths on campus where students can join video calls confidentially.

Secondly, it would be beneficial to keep a small number of same-day appointments for urgent cases, since some illnesses cannot wait for an online booking slot. In addition, clear instructions on using the new system should be sent to international students in several languages.

Thank you for taking the time to consider students' views. I look forward to hearing how the new system develops.

Yours sincerely,

N. Yusupova

TASK 1.2 PRACTICE — now you try, with a new message

You are a member of a local sports centre. You received this email from the centre's manager.

Dear Member,

We are pleased to announce that from 1 March, personal fitness assessments will be carried out by a new AI-powered app rather than by our instructors. The app will analyse your movements through your phone's camera, track your heart rate and create a personalised training plan for you. Instructors will remain available in the gym, but one-to-one sessions will only be offered at an extra cost. Before the change is introduced, we would like to know what you think about it.

Kind regards,

The Manager

Task 1.2 — Write a letter to the Manager. Write about your feelings about the change and what you think the centre should do. Write about 120–150 words.

SPEAKING PART 1.1

QUESTIONS

Answer the questions. Speak for about 30 seconds for each one.

1. How do you usually look after your health?
2. Do you prefer to see a doctor in person or online? Why?
3. When was the last time you visited a doctor or a clinic?
4. Do you think people in your country take too many medicines?
5. Would you trust an app to tell you what is wrong with you?
6. What do you do when you feel unwell before an important exam or event?

SAMPLE ANSWERS

Do you prefer to see a doctor in person or online? Why?

On the whole, I'd rather *see a doctor in person.* ***It's much easier to explain*** *what's wrong when someone can actually examine you.* ***That said****, for something simple like renewing a prescription, an online appointment* ***saves a lot of time****, so I can* ***see the appeal*** *for busy people.*

Do you think people in your country take too many medicines?

To some extent, yes. *A lot of people* ***reach for*** *painkillers or antibiotics the moment they feel slightly* ***under the weather****, often without asking a doctor first.* ***I think it's partly because*** *medicines are so easy to buy here, but* ***it would be better if*** *people tried rest and fluids first.*

Would you trust an app to tell you what is wrong with you?

Only up to a point. *I'd happily use one to* ***get a rough idea*** *of what might be wrong, but* ***I wouldn't rely on it completely****. Apps can't see the whole picture, so* ***for anything serious*** *I'd definitely want* ***a second opinion*** *from a real doctor.*

USEFUL LANGUAGE	
Giving a preference:	*On the whole, I'd rather... / I tend to prefer... because...*
Talking about habits:	*I usually make a point of... / I try to... whenever I can.*
Partly agreeing:	*To some extent, yes. / Only up to a point. / That depends on...*
Giving reasons:	*I think it's partly because... / The main reason is that...*
Adding a contrast:	*That said, ... / On the other hand, ... / Having said that, ...*
Health vocabulary:	*feel under the weather / get a second opinion / make a full recovery / be on the mend*

AGEING

LIVING LONGER

UNIT 10

WARM-UP SPEAKING

Discuss the questions below with a partner, or answer each one in about 30 seconds.

1. How old are the oldest members of your family? What do they do in a typical day?

2. Why do you think people today live longer than people did a hundred years ago?

3. What is the best age to retire, in your opinion? Why?

4. What can young people learn from older people, and what can older people learn from the young?

UNDERSTANDING KEY VOCABULARY

Article 1

Read the sentences. Match each word in bold to its definition (a–h).

1 The giant tortoise is famous for its remarkable **longevity**, often living for more than 150 years.

2 Poor **sanitation** in the overcrowded camp led to an outbreak of disease within weeks.

3 The doctors explained that the new treatment could **prolong** his life by several years.

4 The village held a celebration for its three **centenarians**, the oldest of whom had just turned 104.

5 Office workers with a **sedentary** lifestyle are advised to stand up and stretch at least once an hour.

6 Rice is the **staple** food for more than half of the world's population.

7 My grandfather loves sweets, but he has always eaten them in **moderation**.

8 Her **resilience** after losing her job impressed everyone; within months she had started her own business.

a (adj) involving a lot of sitting and very little physical activity

b (n) the ability to recover quickly from difficulties, illness or shock

c (n) long life; the fact of living for a long time

d (adj) main or most important; eaten or used regularly

e (n) the quality of avoiding extremes; not having too much of something

f (v) to make something last for a longer time

g (n) people who are a hundred years old or older

h (n) systems that provide clean water and remove waste in order to protect public health

Complete the sentences with the correct form of a word from either article.

1. The _____ of the population is one of the biggest challenges facing European governments. (age)

2. Thanks to improved _____ conditions, cholera has almost disappeared from the city. (sanitation)

3. Children who grow up with grandparents nearby are often more emotionally _____. (resilience)

4. After her _____, my grandmother began volunteering at the local library three days a week. (retirees)

5. Most doctors agree that drinking coffee _____ is unlikely to cause any harm. (moderation)

THE SECRETS OF A LONG LIFE

1 In 1900, a baby born in most European countries could expect to live to around fifty. Today, the same child might reasonably hope to reach eighty or beyond, and in Japan average life expectancy now exceeds eighty-four. This dramatic rise in **longevity** is one of the great achievements of the modern era, yet it happened so gradually that few people noticed it taking place. Understanding where those extra decades came from tells us a great deal about what we might do to gain a few more.

2 Surprisingly, much of the early progress had little to do with hospitals or advanced medicine. Improvements in **sanitation** — clean drinking water, proper sewage systems and safer food storage — sharply reduced deaths from infectious diseases such as cholera and typhoid. Because these illnesses killed so many young children, preventing them raised average life expectancy far more than any later treatment for heart disease or cancer. Vaccination programmes and, later, antibiotics continued the trend, turning once-fatal childhood illnesses into rare and manageable events.

3 In the second half of the twentieth century, the focus shifted towards older adults. Medical advances made it possible to **prolong** the lives of people with conditions that would once have been fatal: blood pressure medication, heart surgery and cancer screening allowed patients to live with illness for years rather than months. Critics point out, however, that adding years to life is not the same as adding life to years. Many people now spend their final decade managing several chronic conditions at once.

4 This distinction led researchers to look for places where people not only live long but also stay healthy. In the early 2000s, a team of scientists identified five regions — Okinawa in Japan, Sardinia in Italy, Nicoya in Costa Rica, Ikaria in Greece and Loma Linda in California — where an unusually high proportion of residents become **centenarians**. They called these regions 'blue zones', after the blue ink first used to circle them on a map, and began to study what made them different.

5 What do these communities have in common? Remarkably, none of them relies on gyms or fashionable diets. Instead, physical activity is built into everyday life: residents garden, walk to visit neighbours and climb hills to reach their fields, so they rarely lead a **sedentary** existence. Their diets are largely plant-based, with beans, whole grains and vegetables forming the **staple** foods, while meat is eaten only occasionally. Many also eat in **moderation**; Okinawans, for instance, traditionally stop eating when they feel about eighty per cent full.

6 Just as important, it seems, are the social and emotional sides of life. Older people in blue zones typically live close to their families, remain respected members of their communities and have a clear sense of purpose — a reason to get up in the morning. Strong social networks appear to build **resilience**, helping people cope with stress, illness and loss. Loneliness, by contrast, has been linked to health risks comparable to those of smoking, which suggests that friendship may be as vital to a long life as diet.

7 Not everyone accepts the blue-zone story uncritically. Some researchers argue that birth records in certain regions are unreliable, and that a few supposed centenarians may simply have been younger than they claimed. Others warn that genes play a part that lifestyle cannot fully replace. Even so, the broader lessons — move naturally, eat mostly plants, stay connected — are supported by a large body of evidence. There is no guarantee that these habits will help anyone outlive their parents, but small daily choices may decide how well we spend the years we have.

Match headings A–H to paragraphs 1–7 of Article 1. There is one extra heading you do not need.

A The power of belonging

B Clean water before clever medicine

C Why genes are the whole story

D Reasons for caution

E An achievement that went almost unnoticed

F Everyday movement and simple food

G Longer lives, but not always healthier ones

H Mapping the world's longest-lived communities

LANGUAGE DEVELOPMENT - Talking about age and later life

VOCABULARY SKILL

Verb + noun collocations

Many verbs combine with a fixed set of nouns when we talk about age, work and family responsibilities — for example **lead an active lifestyle**, **take early retirement** or **shoulder the burden** of caring for a relative. Learning these combinations as complete chunks makes your speaking and writing sound more natural and precise, and helps you avoid unnatural choices such as *make a pension* or *do a lifestyle*.

lead	an active / a sedentary lifestyle, a full life
draw	a pension, benefits
take	early retirement, responsibility for someone
raise	the retirement age, a family
bridge	the generation gap, the divide
shoulder	the burden, the cost

Complete the sentences with the correct verb from the table above. Use the correct form.

1. Although she is eighty, my grandmother still _____ an active lifestyle and swims three times a week.

2. After thirty years at the bank, he decided to _____ early retirement at the age of fifty-five.

3. The government plans to _____ the retirement age to sixty-seven over the next decade.

4. Workplace mentoring schemes can help to _____ the generation gap between younger and older employees.

5. She began to _____ a pension when she turned sixty, although she continued working part-time.

6. In many families, one daughter ends up having to _____ the burden of caring for both parents.

Article 2

2 The sentences below are about ageing societies. Choose the correct meaning (a, b or c) for the word in bold.

1. Falling birth rates are causing a major **demographic** shift in many countries.

a related to the structure of a population
b related to the national economy
c related to political parties

2. Rising healthcare costs are putting a serious **strain** on the national budget.

a a type of tax
b pressure that causes difficulties
c a sudden increase

3. Many employees have decided to **postpone** retirement until they are sixty-eight.

a look forward to
b cancel completely
c delay until a later time

4. Almost a third of the country's **workforce** will be over fifty within a decade.

a all the people who work in a country or organisation
b the managers of a company
c the people who receive pensions

5. Caring for elderly relatives can become a heavy **burden** for a single family member.

a a pleasant duty
b a financial reward
c a difficult responsibility

6. **Intergenerational** programmes bring teenagers and pensioners together to share skills.

a involving people from different countries
b involving people of different ages
c involving people with the same job

7. Shared traditions help to maintain social **cohesion** in a rapidly changing city.

a the state of people being united
b competition between groups
c rules imposed by the government

8. The childcare that grandparents provide is often **undervalued** because no money changes hands.

a paid too much
b completely ignored by law
c not given enough recognition

WHO WILL CARE FOR AN AGEING WORLD?

1. For most of human history, societies were shaped like pyramids: a broad base of children and young adults supported a narrow peak of older people. That shape is now changing rapidly. As people live longer and families have fewer children, the **demographic** pyramid in many countries is turning into something closer to a column. By 2050, according to United Nations estimates, one in six people worldwide will be over sixty-five, and in parts of Europe and East Asia the figure will be closer to one in three. This transformation raises difficult questions about money, work and care.

2. The most immediate concern is pensions. Most state pension systems were designed on a simple principle: today's workers pay for today's retirees. When there were five or six workers for every pensioner, this arrangement worked well. Now that the ratio is falling towards two to one in some countries, it is placing considerable **strain** on public finances. Governments face an uncomfortable choice between raising taxes on a shrinking working population, reducing pension payments, or asking people to work for longer. None of these options is popular, and attempts to introduce them have led to strikes and protests in several countries.

3. Increasingly, the third option is the one being chosen. Many countries have gradually raised the official retirement age, and a growing number of people choose to **postpone** retirement voluntarily. Supporters argue that this makes sense: people are healthier at sixty-five than their grandparents were, and keeping experienced employees in the **workforce** helps companies retain valuable knowledge. Critics, however, point out that not every job can be done at seventy. A university lecturer may happily continue teaching, whereas a builder or a nurse working night shifts may simply be physically unable to carry on.

4. Pensions and work are only part of the picture. As people live longer, the need for long-term care rises too. In much of Western Europe and North America, this care is increasingly provided by professional services such as nursing homes, which are expensive and often short of staff. Where such services are unavailable or unaffordable, the **burden** usually falls on families, and in particular on middle-aged women, who may find themselves looking after their own children and their elderly parents at the same time — a situation often described as being part of the 'sandwich generation'.

5. In Uzbekistan, as in much of Central Asia, the picture looks rather different. Multigenerational households, in which grandparents, parents and children live under one roof, remain common, and it is traditionally the youngest son and his wife who stay in the family home to look after his parents. These **intergenerational** arrangements bring clear benefits: older people are rarely isolated, and grandparents play a central role in raising grandchildren, passing on language, customs and values. Respect for elders, shown in everyday gestures and in the advice sought from the *oqsoqol* of the *mahalla*, helps to strengthen social **cohesion** in the wider community.

6. Such traditions are not immune to change. Urbanisation and migration for work mean that more young families now live in apartments far from their parents, and the extended household is less universal than it once was. Nevertheless, the multigenerational model offers lessons that other societies are beginning to recognise. The unpaid contribution of older people — caring for grandchildren, volunteering, advising younger relatives — is often **undervalued** because it does not appear in economic statistics. If ageing societies are to thrive, they may need to see older citizens less as a cost to be managed and more as a resource to be valued.

DISCUSSION

Work with a partner. Discuss the questions.

1 What are the advantages and disadvantages of three generations living in the same house? Would you like to live this way when you have your own family?

2 Should people be allowed to choose when they retire, or should there be a fixed retirement age for everyone? Give reasons.

GRAMMAR

Concession and contrast

Academic writers rarely present only one side of an issue. Language of concession allows them to accept that an opposing fact or argument is true, while still showing that their main point is stronger. Language of contrast shows how two facts or situations differ.

The rules below explain how to use the most common expressions of concession and contrast, and which grammatical structures follow them.

1. Use **although / even though** + a full clause (subject + verb). *Even though* is more emphatic. The clause can come first (followed by a comma) or second.

Although life expectancy has risen, many people spend their final years in poor health.

2. Use **despite / in spite of** + a noun, a pronoun or an *-ing* form — never a full clause. To add a clause, use **despite the fact that**.

Despite living alone, my great-aunt rarely feels lonely. / In spite of the fact that she lives alone, she rarely feels lonely.

3. Use **whereas / while** + clause to contrast two different facts or situations, usually within one sentence.

Okinawans eat mainly vegetables, whereas many Western diets are high in red meat.

4. Use **however / nevertheless / nonetheless** to link two separate sentences. They usually come at the start of the second sentence, followed by a comma. *Nevertheless* means 'despite what has just been said'.

Raising the pension age is deeply unpopular. Nevertheless, many governments see no realistic alternative.

5. Use **admittedly / it is true that / granted** to concede a point, then follow it with **but / however / even so** to return to your main argument.

Admittedly, some older workers are slower to adopt new technology. However, they often bring decades of valuable experience.

6. Use **even so / all the same** at the start of a sentence to introduce a surprising contrast with what has just been said.

The village has no hospital and few doctors. Even so, its residents are among the healthiest in the country.

Exercise A — Complete the paragraph with although, despite, admittedly, nevertheless, whereas, in spite of, however and while. Use each word or phrase once.

(1) _____ people are living longer than ever before, few of us plan carefully for old age. Many young people save nothing for retirement, (2) _____ knowing that state pensions may be lower in the future. (3) _____, saving can seem impossible for those on low incomes. (4) _____, financial advisers insist that even small regular amounts make a significant difference over forty years. In some countries, older people depend mainly on their pensions, (5) _____ in Uzbekistan many still rely on support from their children. (6) _____ the rising cost of living, most families continue to care for elderly relatives at home. Multigenerational living has many benefits. (7) _____, it can also create tension when different generations hold very different values. (8) _____ some young couples now prefer to live independently, the majority still see caring for their parents as a duty rather than a burden.

Exercise B — Combine each pair of sentences using the word or phrase in brackets. Make any changes that are necessary.

1. My grandfather is eighty-five. He still works in his garden every day. (although)

2. The village has very few doctors. Its residents live unusually long lives. (despite)

3. Younger workers adapt quickly to new technology. Older workers often have better judgement. (whereas)

4. Raising the retirement age will save the government money. It is deeply unpopular with voters. (nevertheless)

5. There was very little space in the apartment. The family was happy to welcome their grandmother. (in spite of)

ACADEMIC WRITING

Counter-arguments and refutation — answering the other side

ACADEMIC WRITING SKILL

Counter-arguments and refutation

A persuasive argument becomes stronger, not weaker, when the writer acknowledges what opponents believe. First present the opposing view fairly (*Opponents argue that… / It is sometimes claimed that…*), then refute it by showing its weakness (*However, this overlooks… / This argument fails to take into account… / While this may be true, …*). The refutation should be supported by a reason, a fact or an example rather than a simple denial.

COUNTER-ARGUMENT: *Opponents argue that raising the retirement age is unfair to people in physically demanding jobs.*

REFUTATION: *However, this overlooks the fact that flexible schemes could allow such workers to move into lighter roles or reduce their hours, rather than leave work altogether.*

Each sentence below presents a counter-argument. Write a refutation of one or two sentences, using a phrase such as However, this overlooks… / This argument fails to take into account… / While this may be true, …

1 Opponents argue that older employees take jobs away from young people.

_____

2 Some people claim that living with grandparents limits a young couple's independence.

_____

3 It is often argued that pensions should be reduced because the state can no longer afford them.

_____

4 Critics maintain that the blue zones prove nothing, since birth records there are unreliable.

_____

5 Some say that professional care homes provide better care than families can.

_____

6 Many believe that older people are simply unable to learn how to use new technology.

_____

WRITING TASK

Write a short paragraph (80–100 words) arguing that older people should be encouraged to keep working after the official retirement age. Include at least one counter-argument with a clear refutation, and use at least two expressions of concession or contrast from the Grammar section.

READING PRACTICE

PART 3

Read the text and choose the correct heading for each paragraph from the list of headings below. There are more headings than paragraphs, so you will not use all of them. You cannot use any heading more than once.

List of Headings:

A) Young helpers

B) Importance of discipline

C) Home education

D) Awkward moments

E) A warm welcome

F) Daddy day care

G) Sibling rivalry

H) Methods of persuasion

15. Paragraph I
16. Paragraph II
17. Paragraph III
18. Paragraph IV
19. Paragraph V
20. Paragraph VI

I. Family reunions can be a joyous occasion and an opportunity to meet new family members as well. Our parents have aunts, uncles and cousins that we may only meet once or twice in life. Often, though, meeting an extended family member can be like meeting a complete stranger. You're not even sure you will like them, and the whole affair can feel a bit odd or uncomfortable.

II. Traditionally, the mother is the primary caregiver to an infant child. She takes time off work and stays at home to provide round-the-clock attention during the child's early years. But recently, more and more men are taking paternity leave from work to stay at home with their new-borns. Mothers find it a welcome trend, as caring for an infant can be exhausting work.

III. While it's true that kids must carry out their responsibilities just as adults do, sometimes Mum and Dad have to be a bit crafty to make it happen. They have to offer a little reward for a job well done. It may not be good practice to do it all the time, but if they promise a trip to the cinema, for example, the kids will be more likely to clean their rooms.

IV. Many families live far away from one another these days. They may have moved across the country, or even to the other side of the world. It's especially hard for parents to see their children do this, either for work or for adventure. So when special occasions come round, especially the holiday seasons, Mum and Dad are so grateful when their sons and daughters return home.

V. Sometimes a household is nothing but chaos for days on end. This is especially true when kids reach their adolescent phase, and you have a couple of teenagers who compete over every little thing in life. No matter how often parents attempt to stop two kids from fighting each other, nothing seems to work. They just have to wait until one of them moves away to university!

VI. In most families, parents have to constantly remind their children of their responsibilities. But there are times when kids can be especially useful round the house of their own will. They also understand those times when Mum or Dad isn't feeling well, and maybe it would be nice if they made dinner for a change or did a little extra tidying up in the living room.

✔ Wordlist

1. family reunion ▢
2. joyous occasion ▢
3. extended family member ▢
4. a complete stranger ▢
5. the whole affair ▢
6. primary caregiver ▢
7. infant ▢
8. take time off work ▢
9. round-the-clock attention ▢
10. paternity leave ▢
11. new-born ▢
12. a welcome trend ▢
13. exhausting ▢
14. carry out (responsibilities) ▢
15. crafty ▢
16. a job well done ▢
17. good practice ▢
18. be more likely to ▢
19. far away from one another ▢
20. special occasions ▢
21. come round ▢
22. grateful ▢
23. nothing but chaos ▢
24. for days on end ▢
25. adolescent phase ▢
26. compete over ▢
27. no matter how often ▢
28. of their own will ▢
29. for a change ▢
30. tidying up ▢

PART 6

You will hear a part of a lecture. For each question, fill in the missing information in the numbered space.
Write no more than one word for each answer.

THE GRAND UNION CANAL

The canals were built despite the fact that there was no public 30_____ and very little technology.

From 1790 to 1929, there were many canals that were 31_____ with each other but which were not uniform in size.

The new union of canals provided a 32_____ link between major industrial cities.

There are plenty of natural habitats for a variety of wildlife on and around the canals.

Walkers can go to the nearest waterway office to get information on circular 33_____ so they can start and end at the same place.

Anglers can fish in the canals and 34_____.

If you want to go fishing, you must buy a rod license.

The waterways authority request that people are 35_____ towards the canal users.

Phrases & Idioms Wordlist

Over the hill: Past one's best; considered too old for something (often used humorously).

Long in the tooth: Old, especially too old for a particular activity or job.

Young at heart: Feeling and behaving in a youthful, energetic way despite being old.

Age is just a number: A person's age need not limit what they can do or enjoy.

Getting on (in years): Becoming old.

Live to a ripe old age: To live until one is very old.

In one's prime: At the period of life when one is strongest, healthiest or most successful.

Act one's age: To behave in a way that is suitable for someone of one's age, especially more maturely.

Turn back the clock: To return to an earlier time, or to try to make things as they were in the past.

Old habits die hard: It is very difficult to stop doing something you have done for a long time.

Time flies: Time seems to pass very quickly.

One's twilight years: The last part of a person's life.

A senior moment: A short lapse of memory, humorously linked to getting older.

Set in one's ways: Unwilling to change one's habits or opinions, often because of age.

The generation gap: The differences in attitudes, values and tastes between younger and older people.

USEFUL LANGUAGE — Opinion essay

Function	Useful phrases
Introducing the topic	*It is often argued that… / In recent years, there has been growing debate about…*
Stating your opinion	*In my view, … / I would argue that… / I am firmly convinced that…*
Adding supporting points	*Furthermore, … / In addition, … / Another key advantage is that…*
Presenting a counter-argument	*Opponents argue that… / It is sometimes claimed that… / Admittedly, …*
Refuting	*However, this overlooks… / This argument fails to take into account… / Nevertheless, …*
Concluding	*To conclude, … / On balance, … / All things considered, …*

THE TASK

You are writing an essay for an academic audience. Write 180–200 words.

In many countries, people are now expected to work until they are sixty-five or even older. Some people think this is a positive development, while others believe it is unfair. Discuss both views and give your own opinion.

PARAGRAPH PLAN

Introduction	paraphrase the question + state your position briefly
Body paragraph 1	arguments in favour of working longer, with reasons and examples
Body paragraph 2	the opposing view (counter-argument) + your refutation
Conclusion	summarise both views and restate your opinion in new words

RULES for Part 2 (essay)

☐ Answer ALL parts of the question — discuss both views AND give your own opinion.

☐ Write 180–200 words in four clear paragraphs: introduction, two body paragraphs and a conclusion.

☐ State your position in the introduction and keep it consistent throughout the essay.

☐ Include at least one counter-argument and refute it with a reason or an example.

☐ Use a formal, academic register: no contractions, no slang, and avoid 'I think' in every sentence.

☐ Show a range of grammar, including concession and contrast (although, despite, whereas, nevertheless).

SAMPLE ANSWER

In many countries, the official retirement age has risen to sixty-five or beyond, and people are increasingly expected to remain in work for longer. While some welcome this trend, others regard it as deeply unfair.

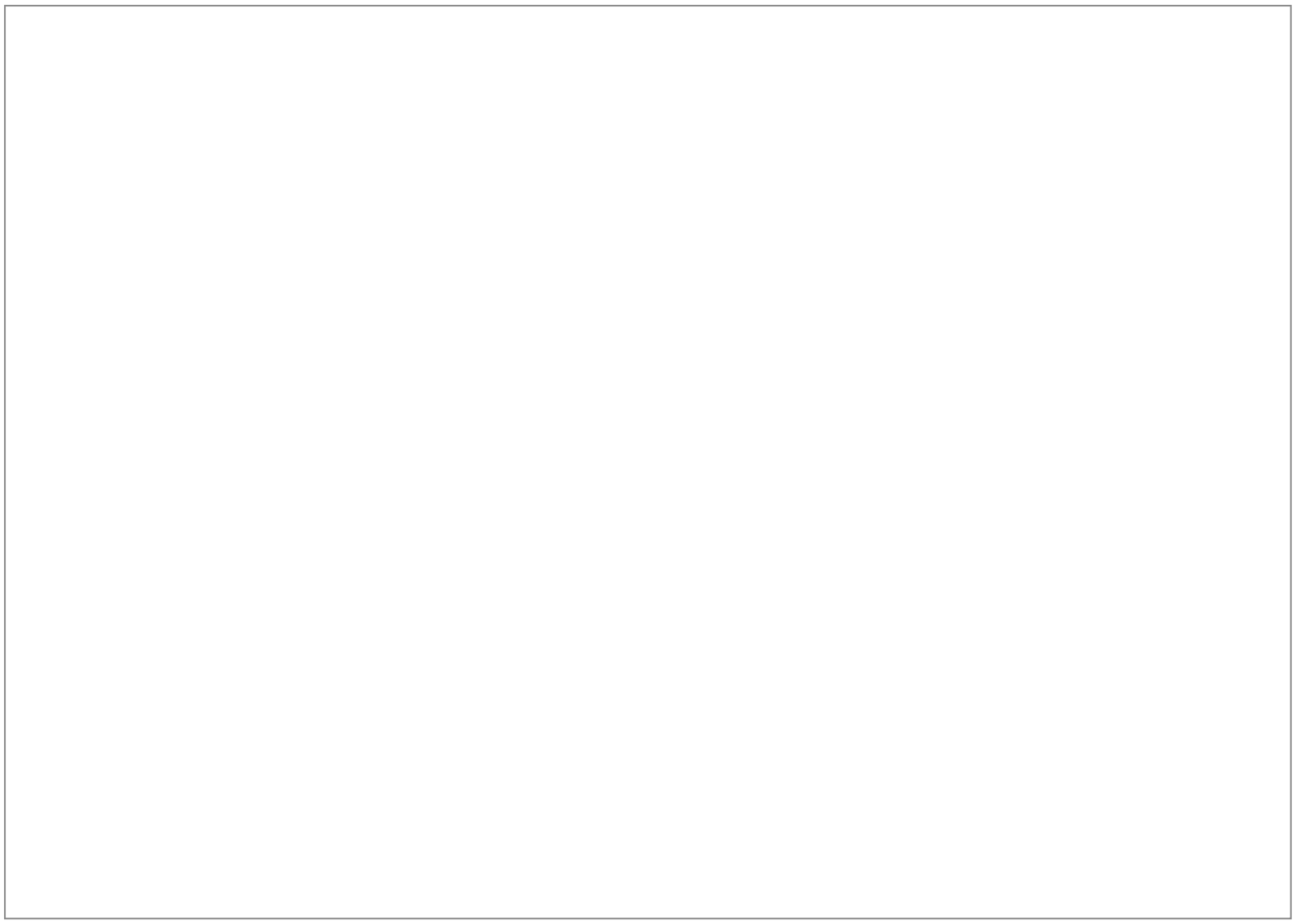
Those in favour point out that people today are considerably healthier than previous generations were at the same age, so many are perfectly capable of continuing to work. Keeping experienced employees in the workforce also allows companies to benefit from their knowledge, and it reduces the pressure on pension systems that are already struggling to cope.

Opponents argue that a higher retirement age punishes people in physically demanding jobs, who may be exhausted long before they reach sixty-five. Admittedly, this is a serious concern. However, it overlooks the possibility of flexible schemes, which could allow such workers to move into lighter roles or reduce their hours gradually rather than stop altogether.

In my view, working longer is a sensible response to longer lives, provided that it is not imposed in the same way on everyone. Although no system will satisfy every worker, a flexible approach would protect both public finances and the dignity of older people.

PART 2 PRACTICE — now you try

Some people believe that elderly parents should live with their adult children, while others think they are better cared for in professional care homes. Discuss both views and give your own opinion.



SPEAKING PART 3

STATEMENT

Elderly people should live with their families rather than in care homes.

Discuss both sides of the statement and give your own opinion. Use the points below to help you.

FOR	AGAINST
older people feel loved and are rarely lonely	families may lack the medical skills needed
grandparents help raise children and pass on traditions	carers, often women, can become exhausted
it is cheaper than paying for professional care	small city apartments may not have enough space

SAMPLE ANSWER

*I'd say I **largely agree with this statement**, although I can see that it isn't the right solution for every family.*

*The strongest argument in favour is emotional. **There's no doubt that** older people who live with their children and grandchildren feel valued and are far less likely to suffer from loneliness, which, as we know, can seriously damage health. **On top of that**, grandparents play a huge role in bringing up children — in my own family, my grandmother taught us our first songs and recipes. It's also **worth mentioning** that family care is usually much cheaper than a care home.*

***Having said that**, there are some convincing arguments on the other side. **Admittedly**, a family can't always provide the medical care that someone with a serious illness needs, and the person who does most of the caring — usually a daughter or daughter-in-law — can become completely exhausted. **What's more**, many young couples in cities now live in small apartments, so there simply isn't enough room.*

***All things considered**, I think living with the family is the best option **in most cases**, because it benefits every generation. **Nevertheless**, I believe professional care should be available, and families shouldn't feel ashamed of using it when an elderly relative's needs become too great for them to manage at home.*

USEFUL LANGUAGE	
Giving your overall view:	*I largely agree with... / I'm not entirely convinced that...*
Presenting arguments for:	*The strongest argument in favour is... / There's no doubt that...*
Adding points:	*On top of that, ... / What's more, ... / It's also worth mentioning that...*
Moving to the other side:	*Having said that, ... / On the other hand, ... / Admittedly, ...*
Giving examples:	*In my own family, ... / Take my grandparents, for instance...*
Concluding:	*All things considered, ... / On balance, I'd say... / Nevertheless, I believe...*

ANSWER KEY

Unit 1 · Conservation

Article 1 vocabulary	1 b 2 a 3 b 4 b 5 a 6 b 7 c 8 b
Match the headings	1 F 2 B 3 A 4 D 5 E (C not needed)
Compound adjectives	1 energy-efficient 2 long-term 3 out-of-date 4 state-of-the-art
Article 2 vocabulary	1 c 2 f 3 e 4 a 5 d 6 b 7 g 8 h
Word forms (either article)	1 demolished 2 vulnerability 3 adaptation 4 compatibility 5 affluence
Grammar Exercise A	1 a 2 b 3 a
Grammar Exercise B	1 doesn't find 2 will probably face 3 were 4 would turn 5 chose 6 would need 7 changes 8 will lose
Academic writing (model answers)	1 It is often argued that cities spend too much money restoring old buildings. 2 It is widely believed that digital records are just as important as physical ones. 3 It is often suggested that governments should protect more historic sites. 4 It is generally felt that families should preserve old photographs more carefully. 5 It is argued by some critics that not every old building deserves to be saved.
Reading Part 1	1 game 2 available 3 video 4 planet 5 players 6 connect
Reading Part 2 (cafés)	7 B 8 D 9 E 10 J 11 A 12 C 13 H 14 F (G, I not needed)
Listening Part 1	1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____ 7 _____ 8 _____
Listening Part 2	9 _____ 10 _____ 11 _____ 12 _____ 13 _____ 14 _____

Unit 2 · Design

Article 1 vocabulary	1 b 2 d 3 e 4 f 5 g 6 c 7 a 8 h
Word forms	1 sustainability 2 associated 3 refinements 4 reinforcement 5 criterion
Match the headings (Article 1)	1 A 2 B 3 D 4 E 5 F 6 G 7 H (C not needed)
Language development: collocations	1 provoked 2 inspire 3 aroused 4 evoke
Article 2 vocabulary	1 b 2 a 3 c 4 a 5 b 6 b 7 b
Grammar Exercise A	1 who/that 2 – (or that/which) 3 – (or who/that) 4 – (or that/which) 5 that/which 6 – (or that/which) 7 – (or that/which) 8 that/which
Grammar Exercise D	1 The design studio where our designers create new prototypes is in the old power station. 2 The rebrand leading to instant recognition cost the company millions. 3 This is the exhibition hall where visitors can see the winning designs. 4 The new logo appealing to a younger audience has boosted sales significantly.
Academic writing: paraphrasing (model answers)	1 A logo needs to be adaptable enough to work in different places and to stay relevant over the years. 2 At first, people usually react negatively when a logo they know well is changed. 3 Above all, a logo must communicate a narrative. 4 For some firms, a fresh logo is simply a way of signalling that the business has grown up. 5 A logo that is easy to recognise lets a brand get noticed straight away, even among many competitors.
Reading practice Part 3 (Kushan Empire)	15 B 16 H 17 D 18 E 19 F 20 C
Listening podcast: match the two parts	1 b 2 h 3 d 4 c 5 f 6 e 7 g 8 a
Listening podcast – Task 1	1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____
Listening Part 3	15 _____ 16 _____ 17 _____ 18 _____

Unit 3 · Technology

Article 1 vocabulary	1 a 2 b 3 c 4 d 5 e 6 f
Word forms	1 humiliating 2 hostility 3 accountability 4 malice 5 regulation
Match the headings	1 A 2 B 3 C 4 D 5 E 6 F 7 G (H not needed)
Collocations (verb + noun)	1 take responsibility 2 built a reputation 3 suffer/experience abuse 4 exhibit (hostile) behaviour 5 suffer/experience pain 6 lose confidence
Article 2 vocabulary	1 c 2 b 3 c 4 a 5 a 6 b 7 a 8 c
Grammar Exercise A	1 It is believed that these figures underestimate the true scale of the problem. / These figures are believed to underestimate the true scale of the problem. 2 It is claimed that anonymous accounts are responsible for most abusive comments. / Anonymous accounts are claimed to be responsible for most abusive comments. 3 It is expected that victims of harassment will eventually stop using the platform. / Victims of harassment are expected to stop using the platform eventually. 4 It is understood that self-regulation is the most realistic short-term solution. / Self-regulation is understood to be the most realistic short-term solution. 5 Ignoring the harassment is considered (to be) sensible advice for victims. 6 It has been argued that these problems will require a stronger legal response.
Grammar Exercise B	1 Although the account had been posting abusive comments for weeks, it was not suspended until yesterday. 2 A full year after the harassment began, the account holder's real identity was discovered (by a journalist). 3 These trolls were not very careful. Evidence of their activity was found quite quickly. 4 The news usually includes positive stories about technology, but in recent months, it has been dominated by stories of online abuse.
Academic writing (F, S or E)	1 S 2 E 3 S 4 F 5 E 6 E 7 F 8 F
Reading Part 4 (Penguins)	21 A 22 B 23 C 24 C 25 B (False) 26 B (False) 27 A (True) 28 C (No information) 29 C (No information)
Listening Part 4	19 _____ 20 _____ 21 _____ 22 _____ 23 _____

Unit 4 · Risk

Article 1 vocabulary	1 d 2 b 3 e 4 h 5 f 6 a 7 g 8 c
Word forms	1 perception 2 probable / likely 3 intuition 4 mitigation 5 complacency
Match the heading	1 B 2 G 3 E 4 H 5 C 6 A 7 F (D is not needed)
Language development	1 weigh up / assess, risks / options / odds 2 outweighed / outweigh, risks / costs / drawbacks 3 pose, threat / risk / danger 4 reduce / minimise, risk 5 taking / running, risk / chance 6 raises / increases, risk / likelihood
Article 2 vocabulary	1 a 2 b 3 c 4 a 5 b 6 c 7 a 8 b
Grammar – Exercise A	1 Had 2 would be earning 3 wouldn't / would not feel 4 Had 5 Should 6 were / was 7 Were 8 Were
Grammar – Exercise B	1 Had I known about the storm, I would have cancelled the trip. 2 Should you have any problems with your equipment, contact the instructor immediately. 3 Were it not for his helmet, he would be in hospital now. 4 Were the company to invest more in safety, there would be fewer accidents. 5 Had they not checked the weather forecast, they would be stuck on the mountain now. 6 Had it not been for the rescue team, the climbers would have died.
Reading practice	30 figs 31 species 32 normalising 33 negative 34 D 35 A
Listening practice	24 _____ 25 _____ 26 _____ 27 _____ 28 _____ 29 _____

Unit 5 · Manufacturing

Article 1 vocabulary	1 d 2 h 3 f 4 b 5 g 6 e 7 a 8 c
Word forms	1 precisely 2 obsolescence 3 customisation 4 deployment 5 viability
Match the heading	1 B 2 E 3 G 4 C 5 H 6 A 7 F (D not needed)
Language development	1 boost 2 phase out 3 meet 4 cut 5 scale up 6 streamline
Article 2 vocabulary	1 b 2 c 3 b 4 a 5 c 6 a 7 c 8 b
Grammar – Exercise A	1 must be checked 2 can be corrected 3 can be sent 4 might be split 5 had its printers upgraded 6 should have been avoided 7 must be inspected 8 have/get them delivered
Grammar – Exercise B	1 The machines must be cleaned at the end of every shift. 2 We will have our robotic arm repaired (by a technician) tomorrow. 3 The software should have been tested before the new line was launched. 4 Replacement parts can be printed on site in a few hours. 5 We are going to get solar panels installed on our factory roof (by a local company).
Reading practice	TEST 3: 1 senses 2 dolphins 3 electricity 4 current 5 detect 6 seventh TEST 4: 1 itch 2 bacteria 3 skin 4 mice 5 drug 6 eczema
Listening practice	30 _____ 31 _____ 32 _____ 33 _____ 34 _____ 35 _____

Unit 6 · Environment

Article 1 vocabulary	1 d 2 b 3 g 4 f 5 h 6 a 7 c 8 e
Word forms	1 disappearance 2 dependent 3 restoration 4 sceptical 5 resilience
Match the heading	1 B 2 E 3 G 4 A 5 H 6 F 7 C (D is not needed)
Language development	1 depleted 2 pose 3 raise 4 taken 5 reverse 6 conserve
Article 2 vocabulary	1 a 2 b 3 b 4 a 5 c 6 a 7 b 8 c
Grammar – Exercise A	1 stem from 2 led to 3 Owing to 4 The disappearance 5 Consequently 6 result in 7 thanks to 8 reduction
Grammar – Exercise B	1 The diversion of rivers for cotton farming resulted in the shrinkage of the Aral Sea. 2 Many illnesses in the region stem from the pollution of the air with salt and dust. 3 Owing to the sharp decrease in rainfall, the harvest was poor. 4 As a consequence of the reintroduction of wolves, the elk changed their behaviour. 5 The construction of dams by beavers leads to a reduction in flooding downstream.
Reading practice	7 J 8 B 9 G 10 E 11 I 12 F 13 H 14 D (A and C are not needed)
Listening practice	Part 1: 1 _____ 2 _____ 3 _____ 4 _____ 5 _____ 6 _____ 7 _____ 8 _____ Part 2: 9 _____ 10 _____ 11 _____ 12 _____ 13 _____ 14 _____

Unit 7 · Architecture

Article 1 vocabulary	1 c 2 g 3 e 4 b 5 f 6 h 7 d 8 a
Word forms	1 intimacy 2 authentic 3 efficiency 4 demolish 5 pedestrianised
Match the heading	1 C 2 G 3 E 4 B 5 H 6 A 7 F (D is not needed)
Language development	1 revitalise 2 preserve 3 tackle 4 restoring 5 enhance 6 demolish
Article 2 vocabulary	1 b 2 a 3 c 4 a 5 c 6 b 7 b 8 a
Grammar – Exercise A	1 What 2 that/which 3 do 4 also 5 have 6 did 7 What/All 8 why/that
Grammar – Exercise B	1 What gives the old city its unique character is its narrow streets. 2 It was local volunteers who/that rescued the abandoned caravanserai. 3 Not only does the new park reduce traffic noise, but it also attracts wildlife. 4 Only after the old bazaar had been demolished did we understand its value. 5 Seldom have developers shown such respect for a historic site. 6 What I love is the way the afternoon light falls across the courtyard.
Reading practice	15 H 16 G 17 B 18 E 19 C 20 A
Listening practice	15 _____ 16 _____ 17 _____ 18 _____

Unit 8 · Energy

Article 1 vocabulary	1 b 2 h 3 f 4 g 5 c 6 e 7 a 8 d
Word forms	1 viability 2 intermittency 3 insulate 4 fluctuations 5 subsidised
Match the heading	1 B 2 G 3 F 4 C 5 A 6 H 7 E (D is not needed)
Language development	1 generate 2 meet 3 phase out 4 harness 5 cut / reduce 6 store
Article 2 vocabulary	1 a 2 b 3 c 4 a 5 b 6 c 7 a 8 b
Grammar – Exercise A	1 is likely to change 2 will have overtaken 3 is bound to fall 4 is set to open 5 will be supplying 6 will not (won't) disappear 7 will still be using 8 is unlikely to reach
Grammar – Exercise B	1 Electricity prices are bound to rise again this winter. 2 Three new wind farms are set to open next year. 3 More people are likely to buy electric cars in the future. 4 By 2040, the country will have closed all its coal plants. 5 At 7 p.m. tomorrow, the battery will be supplying power to the house.
Reading practice	21 B 22 C 23 D 24 B 25 A 26 B 27 C 28 A 29 C
Listening practice	19 _____ 20 _____ 21 _____ 22 _____ 23 _____

Unit 9 · Medicine

Article 1 vocabulary	1 c 2 e 3 h 4 g 5 b 6 a 7 f 8 d
Word forms	1 resistance 2 convenient 3 accuracy 4 reproduction 5 susceptibility
Match the heading	1 D 2 B 3 F 4 H 5 C 6 A 7 G (E is not needed)
Language development	1 conduct 2 make 3 pose 4 undergo 5 raised 6 develop
Article 2 vocabulary	1 a 2 b 3 c 4 a 5 c 6 b 7 c 8 b
Grammar – Exercise A	1 found / has found 2 point out 3 claims / has claimed 4 argue 5 acknowledge 6 warned 7 suggests 8 concludes / concluded
Grammar – Exercise B	1 It has been found that resistant infections lengthen hospital stays. 2 Overuse in farming is believed to be a major cause of resistance. 3 It has been shown that video consultations save patients time and money. 4 The new drug is thought to be safer than older treatments. 5 It has been suggested that hospitals should share patient data more openly.
Reading practice	30 social 31 tags 32 equipment 33 water 34 B 35 C
Listening practice	24 _____ 25 _____ 26 _____ 27 _____ 28 _____ 29 _____

Unit 10 · Ageing

Article 1 vocabulary	1 c 2 h 3 f 4 g 5 a 6 d 7 e 8 b
Word forms	1 ageing 2 sanitary 3 resilient 4 retirement 5 moderately
Match the heading	1 E 2 B 3 G 4 H 5 F 6 A 7 D (C is not used)
Language development	1 leads 2 take 3 raise 4 bridge 5 draw 6 shoulder
Article 2 vocabulary	1 a 2 b 3 c 4 a 5 c 6 b 7 a 8 c
Grammar – Exercise A	1 Although 2 despite 3 Admittedly 4 Nevertheless 5 whereas 6 In spite of 7 However 8 While
Grammar – Exercise B	1 Although my grandfather is eighty-five, he still works in his garden every day. 2 Despite the village having very few doctors, its residents live unusually long lives. / Despite the fact that the village has very few doctors, its residents live unusually long lives. 3 Younger workers adapt quickly to new technology, whereas older workers often have better judgement. 4 Raising the retirement age will save the government money. Nevertheless, it is deeply unpopular with voters. 5 In spite of the lack of space in the apartment, the family was happy to welcome their grandmother.
Reading practice	15 D 16 F 17 H 18 E 19 G 20 A
Listening practice	30 _____ 31 _____ 32 _____ 33 _____ 34 _____ 35 _____

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